

HE2001 Microeconomics II

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Abstract

Comprehensive revision notes for HE2001 Microeconomics II, covering consumer choice, Slutsky decomposition, endowments, intertemporal choice, uncertainty, welfare measurement, competitive equilibrium, social choice, and utilitarian welfare. The chapters combine core theory, key formulas, worked examples, and exam-focused synthesis.

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Course Overview & Topic Map

Topic Area	Course Part	Related Lectures	Tutorials	Role in Notes
Chapter 1: Consumer Theory Foundations	Part I	Lecture 1	Tutorial 1	Full theory chapter on mathematical setup, consumption, preferences, utility, and revealed preference.
Chapter 2	Part I	Lecture 2	Tutorial 2	Full theory chapter.
Chapter 3	Part I	Lecture 3	Tutorial 3	Full theory chapter.
Chapter 4	Part I	Lecture 4	Tutorial 4	Full theory chapter.
Chapter 5	Part I	Lecture 5	Tutorial 5	Full theory chapter.
Chapter 6: Part I Summary	Part I	Lecture 6	Tutorial 6	Synthesis chapter summarising and connecting Part I material rather than introducing a normal new-content chapter.
Chapter 7	Part II	Lecture 7	Tutorial 7	Full theory chapter.
Chapter 8	Part II	Lecture 8	Tutorial 8	Full theory chapter.
Chapter 9	Part II	Lecture 9	Tutorial 9	Full theory chapter.
Chapter 10	Part II	Lecture 10	Tutorial 10	Full theory chapter.
Chapter 11	Part II	Lecture 11	—	Lecture-only chapter.
Chapter 12: Part II Summary	Part II	Lectures 7–11	Tutorials 7–10	Synthesis chapter consolidating the second half of the course, including Lecture 11 as lecture-only material.

“This revision guide is organized into chapter notes plus part-level synthesis chapters, preserving lecture traceability while improving revision efficiency.”

How to Use This Document

Lectures provide the core theory, definitions, models, and examinable derivations. Tutorials are used selectively as worked examples only when they sharpen technique or clarify a central idea. The summary chapters are designed to compress, compare, and connect topics within each course part rather than to function as ordinary theory chapters.

Box Legend

- **Definition / Theorem** boxes state formal concepts and results.
- **Key Formula** boxes collect expressions, identities, and compact derivations worth memorising.
- **Worked Example** boxes show representative tutorial-style applications.
- **Common Mistake** boxes flag errors that often appear under exam pressure.
- **Policy Application** boxes connect the theory to interpretation and welfare reasoning.
- **Exam Focus** boxes identify the highest-yield revision targets.

Suggested Revision Flow

1. Read the chapter prose and Definition/Theorem boxes to fix the conceptual structure.
2. Memorise the Key Formula boxes and reconstruct the algebra without notes.
3. Attempt each Worked Example before reading the solution.
4. Use the summary chapters for cross-topic comparison and final consolidation.
5. Review Common Mistake and Exam Focus boxes immediately before quizzes and tests.

Part I

Main Notes

1 Consumer Theory Foundations

This chapter establishes the formal language used throughout HE2001. The lecture begins with minimal mathematical setup, then builds the consumption model, formalises preferences, and finally shows how observed choices can be used to recover information about welfare through revealed preference. Conceptually, the chapter answers three basic questions:

1. What objects describe a consumer's choice problem?
2. What properties make a binary relation behave like a rational preference?
3. What restrictions must observed choices satisfy if they come from maximisation of an increasing preference?

1.1 Mathematical Setup

The course treats a consumer as choosing a vector of goods from a feasible set. That requires only a small amount of notation.

Definition 1.1: Basic sets, vectors, and functions

For an integer $L \geq 2$:

- $\mathbb{R}$ is the set of real numbers.
- $\mathbb{R}_+ = \{x \in \mathbb{R} : x \geq 0\}$ and $\mathbb{R}_{++} = \{x \in \mathbb{R} : x > 0\}$.
- $\mathbb{R}_+^L$ is the set of all L -dimensional nonnegative vectors.
- $\mathbb{R}_{++}^L$ is the set of all L -dimensional strictly positive vectors.
- A function $f : X \rightarrow Y$ maps each element of a set X to an element of a set Y .

In this course, typical functions include utility functions, demand functions, and cost functions.

A set records the objects under discussion; a vector records several quantities at once; a function translates one kind of economic object into another. These are the only preliminaries needed for Lecture 1.

Key Formula: Dot Product and Expenditure

If $p = (p_1, \dots, p_L) \in \mathbb{R}_{++}^L$ and $x = (x_1, \dots, x_L) \in \mathbb{R}_+^L$, then the dot product is

$$p \cdot x = \sum_{\ell=1}^L p_{\ell} x_{\ell}.$$

Economically, $p \cdot x$ is the total expenditure required to purchase bundle x at prices p .

1.2 Consumption Bundles, Prices, and Budget Sets

The primitive objects of consumer theory are the consumption bundle, the price vector, and the consumer's expenditure level.

Definition 1.2: Consumption bundle and price vector

Suppose there are L goods.

- A *consumption bundle* is a vector $x = (x_1, \dots, x_L) \in \mathbb{R}_+^L$, where x_ℓ is the amount of good ℓ .
- A *price vector* is a vector $p = (p_1, \dots, p_L) \in \mathbb{R}_{++}^L$, where p_ℓ is the price of good ℓ .

The restriction $x \in \mathbb{R}_+^L$ says negative consumption is excluded. The restriction $p \in \mathbb{R}_{++}^L$ says every good has a strictly positive price, which matters later when we convert unused expenditure into more of some good.

Definition 1.3: Budget set and budget line

If the consumer has expenditure level $m > 0$ and faces prices $p \in \mathbb{R}_{++}^L$, the *budget set* is

$$B(p, m) = \{x \in \mathbb{R}_+^L : p \cdot x \leq m\}.$$

In the two-good case, the *budget line* is the boundary

$$p_1x_1 + p_2x_2 = m.$$

The budget set contains every affordable bundle. The budget line contains bundles that exactly exhaust expenditure. In two dimensions, the budget set is the region on or below the line in the nonnegative quadrant.

Key Formula: Two-Good Budget Geometry

For $L = 2$, the budget line is

$$p_1x_1 + p_2x_2 = m.$$

Its intercepts are

$$x_1 = \frac{m}{p_1} \quad \text{when } x_2 = 0, \quad x_2 = \frac{m}{p_2} \quad \text{when } x_1 = 0.$$

The slope is

$$-\frac{p_1}{p_2}.$$

Thus relative prices determine the steepness of the line, while m shifts the feasible set outward or inward.

Theorem 1.1: Why increasing preferences push choice to the boundary

Suppose preferences are increasing and the consumer chooses $x \in B(p, m)$. Then any optimal bundle must satisfy

$$p \cdot x = m.$$

Proof. If instead $p \cdot x < m$, then because each price is strictly positive, there exists some good i and some $\varepsilon > 0$ such that $p_i \varepsilon \leq m - p \cdot x$. The bundle $y = x + \varepsilon e_i$ is still affordable and satisfies $y \succ x$. Increasing preferences imply $y \succ x$, contradicting optimality of x . $\square$

Worked Example**Budget affordability and the budget line.**

Take two goods, expenditure $m = 2$, and consider the two price vectors

$$p^{(1)} = (1, 2), \quad p^{(2)} = (2, 1).$$

The corresponding budget lines are

$$x_1 + 2x_2 = 2, \quad 2x_1 + x_2 = 2.$$

Now test the bundle $x = (1, 1)$.

$$p^{(1)} \cdot x = 1(1) + 2(1) = 3 > 2, \quad p^{(2)} \cdot x = 2(1) + 1(1) = 3 > 2.$$

Therefore $x = (1, 1)$ is not affordable under either price system.

This is the fastest algebraic way to check feasibility in an exam: compute $p \cdot x$ and compare it with m .

Common Mistake

Do not confuse the *budget set* with the *budget line*. The line is only the boundary $p \cdot x = m$; the set is the full collection of affordable bundles $p \cdot x \leq m$ together with nonnegativity.

1.3 Preferences

A consumer theory model needs a way to rank bundles. Lecture 1 uses a binary relation to represent this ranking.

Definition 1.4: Preference relation

A binary relation $\succeq$ on $\mathbb{R}_+^L$ assigns to each pair (x, x') a statement of whether x is weakly preferred to x' . The notation means:

$$x \succeq x' \quad \text{“}x \text{ is weakly preferred to } x' \text{.”}$$

From $\succeq$ we derive:

- *strict preference*: $x \succ x'$ means $x \succeq x'$ and not $x' \succeq x$;
- *indifference*: $x \sim x'$ means $x \succeq x'$ and $x' \succeq x$.

A binary relation is called a *preference* if it is complete and transitive.

Definition 1.5: Completeness and transitivity

A binary relation $\succeq$ on $\mathbb{R}_+^L$ is:

- *complete* if for every x, x' , either $x \succeq x'$ or $x' \succeq x$ (or both);
- *transitive* if $x \succeq x'$ and $x' \succeq x''$ imply $x \succeq x''$.

Completeness says the consumer can compare any two bundles. Transitivity says the ranking is logically consistent across chains of comparisons. Without completeness, some bundles may be incomparable; without transitivity, circular rankings can arise.

Definition 1.6: Increasing preferences

For bundles $x, x' \in \mathbb{R}_+^L$:

$$\begin{aligned} x \geq x' &\iff x_\ell \geq x'_\ell \text{ for all } \ell, \\ x > x' &\iff x \geq x' \text{ and } x_\ell > x'_\ell \text{ for some } \ell. \end{aligned}$$

A preference relation $\succeq$ is *increasing* if

$$x > x' \implies x \succ x'.$$

Increasingness is the formal version of “more is better.” If bundle x gives at least as much of every good and strictly more of at least one good, then it must be strictly preferred.

Worked Example**Checking completeness, transitivity, and increasingness.**

Consider the relation on $\mathbb{R}_+^2$ defined by

$$(x_1, x_2) \succeq (y_1, y_2) \iff \text{either } x_1 > y_1, \text{ or } x_1 = y_1 \text{ and } x_2 \geq y_2.$$

This is a lexicographic-type ranking.

- **Complete:** for any two bundles, either the first coordinate differs, or the first coordinates are equal and the second coordinate decides the comparison.
- **Transitive:** if x beats y by the first coordinate, and y beats z , then x also beats z ; if first coordinates tie, the second coordinate preserves the ordering.
- **Increasing:** whenever $x \geq y$ and $x \neq y$, either $x_1 > y_1$, or $x_1 = y_1$ and $x_2 > y_2$, so $x \succ y$.

Hence this relation is complete, transitive, and increasing.

Common Mistake

Increasing does *not* mean “strictly more of every good.” The condition $x > x'$ only requires weakly more of every good and strictly more of at least one good.

1.4 Utility Representation

Economists often work with utility functions because optimisation and algebra are easier to express with numbers than with a raw binary relation.

Definition 1.7: Utility representation

A utility function is a map $U : \mathbb{R}_+^L \rightarrow \mathbb{R}$. It *represents* a preference relation $\succeq$ if

$$x \succeq x' \iff U(x) \geq U(x').$$

If U is increasing in the sense that $x > x'$ implies $U(x) > U(x')$, then the induced preference relation is increasing.

Utility is therefore a numerical encoding of a ranking. In Lecture 1, the important point is representational: utility is useful because maximising $U(x)$ is equivalent to choosing the most preferred bundle whenever U represents the underlying preference.

Worked Example

A utility function that generates an increasing preference.

Let

$$U(x_1, x_2) = \sqrt{x_1} + \sqrt{x_2}.$$

Then for any two bundles x and y ,

$$x \succeq y \iff U(x) \geq U(y).$$

Because the square-root function is increasing on $\mathbb{R}_+$, if $x > y$ then

$$\sqrt{x_1} \geq \sqrt{y_1}, \quad \sqrt{x_2} \geq \sqrt{y_2},$$

with strict inequality in at least one coordinate. Hence

$$U(x) = \sqrt{x_1} + \sqrt{x_2} > \sqrt{y_1} + \sqrt{y_2} = U(y).$$

So the represented preference is increasing, complete, and transitive.

Exam Focus

When asked whether a utility function represents preferences, the core test is always the ranking statement

$$x \succeq y \iff U(x) \geq U(y).$$

Do not interpret utility levels cardinally unless the question explicitly introduces that structure. In this chapter, utility is used only as a ranking device.

1.5 Revealed Preference

The lecture then asks a more empirical question: if we only observe prices and chosen bundles, what can we infer about preferences?

Core Principles Behind Revealed Preference

Classic consumer theory in Lecture 1 relies on three connected ideas:

1. What is chosen is preferred to what could have been chosen.
2. More is preferred.
3. What is preferred is best from the consumer's own point of view.

Together, these principles allow observed choices to reveal welfare information.

Suppose an observation consists of prices $p^t \in \mathbb{R}_{++}^L$ and a chosen bundle $x^t \in \mathbb{R}_+^L$. Let expenditure in that observation be $m^t = p^t \cdot x^t$.

Definition 1.8: Directly and strictly directly revealed preference

Given an observation (p^t, x^t) :

- x^t is *directly revealed preferred* to x , written $x^t R x$, if

$$p^t \cdot x \leq p^t \cdot x^t.$$

That is, bundle x was affordable when x^t was chosen.

- x^t is *strictly directly revealed preferred* to x , written $x^t P x$, if

$$p^t \cdot x < p^t \cdot x^t.$$

That is, bundle x was strictly cheaper than the chosen bundle.

Economically, R is an observed revealed ranking and P is its strict version. The distinction matters because strict affordability combined with increasingness is strong enough to generate strict welfare conclusions.

Theorem 1.2: From revealed preference to actual preference

Suppose the consumer chooses each observed bundle by maximising an increasing preference relation $\succeq$.

1. If $x R y$, then $x \succeq y$.
2. If $x P y$, then $x \succ y$.

Proof. If $x R y$, then y was affordable when x was chosen. By optimality of x , we must have $x \succeq y$.

If $x P y$, then y was strictly cheaper than x at the prices where x was chosen. Because prices are strictly positive, some bundle z exists with $z \succ y$ and still affordable at those prices. Increasingness implies $z \succ y$. Since x was chosen when z was affordable, optimality gives $x \succeq z$. By transitivity, $x \succ y$. $\square$

Worked Example

Comparing welfare across two periods.

Suppose a consumer chooses

$$x^1 = (2, 2) \text{ at } p^1 = (1, 1), \quad x^2 = (0, 3) \text{ at } p^2 = (2, 1).$$

Compute expenditures:

$$p^1 \cdot x^1 = 4, \quad p^2 \cdot x^2 = 3.$$

Now check whether period-2 bundle x^2 was affordable in period 1:

$$p^1 \cdot x^2 = 1(0) + 1(3) = 3 \leq 4.$$

So $x^1 R x^2$, which implies

$$x^1 \succeq x^2.$$

Next check whether x^1 was affordable in period 2:

$$p^2 \cdot x^1 = 2(2) + 1(2) = 6 > 3.$$

So period 2 does *not* reveal $x^2 \succeq x^1$.

Hence the data support the welfare conclusion

$$x^1 \succ x^2.$$

The consumer is weakly better off in period 1.

Policy Application

Revealed preference gives a practical welfare test after policy changes. If an old bundle remains affordable after a tax or price change but the consumer switches away from it, the new choice reveals at least weak improvement relative to the old affordable alternatives. Conversely, if the new bundle was affordable before the change but the old bundle was chosen instead, the pre-change situation is revealed to be weakly better. This is the logic behind welfare comparisons from observed demand rather than from directly measured happiness.

1.6 WARP

Revealed preference yields testable restrictions. The main one in Lecture 1 is WARP.

Definition 1.9: Weak Axiom of Revealed Preference (WARP)

A dataset satisfies *WARP* if

$$xRy \implies \text{not } (yPx).$$

Equivalently: once x is chosen when y is affordable, it cannot happen that y is later chosen from a budget where x is strictly cheaper.

WARP is a consistency condition on observable choices. It does not require knowledge of the utility function itself. It only checks whether observed decisions could have come from maximisation of an increasing preference.

Theorem 1.3: Increasing preference maximisation implies WARP

If each observed choice maximises an increasing preference relation $\succeq$, then the resulting data satisfy WARP.

Proof. Assume xRy . By Theorem 1.2, this implies $x \succeq y$. Suppose for contradiction that yPx also holds. Then Theorem 1.2 implies $y \succ x$. But $y \succ x$ means it is not true that $x \succeq y$, contradicting $x \succeq y$. Therefore yPx cannot occur, so WARP must hold. $\square$

Worked Example

A clean WARP violation.

Observation 1: prices are $p = (2, 1)$ and the consumer chooses

$$x = (1, 0).$$

Observation 2: prices are $p' = (1, 2)$ and the consumer chooses

$$x' = (0, 1).$$

Now compare costs crosswise:

$$p \cdot x = 2, \quad p \cdot x' = 1.$$

So under prices p , bundle x' is strictly cheaper than the chosen bundle x , hence

$$xPx'.$$

Similarly,

$$p' \cdot x' = 2, \quad p' \cdot x = 1,$$

so under prices p' , bundle x is strictly cheaper than the chosen bundle x' , hence

$$x'Px.$$

Thus both strict revealed rankings hold in opposite directions, which violates WARP.

Conclusion: these two observations cannot be rationalised by maximisation of an increasing preference.

Common Mistake

Do not say that xRy and yRx automatically violate WARP. Mutual *weak* direct revealed preference can occur when each bundle is merely affordable at the other observation. WARP is violated only when the reverse comparison is *strict*, i.e. when yPx also holds.

Exam Focus

For any revealed preference question, use the same checklist:

1. Write down each observed expenditure $m^t = p^t \cdot x^t$.
2. For every pair of bundles, test affordability by computing cross-costs $p^t \cdot x^s$.
3. Convert affordability into R or P statements.
4. Use $R \Rightarrow \succeq$ and $P \Rightarrow \succ$ only when the choice is interpreted as maximisation of an increasing preference.
5. Check WARP by ruling out the pattern xRy together with yPx .

1.7 Chapter Summary

Key Formula: Chapter 1 Summary

$$\begin{aligned}
 B(p, m) &= \{x \in \mathbb{R}_+^L : p \cdot x \leq m\}, \\
 xRy &\iff y \text{ was affordable when } x \text{ was chosen,} \\
 xPy &\iff y \text{ was strictly cheaper when } x \text{ was chosen,} \\
 \text{WARP: } xRy &\implies \text{not } (yPx).
 \end{aligned}$$

Exam Focus

High-yield takeaways from Lecture 1:

- Know the definitions of completeness, transitivity, increasingness, strict preference, and indifference precisely.
- Be able to move fluently between verbal statements and formal notation.
- Always interpret $p \cdot x$ as expenditure.
- For welfare comparisons across periods, affordability is the central test.
- WARP is the first observable implication of rational choice with increasing preferences.

2 Slutsky Equation and Demand Response

Lecture 2 extends the revealed-preference framework from Chapter 1 and then turns to comparative statics of consumer demand. The chapter has two main aims. First, it strengthens WARP into a multi-observation consistency condition, GARP, which uses the transitivity of preferences. Second, it studies how Marshallian demand responds to changes in prices and expenditure, culminating in the Slutsky decomposition of a price effect into substitution and income effects.

2.1 Generalized Revealed Preference

Recall from Chapter 1 that if a bundle x is chosen at prices p , then for any bundle x' :

$$xRx' \iff p \cdot x \geq p \cdot x', \quad xPx' \iff p \cdot x > p \cdot x'.$$

If choices come from maximization of an increasing preference relation $\succeq$, then

$$xRx' \implies x \succeq x', \quad xPx' \implies x \succ x'.$$

Because preferences are transitive, revealed preference should also propagate through chains of observations.

Definition 2.1: Revealed and strictly revealed preference through chains

Let R and P denote direct and strict direct revealed preference.

- x is *revealed preferred* to x' , written xR^*x' , if there exist bundles

$$x^1, \dots, x^N$$

such that

$$xRx^1Rx^2R \cdots Rx^Nx'.$$

- x is *strictly revealed preferred* to x' , written xP^*x' , if such a chain exists and at least one link is strict, i.e. at least one R can be replaced by P .

Theorem 2.1: Behavioral meaning of R^* and P^*

Suppose observed choices maximize an increasing preference relation $\succeq$. Then

$$xR^*x' \implies x \succeq x', \quad xP^*x' \implies x \succ x'.$$

Proof. If

$$xRx^1Rx^2R \cdots Rx^Nx',$$

then each link implies a weak preference comparison under utility maximization:

$$x \succeq x^1, \quad x^1 \succeq x^2, \quad \dots, \quad x^N \succeq x'.$$

By transitivity, $x \succeq x'$.

If at least one link is strict, then at least one strict preference comparison appears in the chain. Combining weak and strict comparisons by transitivity yields $x \succ x'$. $\square$

Definition 2.2: Generalized Axiom of Revealed Preference (GARP)

A dataset satisfies *GARP* if

$$xR^*x' \implies \text{not } (x'Px).$$

Equivalently: once x is revealed preferred to x' through a chain of affordable choices, it cannot be that x' is later chosen when x was strictly cheaper.

GARP is stronger than WARP because it checks for contradictions generated not only by one-step comparisons, but also by multi-step revealed-preference chains.

Theorem 2.2: Afriat's theorem, as used in this course

For a finite dataset of consumer choices:

1. If the data are generated by maximization of an increasing utility function, then the data satisfy GARP.
2. If the data satisfy GARP, then there exists an increasing utility function that rationalizes the data.

Hence GARP is the correct revealed-preference test for rationalizability by an increasing utility function.

Exam Focus

In two-good settings, Lecture 2 notes that WARP and GARP are equivalent. In three or more goods, GARP is strictly stronger because cycles can arise through multiple observations even when no strict two-observation cycle exists.

Worked Example

WARP can hold while GARP fails in three goods.

Consider three observations:

$$x^1 = (1, 0, 0), \quad x^2 = (0, 1, 0), \quad x^3 = (0, 0, 1),$$

at prices

$$p^1 = (2, 4, 1), \quad p^2 = (1, 2, 4), \quad p^3 = (4, 1, 2).$$

Since each chosen bundle is a unit vector,

$$p^1 \cdot x^1 = 2, \quad p^2 \cdot x^2 = 2, \quad p^3 \cdot x^3 = 2.$$

Now compute cross-costs:

$$p^1 \cdot x^3 = 1 \leq 2 \implies x^1 R x^3,$$

$$p^3 \cdot x^2 = 1 \leq 2 \implies x^3 R x^2,$$

$$p^2 \cdot x^1 = 1 \leq 2 \implies x^2 R x^1.$$

So

$$x^1 R x^3 R x^2 R x^1,$$

which gives a revealed-preference cycle. Moreover,

$$p^2 \cdot x^1 = 1 < 2 = p^2 \cdot x^2,$$

so in fact

$$x^2 P x^1.$$

Thus GARP is violated because $x^1 R^* x^1$ through a chain that ends with a strict reversal. However, there is no strict *two-observation* cycle, so WARP is not violated. This shows that GARP is strictly stronger than WARP in dimensions $L \geq 3$.

Common Mistake

Do not test GARP by looking only for pairs xRy and yPx . That checks WARP. GARP requires checking whether a contradiction appears *after a chain* of revealed-preference comparisons.

2.2 Budgets, Utility Maximization, and Demand

The second part of the lecture returns to standard consumer theory. The two core objects are the utility function and the budget set.

Definition 2.3: Indifference curve and demand function

Let $U : \mathbb{R}_+^L \rightarrow \mathbb{R}$ be a utility function.

- An *indifference curve* is a set of bundles with the same utility level:

$$\{x \in \mathbb{R}_+^L : U(x) = \bar{u}\}.$$

- A *demand function* is a map

$$x : \mathbb{R}_{++}^L \times \mathbb{R}_{++} \rightarrow \mathbb{R}_+^L$$

such that $x(p, m)$ is affordable and utility-maximizing at prices p and expenditure m :

$$p \cdot x(p, m) \leq m, \quad U(x(p, m)) \geq U(\tilde{x}) \quad \forall \tilde{x} \in B(p, m).$$

The demand function gives the consumer's optimal bundle as a function of prices and expenditure. Its ℓ -th component, $x_\ell(p, m)$, is demand for good ℓ .

Theorem 2.3: Exhausting the budget under increasing utility

If U is increasing and $x(p, m)$ maximizes U over $B(p, m)$, then

$$p \cdot x(p, m) = m.$$

Proof. If $p \cdot x(p, m) < m$, then some strictly positive amount of some good can be added while remaining in the budget set because each price is strictly positive. The resulting bundle is strictly larger componentwise, so increasing utility implies it yields higher utility, contradicting optimality. $\square$

Exam Focus

When the question gives a Marshallian demand function $x(p, m)$, always verify two things first:

1. affordability: $p \cdot x(p, m) \leq m$;
2. if utility is increasing, budget exhaustion: $p \cdot x(p, m) = m$.

These are the fastest internal consistency checks.

2.3 Demand Response to Price Changes

Let $x(p, m) = (x_1(p, m), x_2(p, m))$ denote Marshallian demand in the two-good case. The lecture studies what happens when the price of good 1 increases from p_1 to $p'_1 > p_1$, holding p_2 and m fixed.

Definition 2.4: Gross substitutes, gross complements, ordinary goods, and Giffen goods

Suppose the price of good 1 rises from p_1 to p'_1 .

- If

$$x_2(p'_1, p_2, m) > x_2(p_1, p_2, m),$$

then goods 1 and 2 are *gross substitutes*.

- If

$$x_2(p'_1, p_2, m) < x_2(p_1, p_2, m),$$

then goods 1 and 2 are *gross complements*.

- If

$$x_1(p'_1, p_2, m) < x_1(p_1, p_2, m),$$

then good 1 is *ordinary*.

- If

$$x_1(p'_1, p_2, m) > x_1(p_1, p_2, m),$$

then good 1 is *Giffen*.

These are purely Marshallian concepts: they describe the *total* demand response to a price change before any decomposition into substitution and income effects.

Worked Example

An explicit demand system: ordinary own-price effect and positive cross-price effect.

Suppose

$$U(x_1, x_2) = \sqrt{x_1} + \sqrt{x_2},$$

which generates the Marshallian demand

$$x(p_1, p_2, m) = \frac{m}{p_1 + p_2} \left(\frac{p_2}{p_1}, \frac{p_1}{p_2} \right).$$

Hence

$$x_1(p_1, p_2, m) = \frac{mp_2}{p_1(p_1 + p_2)}, \quad x_2(p_1, p_2, m) = \frac{mp_1}{p_2(p_1 + p_2)}.$$

For good 2, holding p_2 and m fixed,

$$\frac{\partial x_2}{\partial p_1} = \frac{m}{p_2} \cdot \frac{p_2}{(p_1 + p_2)^2} = \frac{m}{(p_1 + p_2)^2} > 0.$$

So when the price of good 1 rises, demand for good 2 rises: the goods are gross substitutes.

For good 1,

$$\frac{\partial x_1}{\partial p_1} = -\frac{mp_2(2p_1 + p_2)}{p_1^2(p_1 + p_2)^2} < 0,$$

so good 1 is ordinary, not Giffen.

Policy Application

The lecture's "meat and potatoes" example shows why Giffen behavior is theoretically possible. When a staple good becomes more expensive, the loss of real purchasing power may be so severe that the consumer buys *more* of the staple and cuts back on the more expensive alternative. This is unusual, but it clarifies that the total Marshallian price effect need not always be negative.

2.4 Demand Response to Expenditure Changes

Now hold prices fixed and increase expenditure from m to $m' > m$.

Definition 2.5: Normal and inferior goods

A good ℓ is:

- *normal* if

$$x_\ell(p, m') > x_\ell(p, m) \quad \text{whenever } m' > m;$$

- *inferior* if

$$x_\ell(p, m') < x_\ell(p, m) \quad \text{for some increase in expenditure } m' > m.$$

Normality and inferiority concern income or expenditure effects, not substitution effects. A good can be ordinary or Giffen in price-response language, and normal or inferior in expenditure-response language.

Worked Example

Normal goods in the square-root demand system.

Continuing the example

$$x_1(p_1, p_2, m) = \frac{mp_2}{p_1(p_1 + p_2)}, \quad x_2(p_1, p_2, m) = \frac{mp_1}{p_2(p_1 + p_2)},$$

we compute

$$\frac{\partial x_1}{\partial m} = \frac{p_2}{p_1(p_1 + p_2)} > 0, \quad \frac{\partial x_2}{\partial m} = \frac{p_1}{p_2(p_1 + p_2)} > 0.$$

Therefore both goods are normal.

Common Mistake

Do not infer that a good is normal merely because its demand curve slopes downward. Downward-sloping Marshallian demand describes a price effect; normality describes the sign of the expenditure effect.

2.5 Compensated Price Changes and the Slutsky Identity

The central technical result of Lecture 2 is the decomposition of a price change into substitution and income effects.

Suppose prices change from

$$p = (p_1, p_2) \quad \text{to} \quad p' = (p'_1, p_2), \quad p'_1 > p_1,$$

while original expenditure is m .

Definition 2.6: Slutsky compensated expenditure

Let $x(p, m)$ be the original Marshallian demand. The *Slutsky compensated expenditure* is

$$m^s = p' \cdot x(p, m).$$

This is the amount of money needed at the new prices to just afford the original bundle.

So the consumer is “compensated” enough to remain able to purchase the original bundle after the price increase. This isolates a pure substitution comparison between the old chosen bundle and the new chosen bundle on a budget that passes through the old choice.

Theorem 2.4: The Slutsky substitution effect is nonpositive for the good whose price rises

If the price of good 1 rises from p_1 to $p'_1 > p_1$, then

$$x_1(p', m^s) - x_1(p, m) \leq 0.$$

Proof. By construction, the old bundle $x(p, m)$ is affordable at the new prices with compensated expenditure m^s because

$$p' \cdot x(p, m) = m^s.$$

Hence $x(p', m^s)$ is chosen from a budget set that contains $x(p, m)$.

Suppose, for contradiction, that

$$x_1(p', m^s) > x_1(p, m).$$

Since the price of good 2 is unchanged, compare the old prices:

$$\begin{aligned} p \cdot x(p', m^s) &= p_1 x_1(p', m^s) + p_2 x_2(p', m^s) \\ &< p'_1 x_1(p', m^s) + p_2 x_2(p', m^s) \\ &= p' \cdot x(p', m^s) \\ &\leq m^s \\ &= p' \cdot x(p, m) \\ &= p'_1 x_1(p, m) + p_2 x_2(p, m). \end{aligned}$$

Because $x_1(p', m^s) > x_1(p, m)$ and $p'_1 > p_1$, the only way this can hold is if the new choice is cheaper at old prices than the old choice, implying

$$x(p, m) P x(p', m^s).$$

But at the new compensated prices, the new choice is chosen while the old one is affordable, so

$$x(p', m^s) R x(p, m).$$

This violates WARP, and therefore also GARP, for utility-maximizing behavior. Hence the supposition was false, so

$$x_1(p', m^s) \leq x_1(p, m).$$

□

The lecture packages this into three objects.

Key Formula: Slutsky Decomposition

Let

$$\Delta x_1 = x_1(p', m) - x_1(p, m)$$

denote the total Marshallian change in demand for good 1 after the price increase. Define

$$\Delta x_1^s = x_1(p', m^s) - x_1(p, m)$$

as the *Slutsky substitution effect*, and

$$\Delta x_1^n = x_1(p', m) - x_1(p', m^s)$$

as the *Slutsky income effect*. Then

$$\Delta x_1 = \Delta x_1^s + \Delta x_1^n.$$

This is the *Slutsky identity*.

The identity is purely algebraic: add and subtract $x_1(p', m^s)$. Its power comes from the fact that the substitution effect has a definite sign, while the income effect depends on whether the good is normal or inferior.

Theorem 2.5: Law of demand for normal goods

Suppose good 1 is normal. If its price rises from p_1 to $p'_1 > p_1$, holding p_2 and m fixed, then

$$x_1(p'_1, p_2, m) - x_1(p_1, p_2, m) \leq 0.$$

Proof. First,

$$m^s = p' \cdot x(p, m) = p'_1 x_1(p, m) + p_2 x_2(p, m) \geq p_1 x_1(p, m) + p_2 x_2(p, m) = m.$$

So the compensated expenditure is at least as large as the original expenditure.

If good 1 is normal, then at the new prices p' a higher expenditure raises demand for good 1:

$$m^s \geq m \implies x_1(p', m^s) \geq x_1(p', m),$$

which is equivalent to

$$x_1(p', m) - x_1(p', m^s) \leq 0.$$

Hence the income effect is nonpositive:

$$\Delta x_1^n \leq 0.$$

From Theorem 2.4, the substitution effect also satisfies

$$\Delta x_1^s \leq 0.$$

Therefore, by the Slutsky identity,

$$\Delta x_1 = \Delta x_1^s + \Delta x_1^n \leq 0.$$

□

Worked Example

Computing the Slutsky effects explicitly.

Use the demand system

$$x_1(p_1, p_2, m) = \frac{mp_2}{p_1(p_1 + p_2)}.$$

Take

$$p = (1, 1), \quad m = 2, \quad p' = (2, 1).$$

Original demand:

$$x(p, m) = (1, 1).$$

Compensated expenditure:

$$m^s = p' \cdot x(p, m) = 2(1) + 1(1) = 3.$$

Compensated demand for good 1:

$$x_1(p', m^s) = \frac{3 \cdot 1}{2(2 + 1)} = \frac{1}{2}.$$

Uncompensated new demand for good 1:

$$x_1(p', m) = \frac{2 \cdot 1}{2(2 + 1)} = \frac{1}{3}.$$

So:

$$\Delta x_1^s = \frac{1}{2} - 1 = -\frac{1}{2},$$

$$\Delta x_1^n = \frac{1}{3} - \frac{1}{2} = -\frac{1}{6},$$

and hence

$$\Delta x_1 = \Delta x_1^s + \Delta x_1^n = -\frac{1}{2} - \frac{1}{6} = -\frac{2}{3}.$$

Interpretation:

- the substitution effect is negative, as theory predicts;
- the income effect is also negative because good 1 is normal;
- therefore the total demand response is negative.

Exam Focus

For price-change questions, keep the order of objects clear:

1. old choice $x(p, m)$;
2. new prices p' ;
3. compensated expenditure $m^s = p' \cdot x(p, m)$;
4. compensated demand $x(p', m^s)$;
5. uncompensated new demand $x(p', m)$.

Most sign errors come from mixing up m and m^s .

2.6 Chapter Summary

Key Formula: Lecture 2 Summary

$xR^*x' \iff x$ is revealed preferred to x' through a chain of direct revealed preferences,

$xP^*x' \iff$ the chain contains at least one strict link,

GARP: $xR^*x' \implies \text{not } (x'Px)$,

$$m^s = p' \cdot x(p, m),$$

$$\Delta x_1 = x_1(p', m) - x_1(p, m),$$

$$\Delta x_1^s = x_1(p', m^s) - x_1(p, m),$$

$$\Delta x_1^n = x_1(p', m) - x_1(p', m^s),$$

$$\Delta x_1 = \Delta x_1^s + \Delta x_1^n.$$

If good 1 is normal and its price rises, then both $\Delta x_1^s \leq 0$ and $\Delta x_1^n \leq 0$, hence

$$\Delta x_1 \leq 0.$$

Exam Focus

The highest-yield ideas from Lecture 2 are:

- GARP is the multi-observation revealed-preference test and rationalizability condition.
- Marshallian demand describes the *total* response to price and expenditure changes.

- Gross substitutes/complements refer to cross-price effects; ordinary/Giffen refer to own-price effects.
- Normal/inferior refer to expenditure effects.
- The Slutsky identity decomposes a price effect into substitution and income effects.
- For normal goods, the law of demand follows immediately because both components are nonpositive.

3 Hicksian Substitution, MRS, and Endowments

Chapter 2 decomposed a Marshallian price effect using Slutsky compensation: adjust income just enough so that the original bundle remains affordable at the new prices. Lecture 3 introduces a second compensation concept, Hicksian compensation, which instead keeps utility constant. The chapter then develops marginal utility and the marginal rate of substitution, and finally studies consumer choice when expenditure comes from an endowment rather than exogenous income.

3.1 Hicksian Substitution

Let the initial price vector be $p = (p_1, p_2)$, expenditure be m , and initial demand be

$$x^0 := x(p, m).$$

Suppose the price of good 1 rises to

$$p' = (p'_1, p_2), \quad p'_1 > p_1.$$

Definition 3.1: Slutsky and Hicksian compensation

Let $u_0 := U(x^0)$ denote the utility level at the original optimum.

- The *Slutsky compensated expenditure* is

$$m^s := p' \cdot x^0.$$

This is the income needed at the new prices to still afford the original bundle.

- The *Hicksian compensated expenditure* is

$$m^h := \min \{ p' \cdot y : y \in \mathbb{R}_+^2, U(y) \geq u_0 \}.$$

This is the minimum expenditure needed at the new prices to reach the original utility level.

The difference is conceptual:

- Slutsky compensation keeps the *old bundle* feasible.
- Hicksian compensation keeps the *old utility level* feasible at minimum cost.

Theorem 3.1: Hicksian compensation is weakly below Slutsky compensation

For a price increase from p to p' , the compensation levels satisfy

$$m^h \leq m^s.$$

Equivalently, Slutsky compensation is weakly larger than Hicksian compensation.

Proof. The original bundle x^0 delivers utility u_0 , so it is feasible in the expenditure-minimization problem defining m^h . Therefore

$$m^h \leq p' \cdot x^0 = m^s.$$

□

Exam Focus

The intuition is simple: the old bundle is one way to attain the old utility level at the new prices, but it need not be the *cheapest* way. Hence Hicksian compensation is weakly smaller.

To isolate substitution under constant utility, let $h(p', u_0)$ denote Hicksian demand at the new prices and old utility. In the lecture's notation, this is often written as $x(p', m^h)$.

Definition 3.2: Hicksian substitution and income effects

For a price increase in good 1, define

$$\Delta x_1 := x_1(p', m) - x_1(p, m)$$

as the total Marshallian change in demand for good 1. The Hicksian decomposition is

$$\Delta x_1 = \Delta x_1^h + \Delta x_1^q,$$

where

$$\Delta x_1^h := h_1(p', u_0) - x_1(p, m)$$

is the *Hicksian substitution effect*, and

$$\Delta x_1^q := x_1(p', m) - h_1(p', u_0)$$

is the *Hicksian income effect*.

Theorem 3.2: Hicksian compensated demand for the good whose price rises is nonincreasing

If the price of good 1 rises from p_1 to $p'_1 > p_1$, then

$$\Delta x_1^h = h_1(p', u_0) - x_1(p, m) \leq 0.$$

Proof. Let

$$x^h := h(p', u_0).$$

Because x^0 and x^h both deliver utility level u_0 , each solves an expenditure-minimization problem at its own price vector:

$$p \cdot x^0 \leq p \cdot x^h, \quad p' \cdot x^h \leq p' \cdot x^0.$$

Subtracting the first inequality from the second gives

$$(p' - p) \cdot (x^h - x^0) \leq 0.$$

Only the price of good 1 changes, so

$$(p'_1 - p_1)(x_1^h - x_1^0) \leq 0.$$

Since $p'_1 - p_1 > 0$, it follows that

$$x_1^h - x_1^0 \leq 0.$$

Hence

$$\Delta x_1^h \leq 0.$$

□

Worked Example: Slutsky versus Hicksian compensation

Suppose the original choice is $x^0 = x(p, m)$ and utility is $u_0 = U(x^0)$. Let new prices be $p' = (p'_1, p_2)$ with $p'_1 > p_1$.

The two compensation levels are:

$$m^s = p' \cdot x^0, \quad m^h = \min\{p' \cdot y : U(y) \geq u_0\}.$$

Since $U(x^0) = u_0$, the original bundle x^0 is feasible for the Hicksian minimization problem. Therefore

$$m^h \leq p' \cdot x^0 = m^s.$$

So the correct ranking is

$$m^h \leq m^s.$$

Interpretation:

- Slutsky gives enough money to buy the old bundle.
- Hicks only gives enough money to reach the old utility level as cheaply as possible.

Hence Hicksian compensation is more economical.

Common Mistake

Do not reverse the inequality. After a price increase, Slutsky compensation is *not* smaller than Hicksian compensation. Keeping the exact old bundle affordable is generally more expensive than keeping only the old utility level attainable.

3.2 Marginal Utility and the Marginal Rate of Substitution

Lecture 3 then formalizes local trade-offs using derivatives.

Definition 3.3: Derivative and partial derivatives

If $f : \mathbb{R} \rightarrow \mathbb{R}$, the derivative of f at x is

$$f'(x) = \lim_{\alpha \rightarrow 0} \frac{f(x + \alpha) - f(x)}{\alpha},$$

when the limit exists.

If $f : \mathbb{R}_+^2 \rightarrow \mathbb{R}$, the partial derivative with respect to the first input is

$$\frac{\partial f(x_1, x_2)}{\partial x_1} = \lim_{\alpha \rightarrow 0} \frac{f(x_1 + \alpha, x_2) - f(x_1, x_2)}{\alpha},$$

and similarly for $\partial f / \partial x_2$.

In consumer theory, the function of interest is utility.

Definition 3.4: Marginal utility

For a differentiable utility function $U(x_1, \dots, x_L)$, the marginal utility of good k is

$$MU_k := \frac{\partial U(x_1, \dots, x_L)}{\partial x_k}.$$

It measures the rate at which utility changes when the quantity of good k increases by a small amount, holding other goods fixed.

Definition 3.5: Marginal rate of substitution

In the two-good case, fix a utility level u and write the indifference curve locally as

$$x_2 = f(x_1, u) \quad \text{so that} \quad U(x_1, f(x_1, u)) = u.$$

The *marginal rate of substitution of good 1 for good 2* is

$$MRS_{12} := -\frac{\partial f(x_1, u)}{\partial x_1}.$$

It is the amount of good 2 the consumer is willing to give up for a small increase in good 1 while holding utility constant.

Theorem 3.3: MRS equals the ratio of marginal utilities

Suppose U is differentiable and $MU_2 > 0$. Then along an indifference curve,

$$MRS_{12} = \frac{MU_1}{MU_2}.$$

Proof. Since utility is constant along the indifference curve,

$$U(x_1, f(x_1, u)) = u.$$

Differentiate both sides with respect to x_1 :

$$\frac{\partial U}{\partial x_1} + \frac{\partial U}{\partial x_2} \cdot \frac{\partial f(x_1, u)}{\partial x_1} = 0.$$

Rearranging gives

$$-\frac{\partial f(x_1, u)}{\partial x_1} = \frac{\frac{\partial U}{\partial x_1}}{\frac{\partial U}{\partial x_2}} = \frac{MU_1}{MU_2}.$$

By definition of MRS_{12} , this proves the result. □

Theorem 3.4: Interior tangency condition

If utility is differentiable and the consumer chooses an interior optimum from the budget set

$$p_1 x_1 + p_2 x_2 = m,$$

then

$$MRS_{12} = \frac{MU_1}{MU_2} = \frac{p_1}{p_2}.$$

Proof. At an interior optimum, the indifference curve is tangent to the budget line. The slope of the budget line is $-p_1/p_2$, while the slope of the indifference curve is $-MRS_{12}$. Equality of slopes gives

$$MRS_{12} = \frac{p_1}{p_2}.$$

Combining with Theorem 3.3 yields

$$\frac{MU_1}{MU_2} = \frac{p_1}{p_2}.$$

□

Worked Example: Cobb–Douglas utility

Let

$$U(x_1, x_2) = x_1 x_2.$$

Then

$$MU_1 = \frac{\partial U}{\partial x_1} = x_2, \quad MU_2 = \frac{\partial U}{\partial x_2} = x_1.$$

Hence

$$MRS_{12} = \frac{MU_1}{MU_2} = \frac{x_2}{x_1}.$$

At an interior optimum,

$$\frac{x_2}{x_1} = \frac{p_1}{p_2} \implies x_2 = \frac{p_1}{p_2} x_1.$$

Substitute into the budget constraint:

$$p_1 x_1 + p_2 x_2 = p_1 x_1 + p_2 \left(\frac{p_1}{p_2} x_1 \right) = 2p_1 x_1 = m.$$

So

$$x_1 = \frac{m}{2p_1}, \quad x_2 = \frac{m}{2p_2}.$$

Therefore the Marshallian demand is

$$x(p_1, p_2, m) = \left(\frac{m}{2p_1}, \frac{m}{2p_2} \right).$$

If the endowment is $\omega = (3, 1)$ and prices are $(p_1, p_2) = (1, 3)$, then

$$m = p \cdot \omega = 1 \cdot 3 + 3 \cdot 1 = 6,$$

so

$$x(1, 3, 6) = \left(\frac{6}{2}, \frac{6}{6} \right) = (3, 1).$$

Thus the consumer chooses exactly the endowment.

Worked Example: quasilinear utility and corner logic

Let

$$U(x_1, x_2) = x_1 + 2\sqrt{x_2}.$$

Then

$$MU_1 = 1, \quad MU_2 = \frac{1}{\sqrt{x_2}}, \quad MRS_{12} = \frac{MU_1}{MU_2} = \sqrt{x_2}.$$

Consider prices $(p_1, p_2) = (2, 1)$ and expenditure $m = 1$. An interior tangency would require

$$\sqrt{x_2} = \frac{p_1}{p_2} = 2 \implies x_2 = 4,$$

but this is impossible because the budget set satisfies

$$2x_1 + x_2 \leq 1.$$

So the tangency candidate is infeasible, and the optimum must be a corner. Because utility is increasing in both goods, the budget binds:

$$2x_1 + x_2 = 1 \implies x_1 = \frac{1 - x_2}{2}.$$

Hence maximize

$$f(x_2) = \frac{1 - x_2}{2} + 2\sqrt{x_2}, \quad x_2 \in [0, 1].$$

Then

$$f'(x_2) = -\frac{1}{2} + \frac{1}{\sqrt{x_2}} > 0 \quad \text{for } x_2 \in (0, 1].$$

So f is increasing on $[0, 1]$, and the optimum is

$$x_2^* = 1, \quad x_1^* = 0.$$

Thus

$$x(2, 1, 1) = (0, 1).$$

This is a standard exam pattern: the tangency condition gives a candidate, but feasibility and corner checks still matter.

Common Mistake

The tangency condition

$$\frac{MU_1}{MU_2} = \frac{p_1}{p_2}$$

applies only at an *interior* optimum. If the tangency candidate is infeasible or gives a negative quantity, the true optimum is a corner and must be checked separately.

3.3 Endowments and Net Demand

Up to this point, expenditure m was taken as given. Lecture 3 now studies the case where expenditure comes from selling an initial endowment.

Definition 3.6: Endowment economy budget set

Let the consumer's endowment be

$$\omega = (\omega_1, \dots, \omega_L) \in \mathbb{R}_+^L.$$

At prices p , the value of the endowment is

$$m = p \cdot \omega.$$

The consumer solves

$$\max_{x \in \mathbb{R}_+^L} U(x) \quad \text{subject to} \quad p \cdot x \leq p \cdot \omega.$$

If preferences are increasing, the budget binds:

$$p \cdot x(p, p \cdot \omega) = p \cdot \omega.$$

Definition 3.7: Net demand

Given endowment ω and Marshallian demand $x(p, m)$, the *net demand function* is

$$z(p) := x(p, p \cdot \omega) - \omega.$$

For each good ℓ ,

$$z_\ell(p) = x_\ell(p, p \cdot \omega) - \omega_\ell.$$

Interpretation:

- $z_\ell(p) > 0$: the consumer is a net buyer of good ℓ ;
- $z_\ell(p) < 0$: the consumer is a net seller of good ℓ .

Theorem 3.5: Walras' law for net demand

For any endowment ω and increasing utility,

$$p \cdot z(p) = 0.$$

Proof. By definition,

$$z(p) = x(p, p \cdot \omega) - \omega.$$

Taking the dot product with p gives

$$p \cdot z(p) = p \cdot x(p, p \cdot \omega) - p \cdot \omega.$$

Because the chosen bundle exhausts the budget,

$$p \cdot x(p, p \cdot \omega) = p \cdot \omega.$$

Hence

$$p \cdot z(p) = 0.$$

□

In the two-good case, positive prices imply the two net demands must have opposite signs whenever one is nonzero. Thus a consumer cannot be a net buyer of both goods simultaneously.

Worked Example: net demand and Walras' law

Suppose

$$U(x_1, x_2) = \sqrt{x_1} + \sqrt{x_2},$$

which generates the Marshallian demand

$$x(p_1, p_2, m) = \frac{m}{p_1 + p_2} \left(\frac{p_2}{p_1}, \frac{p_1}{p_2} \right).$$

Let the endowment be $\omega = (3, 1)$.

At prices $p = (1, 1)$, endowment income is

$$m = p \cdot \omega = 1 \cdot 3 + 1 \cdot 1 = 4.$$

So

$$x(1, 1, 4) = \frac{4}{2}(1, 1) = (2, 2).$$

Hence net demand is

$$z(1, 1) = (2, 2) - (3, 1) = (-1, 1).$$

Therefore:

- the consumer is a net seller of good 1;
- the consumer is a net buyer of good 2.

Walras' law holds:

$$(1, 1) \cdot (-1, 1) = 0.$$

So

$$z(1, 1) = (-1, 1), \quad p \cdot z(p) = 0.$$

3.4 Price Changes with Endowments

Price changes have a new feature when wealth depends on prices: the consumer's budget changes not only because the slope of the budget line changes, but also because the value of the endowment changes.

Suppose the price of good 1 rises from p_1 to $p'_1 > p_1$, with p_2 unchanged. Let

$$\tilde{x} := x(p, p \cdot \omega)$$

denote the original chosen bundle, and define the Slutsky compensation relative to that bundle as

$$m^s := p' \cdot \tilde{x}.$$

Definition 3.8: Slutsky decomposition with endowments

Define

$$\Delta x_1 := x_1(p', p' \cdot \omega) - x_1(p, p \cdot \omega),$$

$$\Delta x_1^s := x_1(p', m^s) - x_1(p, p \cdot \omega),$$

$$\Delta x_1^n := x_1(p', p' \cdot \omega) - x_1(p', m^s).$$

Then

$$\Delta x_1 = \Delta x_1^s + \Delta x_1^n.$$

As in Chapter 2, the substitution effect Δx_1^s is always nonpositive when the price of good 1 rises. The new issue is the sign of the income effect Δx_1^n .

Theorem 3.6: Endowment-value identity

Let $z_1(p) = x_1(p, p \cdot \omega) - \omega_1$ denote net demand for good 1 at the original prices. Then

$$p' \cdot \omega = m^s - (p'_1 - p_1)z_1(p).$$

Proof. Let $\tilde{x} = x(p, p \cdot \omega)$. Since the original bundle exhausts the original budget,

$$p \cdot \tilde{x} = p \cdot \omega.$$

Now

$$\begin{aligned} p' \cdot \omega &= p' \cdot \omega + (p \cdot \tilde{x} - p \cdot \omega) + (p' \cdot \tilde{x} - p' \cdot \tilde{x}) \\ &= p' \cdot (\omega - \tilde{x}) + p \cdot (\tilde{x} - \omega) + p' \cdot \tilde{x} \\ &= p' \cdot \tilde{x} - (p' - p) \cdot (\tilde{x} - \omega). \end{aligned}$$

Only the price of good 1 changes, so

$$(p' - p) \cdot (\tilde{x} - \omega) = (p'_1 - p_1)(\tilde{x}_1 - \omega_1) = (p'_1 - p_1)z_1(p).$$

Since $m^s = p' \cdot \tilde{x}$, the result follows:

$$p' \cdot \omega = m^s - (p'_1 - p_1)z_1(p).$$

□

This identity immediately explains the sign of the income effect:

- If $z_1(p) > 0$ (the consumer is a net buyer of good 1), then

$$p' \cdot \omega < m^s.$$

So the consumer is poorer than under Slutsky compensation.

- If $z_1(p) < 0$ (the consumer is a net seller of good 1), then

$$p' \cdot \omega > m^s.$$

So the consumer is richer than under Slutsky compensation.

Theorem 3.7: Welfare and direction of the budget shift

Suppose the price of good 1 rises.

1. If the consumer is a net buyer of good 1 at the original prices, then welfare falls.
2. If the consumer is a net seller of good 1 at the original prices, then welfare rises.

Proof. Let $\tilde{x} = x(p, p \cdot \omega)$.

If the consumer is a net buyer of good 1, then $\tilde{x}_1 > \omega_1$. Hence

$$p' \cdot \tilde{x} - p' \cdot \omega = (p'_1 - p_1)(\tilde{x}_1 - \omega_1) > 0.$$

So the old bundle $\tilde{x}$ is no longer affordable at the new prices and new wealth. Therefore the consumer cannot attain the old optimum, so welfare falls.

If the consumer is a net seller of good 1, then $\tilde{x}_1 < \omega_1$. Hence

$$p' \cdot \tilde{x} - p' \cdot \omega = (p'_1 - p_1)(\tilde{x}_1 - \omega_1) < 0.$$

So the old bundle is affordable at the new prices and new wealth, with money left over. Since preferences are increasing, the consumer can buy something strictly better. Therefore welfare rises. □

Theorem 3.8: A seller of the good whose price rises cannot become a buyer

Suppose the consumer is a net seller of good 1 at the original prices:

$$x_1(p, p \cdot \omega) < \omega_1.$$

If the price of good 1 rises, then at the new prices the consumer cannot choose a bundle with

$$x_1(p', p' \cdot \omega) > \omega_1.$$

Proof. Let $\tilde{x} = x(p, p \cdot \omega)$ and suppose for contradiction that the new choice x' satisfies $x'_1 > \omega_1$.

Since $\tilde{x}_1 < \omega_1$, at the new prices

$$p' \cdot \tilde{x} < p' \cdot \omega = p' \cdot x'.$$

So x' is chosen when $\tilde{x}$ is strictly cheaper, hence

$$x' P \tilde{x}.$$

On the other hand, because $x'_1 > \omega_1$,

$$p \cdot x' = p' \cdot x' - (p'_1 - p_1)x'_1 < p' \cdot \omega - (p'_1 - p_1)\omega_1 = p \cdot \omega = p \cdot \tilde{x}.$$

Thus under the original prices, x' was strictly cheaper than the old chosen bundle, so

$$\tilde{x} P x'.$$

This violates WARP. Therefore the new choice cannot satisfy $x'_1 > \omega_1$. □

Theorem 3.9: Law of demand with endowments for a normal good and a net buyer

Suppose good 1 is normal, and the consumer is a net buyer of good 1 at the original prices:

$$z_1(p) \geq 0.$$

If the price of good 1 rises from p_1 to $p'_1 > p_1$, then

$$x_1(p', p' \cdot \omega) \leq x_1(p, p \cdot \omega).$$

Proof. By Definition 3.8,

$$\Delta x_1 = \Delta x_1^s + \Delta x_1^n.$$

The substitution effect satisfies

$$\Delta x_1^s \leq 0.$$

By Theorem 3.6,

$$p' \cdot \omega = m^s - (p'_1 - p_1)z_1(p) \leq m^s$$

because $z_1(p) \geq 0$. Since good 1 is normal, a lower income at fixed prices p' weakly lowers its demand. Therefore

$$\Delta x_1^n = x_1(p', p' \cdot \omega) - x_1(p', m^s) \leq 0.$$

So

$$\Delta x_1 = \Delta x_1^s + \Delta x_1^n \leq 0.$$

This is exactly

$$x_1(p', p' \cdot \omega) \leq x_1(p, p \cdot \omega).$$

□

Worked Example: seller of good 1 and the sign of the income effect

Let

$$U(x_1, x_2) = \sqrt{x_1} + \sqrt{x_2}, \quad x(p_1, p_2, m) = \frac{m}{p_1 + p_2} \left(\frac{p_2}{p_1}, \frac{p_1}{p_2} \right),$$

and let the endowment be

$$\omega = (3, 1).$$

Step 1: original prices. At

$$p = (1, 1),$$

income is

$$m_0 = (1, 1) \cdot (3, 1) = 4.$$

Hence

$$x^0 = x(1, 1, 4) = \frac{4}{2}(1, 1) = (2, 2).$$

So

$$z(1, 1) = x^0 - \omega = (2, 2) - (3, 1) = (-1, 1).$$

The consumer is a net seller of good 1.

Step 2: new prices. Let

$$p' = (2, 1).$$

New endowment income is

$$m_1 = (2, 1) \cdot (3, 1) = 7.$$

So

$$x^1 = x(2, 1, 7) = \frac{7}{3} \left(\frac{1}{2}, 2 \right) = \left(\frac{7}{6}, \frac{14}{3} \right).$$

Step 3: Slutsky compensation. The compensated income needed to still afford the original bundle is

$$m^s = p' \cdot x^0 = (2, 1) \cdot (2, 2) = 6.$$

So the compensated demand is

$$x^c = x(2, 1, 6) = \frac{6}{3} \left(\frac{1}{2}, 2 \right) = (1, 4).$$

Step 4: decomposition for good 1.

$$\Delta x_1^s = x_1^c - x_1^0 = 1 - 2 = -1,$$

$$\Delta x_1^n = x_1^1 - x_1^c = \frac{7}{6} - 1 = \frac{1}{6}.$$

Thus

$$\Delta x_1 = \Delta x_1^s + \Delta x_1^n = -1 + \frac{1}{6} = -\frac{5}{6}.$$

So

$$\Delta x_1^s = -1, \quad \Delta x_1^n = \frac{1}{6}.$$

Interpretation:

- the substitution effect is negative, as usual;
- because the consumer is a seller of good 1, the price increase raises wealth, so the income effect is positive;
- the total change is still negative here because substitution dominates.

Policy Application: wages and labour supply

Lecture 3 interprets an endowment model as a labour-supply problem. Let good 1 be *time/leisure* and let the endowment be

$$\omega = (24, 0),$$

meaning the consumer begins each day with 24 hours of time and no composite consumption good. If the price vector is

$$p = (w, 1),$$

then w is the wage. A worker who supplies labour is a net seller of leisure/time, so a wage increase raises the value of the endowment.

Two forces then conflict:

- the substitution effect makes leisure more expensive, pushing the consumer toward less leisure and more labour;
- the income effect from higher wages makes the consumer richer, which can increase demand for leisure if leisure is normal.

Hence a higher wage can increase or decrease labour supplied. This is the standard intuition behind backward-bending labour supply.

3.5 Chapter Summary

Key Formula: Lecture 3 Summary

$$\begin{aligned}
 m^s &= p' \cdot x(p, m), \\
 m^h &= \min\{p' \cdot y : U(y) \geq U(x(p, m))\}, \\
 m^h &\leq m^s, \\
 \Delta x_1 &= \Delta x_1^h + \Delta x_1^q, \\
 MRS_{12} &= -\frac{\partial f(x_1, u)}{\partial x_1} = \frac{MU_1}{MU_2}, \\
 \text{interior optimum: } \frac{MU_1}{MU_2} &= \frac{p_1}{p_2}, \\
 z(p) &= x(p, p \cdot \omega) - \omega, \\
 p \cdot z(p) &= 0, \\
 p' \cdot \omega &= m^s - (p'_1 - p_1)z_1(p).
 \end{aligned}$$

If good 1 is normal and the consumer is a net buyer of good 1, then a rise in p_1 implies

$$x_1(p', p' \cdot \omega) \leq x_1(p, p \cdot \omega).$$

Exam Focus

The highest-yield points from Lecture 3 are:

- Slutsky and Hicksian compensation are different; Hicksian compensation is weakly smaller.
- Hicksian substitution always reduces compensated demand for the good whose price rises.
- Know how to compute MU_1 , MU_2 , MRS_{12} , and the tangency condition.
- In endowment models, income changes with prices because wealth equals $p \cdot \omega$.
- Use net demand

$$z(p) = x(p, p \cdot \omega) - \omega$$

to determine whether the consumer is a buyer or seller.

- For a net buyer, a price increase hurts welfare; for a net seller, it helps.
- Labour supply is ambiguous because the substitution and income effects work in opposite directions.

4 Intertemporal Consumption and Asset Pricing

This chapter studies choice across time. The first part rewrites intertemporal consumption as an ordinary consumer-choice problem with dated consumption goods and relative prices determined by the interest rate. The second part introduces discounted utility and the Euler-style optimality condition for present versus future consumption. The final part applies present-value logic to asset pricing, net present value, perpetuities, installment loans, and asset returns under taxes.

4.1 Intertemporal Consumption: Two-Period Model

Consider a consumer who lives for two periods, $t = 0, 1$. Let

$$x = (x_0, x_1) \in \mathbb{R}_+^2$$

denote the consumption stream, where x_t is expenditure on consumption in period t . Let

$$y = (y_0, y_1) \in \mathbb{R}_+^2$$

denote the income stream. The consumer can save or borrow between the two periods at interest rate $r \geq 0$.

Definition 4.1: Two-period savings formulation

Let $s_0 \in \mathbb{R}$ denote period-0 saving, where

$$s_0 > 0 \text{ means saving,} \quad s_0 < 0 \text{ means borrowing.}$$

The two-period budget equations are

$$x_0 + s_0 = y_0, \quad x_1 = (1 + r)s_0 + y_1.$$

The first equation says current income is split between current consumption and saving. The second says future consumption equals future income plus the gross return on current saving.

Theorem 4.1: Present-value and future-value budget constraints

The two-period savings system

$$x_0 + s_0 = y_0, \quad x_1 = (1 + r)s_0 + y_1$$

is equivalent to either of the following:

$$(1 + r)x_0 + x_1 = (1 + r)y_0 + y_1,$$

or

$$x_0 + \frac{x_1}{1 + r} = y_0 + \frac{y_1}{1 + r}.$$

Proof. From

$$x_0 + s_0 = y_0,$$

multiply by $1 + r$:

$$(1 + r)x_0 + (1 + r)s_0 = (1 + r)y_0.$$

Add the second budget equation:

$$x_1 = (1 + r)s_0 + y_1.$$

Cancelling $(1 + r)s_0$ gives

$$(1 + r)x_0 + x_1 = (1 + r)y_0 + y_1.$$

Dividing through by $1 + r$ yields

$$x_0 + \frac{x_1}{1+r} = y_0 + \frac{y_1}{1+r}.$$

□

Definition 4.2: Present value and future value

For a two-period stream (a_0, a_1) :

- its *future value* at date 1 is

$$FV(a_0, a_1) = (1+r)a_0 + a_1;$$

- its *present value* at date 0 is

$$PV(a_0, a_1) = a_0 + \frac{a_1}{1+r}.$$

Thus an intertemporal consumption plan is affordable if and only if

$$PV(x_0, x_1) = PV(y_0, y_1).$$

This turns intertemporal choice into an ordinary consumer problem with two goods:

$$p = (p_0, p_1) = \left(1, \frac{1}{1+r}\right),$$

so that

$$p \cdot x = p \cdot y.$$

The “price” of future consumption is therefore

$$p_1 = \frac{1}{1+r}.$$

A rise in r lowers the price of future consumption relative to present consumption.

Key Formula: Two-Period Intertemporal Budget

$$x_0 + \frac{x_1}{1+r} = y_0 + \frac{y_1}{1+r}.$$

Equivalently, with

$$p = \left(1, \frac{1}{1+r}\right),$$

the budget set is

$$B(p, p \cdot y) = \{x \in \mathbb{R}_+^2 : p \cdot x \leq p \cdot y\}.$$

Worked Example: present value, future value, and saving

Suppose

$$y = (190, 11), \quad r = 0.1,$$

and utility is

$$U(x_0, x_1) = x_0 x_1.$$

The present value of income is

$$PV(y) = 190 + \frac{11}{1.1} = 190 + 10 = 200.$$

The future value of income is

$$FV(y) = 1.1(190) + 11 = 209 + 11 = 220.$$

For Cobb–Douglas utility $U(x_0, x_1) = x_0 x_1$, the optimal demand with prices

$$\left(1, \frac{1}{1+r}\right)$$

and present-value wealth m is

$$x(r, m) = \left(\frac{m}{2}, \frac{(1+r)m}{2}\right).$$

Hence, with $m = 200$ and $r = 0.1$,

$$x(0.1, 200) = \left(100, \frac{1.1 \cdot 200}{2}\right) = (100, 110).$$

So the consumer saves

$$s_0 = y_0 - x_0 = 190 - 100 = 90$$

in period 0, and this becomes

$$(1+r)s_0 = 1.1 \cdot 90 = 99$$

in period 1. Since future income is 11, future consumption is

$$99 + 11 = 110.$$

Therefore the optimal plan is

$$x = (100, 110), \quad s_0 = 90.$$

4.2 Comparative Statics with Respect to the Interest Rate

Because

$$p_1 = \frac{1}{1+r},$$

a rise in the interest rate lowers the relative price of future consumption. Thus intertemporal comparative statics are a direct application of the consumer theory from Chapters 2 and 3.

Definition 4.3: Saver, borrower, and net demand for future consumption

Given income stream $y = (y_0, y_1)$ and chosen consumption stream $x = (x_0, x_1)$:

- the consumer is a *saver* if

$$x_1 > y_1 \iff s_0 > 0;$$

- the consumer is a *borrower* if

$$x_1 < y_1 \iff s_0 < 0.$$

Thus a saver is a net buyer of future consumption, while a borrower is a net seller of future consumption.

This matches the endowment logic from Chapter 3 with dated consumption as the two goods and income stream y as the endowment.

Theorem 4.2: Effect of an increase in the interest rate on saver/borrower status

Suppose the interest rate rises from r to $r' > r$.

1. A saver stays a saver.
2. A borrower may remain a borrower or may become a saver.

Proof. Future consumption has price

$$q(r) = \frac{1}{1+r}.$$

If $r' > r$, then

$$q(r') < q(r),$$

so the price of future consumption falls. By Chapter 3, when the price of a good falls, a net buyer of that good remains a net buyer. Thus a saver stays a saver. A net seller may remain a seller or may switch to being a buyer. $\square$

To apply the Slutsky decomposition, write Marshallian intertemporal demand as

$$x(r, m) = (x_0(r, m), x_1(r, m)),$$

where m is present-value wealth. If the interest rate changes from r to r' , then the present value of the income stream changes from

$$m = y_0 + \frac{y_1}{1+r}$$

to

$$m' = y_0 + \frac{y_1}{1+r'}.$$

Definition 4.4: Slutsky decomposition for intertemporal choice

Let

$$\Delta x_1 := x_1(r', m') - x_1(r, m)$$

be the total change in future consumption. Define the Slutsky compensated present-value wealth at the new interest rate by

$$m^s := x_0(r, m) + \frac{x_1(r, m)}{1+r'}.$$

Then

$$\Delta x_1 = \Delta x_1^s + \Delta x_1^n,$$

where

$$\Delta x_1^s := x_1(r', m^s) - x_1(r, m)$$

is the substitution effect, and

$$\Delta x_1^n := x_1(r', m') - x_1(r', m^s)$$

is the income effect.

Since $r' > r$ implies the price of future consumption falls, the substitution effect for future consumption is nonnegative:

$$\Delta x_1^s \geq 0.$$

If future consumption is normal and the consumer is a saver, then the income effect is also nonnegative. The reason is that a saver benefits when the interest rate rises: the price of the good she is a net buyer of falls.

Theorem 4.3: Interest-rate comparative statics for savers

Suppose future consumption is normal and the consumer is a saver at the initial interest rate. If r increases, then

$$x_1(r', m') \geq x_1(r, m).$$

Proof. When r rises, the price of future consumption falls, so the substitution effect satisfies

$$\Delta x_1^s \geq 0.$$

Because the consumer is a saver, she is a net buyer of future consumption. Thus the fall in its price raises her welfare and, if future consumption is normal, the associated income effect also satisfies

$$\Delta x_1^n \geq 0.$$

Hence

$$\Delta x_1 = \Delta x_1^s + \Delta x_1^n \geq 0.$$

□

Exam Focus

A rise in r does *not* mean current wealth necessarily rises. The correct logic is:

1. the price of future consumption falls;
2. substitution pushes toward more future consumption;
3. the welfare effect depends on whether the consumer is a saver or borrower;
4. for a saver, both substitution and income effects push toward more future consumption if future consumption is normal.

For a borrower, the overall effect on future consumption is ambiguous.

4.3 Multi-Period Intertemporal Choice

Now suppose the consumer lives for periods

$$t = 0, 1, \dots, T.$$

Let

$$x = (x_0, x_1, \dots, x_T) \in \mathbb{R}_+^{T+1}, \quad y = (y_0, y_1, \dots, y_T) \in \mathbb{R}_+^{T+1}.$$

Let s_t denote assets carried from period t to period $t + 1$, with

$$s_{-1} = 0, \quad s_T = 0.$$

Definition 4.5: Multi-period intertemporal budget equations

For each period $t = 0, 1, \dots, T$,

$$x_t + s_t = (1 + r)s_{t-1} + y_t.$$

Theorem 4.4: Multi-period present-value budget constraint

The multi-period budget system is equivalent to

$$\sum_{t=0}^T \frac{x_t}{(1+r)^t} = \sum_{t=0}^T \frac{y_t}{(1+r)^t}.$$

Equivalently, with price vector

$$p = \left(1, \frac{1}{1+r}, \frac{1}{(1+r)^2}, \dots, \frac{1}{(1+r)^T}\right),$$

the budget constraint is

$$p \cdot x = p \cdot y.$$

Proof. Divide each period budget equation

$$x_t + s_t = (1+r)s_{t-1} + y_t$$

by $(1+r)^t$:

$$\frac{x_t}{(1+r)^t} + \frac{s_t}{(1+r)^t} = \frac{s_{t-1}}{(1+r)^{t-1}} + \frac{y_t}{(1+r)^t}.$$

Summing from $t = 0$ to T causes the asset terms to telescope:

$$\sum_{t=0}^T \frac{s_t}{(1+r)^t} = \sum_{t=0}^T \frac{s_{t-1}}{(1+r)^{t-1}}$$

because $s_{-1} = s_T = 0$. Hence

$$\sum_{t=0}^T \frac{x_t}{(1+r)^t} = \sum_{t=0}^T \frac{y_t}{(1+r)^t}.$$

□

Theorem 4.5: Only present value matters for demand

Suppose the consumer solves

$$\max_{x \in \mathbb{R}_+^{T+1}} U(x) \quad \text{subject to} \quad p \cdot x \leq p \cdot y.$$

If two income streams y and y' satisfy

$$p \cdot y = p \cdot y',$$

then they generate the same budget set and therefore the same demand.

Proof. The feasible set depends only on the scalar wealth level $p \cdot y$. If

$$p \cdot y = p \cdot y',$$

then

$$\{x : p \cdot x \leq p \cdot y\} = \{x : p \cdot x \leq p \cdot y'\}.$$

Hence the maximization problem is identical, so the solution is the same.

□

Worked Example: different income timings with the same present value

Let $T = 2$ and $r = \frac{1}{2}$. Then

$$p = \left(1, \frac{2}{3}, \left(\frac{2}{3}\right)^2\right).$$

Consider the three income streams

$$y = (2, 3, 9), \quad y' = (8, 0, 0), \quad y'' = (0, 0, 18).$$

Their present values are:

$$p \cdot y = 2 + \frac{2}{3} \cdot 3 + \left(\frac{2}{3}\right)^2 \cdot 9 = 2 + 2 + 4 = 8,$$

$$p \cdot y' = 8,$$

$$p \cdot y'' = \left(\frac{2}{3}\right)^2 \cdot 18 = \frac{4}{9} \cdot 18 = 8.$$

Thus all three income streams have the same present value. Therefore they generate the same intertemporal budget set and the same optimal demand.

So the correct conclusion is:

$$(2, 3, 9), (8, 0, 0), \text{ and } (0, 0, 18) \text{ are equivalent for choice purposes at } r = \frac{1}{2}.$$

Common Mistake

Do not say that the *timing* of income never matters. It matters through its effect on present value. What Lecture 4 shows is that if the present value is the same, then the timing alone does not change the budget set under perfect borrowing and lending at rate r .

4.4 Inflation and the Real Interest Rate

Suppose the monetary price of one unit of consumption rises by factor $1 + \pi$ each period. Then one unit of consumption costs:

$$1 \text{ in period } 0, \quad 1 + \pi \text{ in period } 1, \quad (1 + \pi)^2 \text{ in period } 2, \dots$$

Definition 4.6: Intertemporal budget with inflation

With inflation rate π , the period- t budget equation becomes

$$(1 + \pi)^t x_t + s_t = (1 + r)s_{t-1} + (1 + \pi)^t y_t.$$

Dividing by $(1 + r)^t$ and summing over time yields the *real* present-value budget:

Theorem 4.6: Real present-value budget

With inflation rate π and nominal interest rate r ,

$$\sum_{t=0}^T \left(\frac{1 + \pi}{1 + r}\right)^t x_t = \sum_{t=0}^T \left(\frac{1 + \pi}{1 + r}\right)^t y_t.$$

Hence the real price vector is

$$p = \left(1, \frac{1+\pi}{1+r}, \left(\frac{1+\pi}{1+r} \right)^2, \dots, \left(\frac{1+\pi}{1+r} \right)^T \right).$$

Definition 4.7: Real interest rate

The *real interest rate* ρ is defined by

$$1 + \rho = \frac{1+r}{1+\pi}.$$

Equivalently,

$$\rho = \frac{1+r}{1+\pi} - 1 = \frac{r-\pi}{1+\pi} \approx r - \pi.$$

The approximation

$$\rho \approx r - \pi$$

is accurate when inflation is moderate. It is the standard “nominal minus inflation” rule.

Worked Example: computing the real interest rate

Suppose

$$r = 0.32, \quad \pi = 0.20.$$

Then

$$1 + \rho = \frac{1.32}{1.20} = 1.1,$$

so

$$\rho = 0.1.$$

If the income stream is

$$y = (100, 132),$$

then its real present value is

$$100 + \frac{132}{1.1} = 100 + 120 = 220.$$

Hence

$$\rho = 10\%, \quad PV_{\text{real}}(y) = 220.$$

4.5 Discounted Utility and Optimal Consumption Smoothing

Lecture 4 uses the standard discounted-utility specification

$$U(x_0, x_1) = u(x_0) + \delta u(x_1),$$

where u is the felicity function and $\delta \in [0, 1]$ is the discount factor.

Definition 4.8: Discounted utility

A two-period discounted-utility representation is

$$U(x_0, x_1) = u(x_0) + \delta u(x_1),$$

where:

- $u : \mathbb{R}_+ \rightarrow \mathbb{R}$ is increasing;

- u is concave, so marginal utility is diminishing;
- $\delta \in [0, 1]$ measures the value placed on future felicity.

Theorem 4.7: Concavity implies diminishing marginal utility

If u is concave and differentiable, then u' is decreasing.

Proof. Concavity means that for any $x > x'$ and any $\alpha > 0$,

$$u(x + \alpha) - u(x) \leq u(x' + \alpha) - u(x').$$

Divide by α and let $\alpha \rightarrow 0$ to obtain

$$u'(x) \leq u'(x').$$

Hence u' is decreasing. □

Theorem 4.8: Euler-type first-order condition for discounted utility

Suppose the consumer maximizes

$$u(x_0) + \delta u(x_1)$$

subject to

$$x_0 + \frac{x_1}{1+r} = m,$$

and the optimum is interior. Then

$$\frac{u'(x_0)}{\delta u'(x_1)} = 1 + r,$$

equivalently,

$$u'(x_0) = (1+r)\delta u'(x_1).$$

Proof. The marginal utilities are

$$MU_0 = u'(x_0), \quad MU_1 = \delta u'(x_1).$$

By the interior tangency condition from Chapter 3,

$$\frac{MU_0}{MU_1} = \frac{p_0}{p_1}.$$

Here

$$p_0 = 1, \quad p_1 = \frac{1}{1+r},$$

so

$$\frac{p_0}{p_1} = 1 + r.$$

Hence

$$\frac{u'(x_0)}{\delta u'(x_1)} = 1 + r,$$

which rearranges to

$$u'(x_0) = (1+r)\delta u'(x_1).$$

□

This condition compares the marginal value of consuming one more unit today with the discounted marginal value of consuming one more unit tomorrow, adjusted by the gross return from saving.

Theorem 4.9: Consumption pattern under discounted utility

Assume u is differentiable and strictly concave, so u' is strictly decreasing.

1. If $(1+r)\delta = 1$, then $x_0 = x_1$.
2. If $(1+r)\delta > 1$, then $x_1 > x_0$.
3. If $(1+r)\delta < 1$, then $x_0 > x_1$.

Proof. From Theorem 4.8,

$$u'(x_0) = (1+r)\delta u'(x_1).$$

If $(1+r)\delta = 1$, then

$$u'(x_0) = u'(x_1),$$

and strict monotonicity of u' implies

$$x_0 = x_1.$$

If $(1+r)\delta > 1$, then

$$u'(x_0) > u'(x_1).$$

Because u' is decreasing, this implies

$$x_0 < x_1.$$

Similarly, if $(1+r)\delta < 1$, then

$$u'(x_0) < u'(x_1),$$

so

$$x_0 > x_1.$$

□

Worked Example: logarithmic felicity

Suppose

$$u(x) = \ln x.$$

Then

$$u'(x) = \frac{1}{x}.$$

The first-order condition becomes

$$\frac{1}{x_0} = (1+r)\delta \frac{1}{x_1},$$

so

$$x_1 = (1+r)\delta x_0.$$

If

$$r = 0.1, \quad \delta = 0.9,$$

then

$$(1+r)\delta = 1.1 \times 0.9 = 0.99,$$

hence

$$x_1 = 0.99x_0.$$

So optimal future consumption is slightly below present consumption:

$$x_1 < x_0.$$

Exam Focus

The single most important condition in discounted-utility questions is

$$u'(x_0) = (1 + r)\delta u'(x_1).$$

Always interpret the product $(1 + r)\delta$:

- high δ means patience;
- high r makes future consumption relatively cheap;
- both push consumption toward the future.

4.6 Asset Pricing by No-Arbitrage

An asset is a stream of future monetary payoffs. Lecture 4 prices assets by the no-arbitrage principle.

Definition 4.9: Arbitrage

An *arbitrage opportunity* is a trading strategy that yields a nonnegative payoff in every contingency and a strictly positive payoff in at least one contingency, with no net cost or riskless profit at the prevailing prices.

Theorem 4.10: Present-value pricing of a certain asset

Suppose an asset pays deterministic cash flow

$$y = (y_0, y_1, \dots, y_T).$$

If borrowing and lending are possible at interest rate r and there is no arbitrage, then the asset's time-0 price must be

$$P_0 = \sum_{t=0}^T \frac{y_t}{(1 + r)^t}.$$

Proof. Let

$$PV(y) = \sum_{t=0}^T \frac{y_t}{(1 + r)^t}.$$

If $P_0 < PV(y)$, borrow P_0 at time 0 and buy the asset. The asset's cash flows are sufficient to service the debt and still leave a strictly positive surplus in present-value terms, which is an arbitrage.

If $P_0 > PV(y)$, short the asset, receive P_0 at time 0, invest the proceeds, and use the investment returns to cover the promised asset payments. The excess $P_0 - PV(y)$ yields arbitrage.

Hence no-arbitrage requires

$$P_0 = PV(y).$$

□

Definition 4.10: Net present value

If acquiring an asset requires payment stream

$$m = (m_0, m_1, \dots, m_T)$$

and yields payoff stream

$$y = (y_0, y_1, \dots, y_T),$$

then its *net present value* is

$$NPV = \sum_{t=0}^T \frac{y_t - m_t}{(1+r)^t}.$$

A project is attractive at discount rate r if its net present value is positive.

Theorem 4.11: Perpetuity formula

A perpetuity that pays a fixed amount m each period forever, starting in period 1, has present value

$$PV = \frac{m}{r}.$$

Proof. Let

$$PV = \frac{m}{1+r} + \frac{m}{(1+r)^2} + \frac{m}{(1+r)^3} + \dots$$

Multiply both sides by $\frac{1}{1+r}$:

$$\frac{PV}{1+r} = \frac{m}{(1+r)^2} + \frac{m}{(1+r)^3} + \dots$$

Subtract this from the original expression:

$$PV - \frac{PV}{1+r} = \frac{m}{1+r}.$$

So

$$\frac{r}{1+r} PV = \frac{m}{1+r},$$

hence

$$PV = \frac{m}{r}.$$

□

Worked Example: net present value of a delayed perpetuity project

Suppose a project requires payment of 1 in each of the first three periods and then pays 2 forever from period 3 onward. Its net present value is

$$NPV = -1 - \frac{1}{1+r} - \frac{1}{(1+r)^2} + \frac{2}{(1+r)^3} + \frac{2}{(1+r)^4} + \dots$$

The infinite tail is a perpetuity starting at period 3, so

$$\frac{2}{(1+r)^3} + \frac{2}{(1+r)^4} + \dots = \frac{1}{(1+r)^2} \left(\frac{2}{1+r} + \frac{2}{(1+r)^2} + \dots \right) = \frac{1}{(1+r)^2} \cdot \frac{2}{r}.$$

Therefore

$$NPV = \frac{2}{r(1+r)^2} - \left(1 + \frac{1}{1+r} + \frac{1}{(1+r)^2} \right).$$

So the correct compact expression is

$$NPV = \frac{2}{r(1+r)^2} - \left(1 + \frac{1}{1+r} + \frac{1}{(1+r)^2} \right).$$

Worked Example: installment loan interest rate

An installment loan gives $m = 1000$ upfront and requires monthly payments of $y = 100$ for $T = 12$ months. The correct monthly interest rate r solves

$$1000 = \frac{100}{1+r} + \frac{100}{(1+r)^2} + \cdots + \frac{100}{(1+r)^{12}}.$$

This is *not* the same as taking

$$\frac{12 \cdot 100}{1000} - 1 = 20\%.$$

That simple ratio ignores the fact that payments are made over time rather than all at the end. Numerically, the solution is approximately

$$r \approx 0.03$$

per month, i.e. about 3% monthly. The implied annual rate is therefore much larger than 20%; Lecture 4 reports it as roughly 41% per year.

The exam logic is:

1. write the present-value equation for the loan;
2. solve for the periodic interest rate;
3. do not treat the total repayment as if it occurred only in the final period.

4.7 Asset Price Changes and Rates of Return

Suppose an asset does not pay dividends in period 0 and its price rises from p_0 to p_1 with certainty by next period.

Theorem 4.12: No-arbitrage return condition

If the asset pays no interim cash flow and can be traded freely, then no arbitrage implies

$$r = \frac{p_1 - p_0}{p_0}.$$

Equivalently,

$$p_0 = \frac{p_1}{1+r}.$$

Proof. If

$$\frac{p_1 - p_0}{p_0} > r,$$

borrow p_0 , buy the asset, sell it next period for p_1 , and repay $(1+r)p_0$. The surplus

$$p_1 - (1+r)p_0 > 0$$

is an arbitrage.

If

$$\frac{p_1 - p_0}{p_0} < r,$$

short the asset, receive p_0 , invest at rate r , and next period buy back the asset for p_1 . The surplus

$$(1+r)p_0 - p_1 > 0$$

is an arbitrage.

Therefore only

$$\frac{p_1 - p_0}{p_0} = r$$

can hold. □

Policy Application: housing as an asset

A house pays housing services rather than direct cash. If the monetary rental value of current housing services is y_0 and next period's house price is p_1 , then no-arbitrage implies

$$p_0 = y_0 + \frac{p_1}{1+r}.$$

So the value of a house is the value of current housing services plus the discounted resale value.

Policy Application: depletable resources

If a depletable resource such as oil yields no extraction cost and must eventually be replaced by a perfect substitute with cost C at terminal date T , then no-arbitrage implies

$$p_t = \frac{p_{t+1}}{1+r}.$$

Hence

$$p_0 = \frac{p_T}{(1+r)^T} = \frac{C}{(1+r)^T}.$$

This is the basic present-value logic behind the price path of a scarce exhaustible resource.

4.8 Taxes and Asset Returns

Lecture 4 also studies taxes on interest income and asset returns.

Definition 4.11: After-tax interest rate

If the tax rate on interest income is τ , then one dollar saved becomes

$$1 + r(1 - \tau)$$

next period. Thus the effective after-tax interest rate is

$$r(1 - \tau).$$

This means taxes reduce the reward to saving and soften the incentive to transfer consumption into the future.

Now consider two assets:

- asset A is untaxed;
- asset B faces tax rate τ on its capital gain.

Theorem 4.13: Required pre-tax return on a taxed asset

Suppose asset B has price p_0^B today and price p_1^B next period, and the tax rate on its return is τ .

No arbitrage implies

$$p_0^B = \left(\frac{1 - \tau}{1 - \tau + r} \right) p_1^B,$$

equivalently,

$$\frac{p_1^B - p_0^B}{p_0^B} = \frac{r}{1 - \tau}.$$

Proof. If an investor buys asset B at time 0 for p_0^B , then next period the gross proceeds net of tax are

$$p_1^B - \tau(p_1^B - p_0^B).$$

No arbitrage requires this to equal the gross return from investing p_0^B at interest rate r :

$$(1 + r)p_0^B = p_1^B - \tau(p_1^B - p_0^B).$$

Rearranging:

$$(1 + r)p_0^B = (1 - \tau)p_1^B + \tau p_0^B,$$

so

$$(1 - \tau + r)p_0^B = (1 - \tau)p_1^B.$$

Hence

$$p_0^B = \left(\frac{1 - \tau}{1 - \tau + r} \right) p_1^B.$$

Subtract p_0^B from both sides of the equivalent expression

$$(1 - \tau + r)p_0^B = (1 - \tau)p_1^B$$

to get

$$rp_0^B = (1 - \tau)(p_1^B - p_0^B),$$

which gives

$$\frac{p_1^B - p_0^B}{p_0^B} = \frac{r}{1 - \tau}.$$

□

Thus the taxed asset must offer a higher *pre-tax* return than the untaxed asset in order to remain competitive after tax.

4.9 Chapter Summary

Key Formula: Lecture 4 Summary

$$\begin{aligned}
x_0 + s_0 &= y_0, \\
x_1 &= (1 + r)s_0 + y_1, \\
x_0 + \frac{x_1}{1 + r} &= y_0 + \frac{y_1}{1 + r}, \\
p &= \left(1, \frac{1}{1 + r}\right), \\
\sum_{t=0}^T \frac{x_t}{(1 + r)^t} &= \sum_{t=0}^T \frac{y_t}{(1 + r)^t}, \\
1 + \rho &= \frac{1 + r}{1 + \pi}, \\
U(x_0, x_1) &= u(x_0) + \delta u(x_1), \\
u'(x_0) &= (1 + r)\delta u'(x_1), \\
P_0 &= \sum_{t=0}^T \frac{y_t}{(1 + r)^t}, \\
NPV &= \sum_{t=0}^T \frac{y_t - m_t}{(1 + r)^t}, \\
PV_{\text{perpetuity}} &= \frac{m}{r}, \\
r &= \frac{p_1 - p_0}{p_0}, \\
p_0^B &= \left(\frac{1 - \tau}{1 - \tau + r}\right) p_1^B.
\end{aligned}$$

Exam Focus

The highest-yield skills from Lecture 4 are:

- convert intertemporal budgets into present-value form;
- interpret $1/(1 + r)$ as the price of future consumption;
- distinguish savers from borrowers and use price-change logic correctly;
- use

$$u'(x_0) = (1 + r)\delta u'(x_1)$$

for discounted-utility problems;

- compute present values, real interest rates, and net present values cleanly;
- apply no-arbitrage to price assets and derive required returns;
- adjust return formulas correctly when taxes are imposed.

5 Risk, State Prices, and Expected Utility

This chapter extends consumer theory to uncertain environments. Instead of choosing bundles across goods or across time, the consumer now chooses contingent consumption across states of the world. This allows a unified treatment of insurance, asset markets, diversification, and expected utility. The central ideas are:

1. uncertain payoffs can be represented as contingent consumption bundles;
2. state prices and asset prices are linked by no-arbitrage;
3. diversification and risk sharing can smooth consumption across states;
4. expected utility provides a tractable model of risk preferences.

5.1 Risk and Contingent Consumption

Suppose there are S mutually exclusive states of the world. State s occurs with probability π_s , where

$$\pi_s \geq 0 \quad \text{and} \quad \sum_{s=1}^S \pi_s = 1.$$

Definition 5.1: Contingent consumption bundle

A *contingent consumption bundle* is a vector

$$x = (x_1, x_2, \dots, x_S) \in \mathbb{R}_+^S,$$

where x_s is the amount of money or consumption received if state s occurs.

This is the correct representation for risky prospects. A lottery, job outcome, insurance payoff, or asset return can all be written as a contingent bundle.

Definition 5.2: State prices and the contingent-claims budget set

Let

$$p = (p_1, p_2, \dots, p_S) \in \mathbb{R}_{++}^S$$

be the *state price vector*, where p_s is the price today of one unit of consumption delivered only in state s .

If wealth is $m > 0$, the consumer chooses x subject to

$$p \cdot x = \sum_{s=1}^S p_s x_s \leq m.$$

If the consumer has endowment

$$\omega = (\omega_1, \dots, \omega_S),$$

then the budget constraint becomes

$$p \cdot x \leq p \cdot \omega.$$

State prices play exactly the same role here that ordinary prices played in Chapters 1–4. The difference is that the coordinates now refer to states rather than goods or dates.

Worked Example: contingent consumption from an uncertain career outcome

Suppose there are two states:

- state 1: a high-paying job occurs with probability 0.2;
- state 2: onion farming occurs with probability 0.8.

If the corresponding incomes are

$$10^6 \quad \text{and} \quad 10^3,$$

then the risky prospect is represented by

$$x = (10^6, 10^3), \quad \pi = (0.2, 0.8).$$

This is a contingent consumption bundle because the realized payoff depends on which state occurs.

5.2 Insurance and State Prices

Insurance can be interpreted as trading contingent claims.

Worked Example: insurance as contingent consumption

Suppose the consumer is endowed with

$$\omega = (8, 2), \quad \pi = (0.9, 0.1),$$

so state 1 is the good state and state 2 is the bad state. A unit of insurance:

- costs 2 today;
- pays 3 in state 2.

Relative to the endowment, buying one unit of insurance reduces state-1 consumption by 2 and raises state-2 consumption by 1. If θ units are purchased, then the resulting contingent bundle is

$$x(\theta) = (8 - 2\theta, 2 + \theta).$$

If shorting insurance is allowed, then θ may be negative. For example,

$$\theta = -2$$

gives

$$x(-2) = (12, 0).$$

Now observe that

$$x_1 + 2x_2 = (8 - 2\theta) + 2(2 + \theta) = 12$$

for every θ . Since

$$12 = 8 + 2 \cdot 2 = (1, 2) \cdot (8, 2),$$

this budget line is exactly the same as the contingent-claims budget with state price vector

$$p = (1, 2).$$

So insurance trading is equivalent to buying state-contingent consumption at prices $(1, 2)$.

Exam Focus

When an insurance contract changes payoffs by a fixed amount across states, rewrite the set of attainable bundles as a linear equation. That linear equation usually reveals the implied state prices immediately.

5.3 Assets, Portfolios, and Arbitrage

Assets can also be viewed as contingent consumption bundles.

Definition 5.3: Assets, portfolios, and shorting

An asset is a contingent payoff vector

$$a^k = (a_1^k, \dots, a_S^k) \in \mathbb{R}^S.$$

If its price is q_k , then holding one unit of asset k costs q_k today and delivers a_s^k in state s . A *portfolio* is a vector

$$\theta = (\theta_1, \dots, \theta_K),$$

where θ_k is the quantity of asset k held. The portfolio costs

$$q \cdot \theta = \sum_{k=1}^K q_k \theta_k$$

and delivers contingent consumption

$$\sum_{k=1}^K \theta_k a^k.$$

If $\theta_k < 0$ is allowed, then shorting is allowed.

Definition 5.4: Arbitrage

A portfolio θ is an *arbitrage opportunity* if:

1. it costs nothing or yields money upfront:

$$q \cdot \theta \leq 0;$$

2. its payoff is nonnegative in every state:

$$\sum_{k=1}^K \theta_k a^k \geq 0;$$

3. and it is strictly positive in at least one state.

Arbitrage matters because no-arbitrage is the key pricing principle of the chapter: if two portfolios have the same state-contingent payoffs, they must have the same price.

Theorem 5.1: Pricing by replication

Suppose a new asset $\tilde{a}$ can be replicated by an existing portfolio θ , i.e.

$$\tilde{a} = \sum_{k=1}^K \theta_k a^k.$$

If there is no arbitrage, then the price of the new asset must be

$$\tilde{q} = q \cdot \theta.$$

Proof. If $\tilde{q} < q \cdot \theta$, then buy the new asset and short the replicating portfolio. This gives a positive profit today and zero net payoff in every state, which is an arbitrage.

If $\tilde{q} > q \cdot \theta$, then short the new asset and buy the replicating portfolio. Again, the payoffs cancel state by state, leaving a sure profit today.

So no-arbitrage requires

$$\tilde{q} = q \cdot \theta.$$

□

Definition 5.5: State-price representation

If Arrow securities exist, where:

$$e^1 = (1, 0, \dots, 0), e^2 = (0, 1, \dots, 0), \dots, e^S = (0, \dots, 0, 1),$$

and their prices are

$$p = (p_1, \dots, p_S),$$

then any asset

$$a = (a_1, \dots, a_S)$$

has no-arbitrage price

$$q(a) = \sum_{s=1}^S p_s a_s = p \cdot a.$$

So state prices are the building blocks of all asset prices.

Worked Example: extracting state prices from asset prices

Suppose there are two assets:

$$a^1 = (1, 2), \quad a^2 = (1, 1),$$

with prices

$$q_1 = 3, \quad q_2 = 2.$$

To find the price of one unit of state-1 consumption, construct the Arrow security $(1, 0)$. Observe that

$$2a^2 - a^1 = 2(1, 1) - (1, 2) = (1, 0).$$

Hence its no-arbitrage price is

$$2q_2 - q_1 = 2 \cdot 2 - 3 = 1.$$

So

$$p_1 = 1.$$

To find the price of one unit of state-2 consumption, construct

$$a^1 - a^2 = (1, 2) - (1, 1) = (0, 1),$$

whose price is

$$q_1 - q_2 = 3 - 2 = 1.$$

So

$$p_2 = 1.$$

Therefore the state price vector is

$$p = (1, 1).$$

Worked Example: contingent claims from a risky and a risk-free asset

Suppose there are two assets:

- a risk-free asset with payoff

$$a^f = (1, 1)$$

and price

$$q_f = 3;$$

- a risky asset with payoff

$$a^r = (2, 1)$$

and price

$$q_r = 4.$$

If wealth is

$$m = 12,$$

then spending all wealth on the risk-free asset buys

$$\frac{12}{3} = 4$$

units, which yields

$$(4, 4).$$

Spending all wealth on the risky asset buys

$$\frac{12}{4} = 3$$

units, which yields

$$(6, 3).$$

If a fraction α of wealth is spent on the risky asset and $1 - \alpha$ on the risk-free asset, the payoff is

$$(4, 4) + \alpha((6, 3) - (4, 4)) = (4, 4) + \alpha(2, -1).$$

So each increase of 2 units in state-1 payoff costs a reduction of 1 unit in state-2 payoff. This is exactly the tradeoff implied by state prices

$$p = (1, 2).$$

Common Mistake

Do not confuse asset prices with state prices. Asset prices apply to entire payoff vectors; state prices apply to one unit of payoff in a single state. Asset prices are recovered from state prices by

$$q(a) = p \cdot a.$$

5.4 Diversification and Risk Sharing

Asset markets allow agents to reallocate consumption across states. This is valuable when people dislike large payoff fluctuations.

Definition 5.6: Diversification

Diversification means holding a portfolio of assets rather than concentrating wealth in one risky payoff. The aim is to reduce the variance of payoffs across states without sacrificing too much expected return.

The most effective diversification combines assets with offsetting payoffs, ideally negatively correlated assets.

Worked Example: diversification can create a safe payoff

Suppose

$$a = (1, 0), \quad a' = (0, 1), \quad \pi = \left(\frac{1}{2}, \frac{1}{2}\right).$$

Each asset alone is risky:

- a pays only in state 1;
- a' pays only in state 2.

But the portfolio

$$a + a' = (1, 1)$$

is risk-free.

A second example: let

$$\tilde{a} = (1, 0), \quad \tilde{a}' = (2, 1).$$

Both are risky, but if shorting is allowed,

$$\tilde{a}' - \tilde{a} = (2, 1) - (1, 0) = (1, 1),$$

which is again risk-free.

So diversification works not only by buying several assets, but also by combining long and short positions when markets permit it.

Policy Application: village risk sharing

Suppose each villager faces idiosyncratic crop-failure risk. If one villager's crops fail while others' do not, the villagers can insure one another by trading state-contingent claims. This is risk sharing: each person gives up some consumption in good states in exchange for more consumption in bad states.

However, if *everyone* is hit by the same shock, such as a flood or drought, then internal insurance becomes ineffective. This is *correlated risk*. It explains why some risks, especially large-scale natural-disaster risks, are difficult or expensive to insure.

5.5 Expected Value and Expected Utility

To rank contingent bundles, the lecture introduces expected value and expected utility.

Definition 5.7: Expected value

Given contingent bundle

$$x = (x_1, \dots, x_S)$$

with probabilities

$$\pi = (\pi_1, \dots, \pi_S),$$

its expected value is

$$E[x] = \sum_{s=1}^S \pi_s x_s.$$

Expected value is a natural benchmark, but by itself it does not capture attitudes toward risk. Many people prefer a safer bundle with lower expected value to a riskier bundle with higher expected value.

Definition 5.8: Expected utility and Bernoulli utility

Let

$$u : \mathbb{R}_+ \rightarrow \mathbb{R}$$

be a Bernoulli utility function. The expected utility of contingent bundle

$$x = (x_1, \dots, x_S)$$

is

$$U(x) = E[u(x)] = \sum_{s=1}^S \pi_s u(x_s).$$

The lecture assumes that u is increasing, and often also concave.

Definition 5.9: Increasing and concave Bernoulli utility

A Bernoulli utility function u is:

- *increasing* if more money is better:

$$x > x' \implies u(x) > u(x');$$

- *concave* if marginal utility is diminishing.

If u is differentiable and concave, then u' is decreasing.

Theorem 5.2: Interior optimality under expected utility

Suppose the consumer maximizes

$$U(x) = \sum_{s=1}^S \pi_s u(x_s)$$

subject to

$$p \cdot x \leq m,$$

and the optimum is interior. Then for any two states i, j ,

$$\frac{\pi_i u'(x_i)}{\pi_j u'(x_j)} = \frac{p_i}{p_j}.$$

Proof. The marginal utility of state- s consumption is

$$MU_s = \pi_s u'(x_s).$$

At an interior optimum, the tangency condition from earlier chapters gives

$$\frac{MU_i}{MU_j} = \frac{p_i}{p_j}.$$

Substituting the marginal utilities yields

$$\frac{\pi_i u'(x_i)}{\pi_j u'(x_j)} = \frac{p_i}{p_j}.$$

□

This can be rewritten as

$$\frac{u'(x_i)}{u'(x_j)} = \frac{p_i/\pi_i}{p_j/\pi_j}.$$

So what matters is the price of state consumption relative to its probability.

Worked Example: optimal contingent consumption with logarithmic utility

Suppose there are two states with probabilities

$$\pi_1 = \pi_2 = \frac{1}{2},$$

state prices

$$p = (1, 2),$$

wealth

$$m = 12,$$

and Bernoulli utility

$$u(x) = \ln x.$$

Then expected utility is

$$U(x_1, x_2) = \frac{1}{2} \ln x_1 + \frac{1}{2} \ln x_2.$$

At an interior optimum,

$$\frac{\pi_1 u'(x_1)}{\pi_2 u'(x_2)} = \frac{p_1}{p_2}.$$

Since

$$u'(x) = \frac{1}{x} \quad \text{and} \quad \pi_1 = \pi_2,$$

this becomes

$$\frac{1/x_1}{1/x_2} = \frac{1}{2} \implies \frac{x_2}{x_1} = \frac{1}{2}.$$

So

$$x_2 = \frac{x_1}{2}.$$

Using the budget constraint

$$x_1 + 2x_2 = 12,$$

we obtain

$$x_1 + 2\left(\frac{x_1}{2}\right) = 12 \implies 2x_1 = 12 \implies x_1 = 6,$$

and hence

$$x_2 = 3.$$

Therefore the optimal contingent bundle is

$$x = (6, 3).$$

5.6 First-Order Stochastic Dominance

Expected utility with increasing Bernoulli utility respects a natural ranking of risky prospects.

Definition 5.10: Survival probability and first-order stochastic dominance

Given contingent bundle

$$x = (x_1, \dots, x_S)$$

and probability vector π , define

$$\pi(x, \alpha)$$

to be the probability that x pays at least α .

We say that x *first-order stochastically dominates* x' , written

$$x \succeq_{FOSD} x',$$

if

$$\pi(x, \alpha) \geq \pi(x', \alpha) \quad \text{for every } \alpha.$$

This means that for every threshold α , the probability of receiving at least α is weakly higher under x than under x' .

Theorem 5.3: FOSD theorem

Suppose preferences take the expected-utility form

$$U(x) = \sum_{s=1}^S \pi_s u(x_s),$$

where u is increasing. If

$$x \succeq_{FOSD} x',$$

then

$$U(x) \geq U(x').$$

So FOSD is a robust welfare ranking: every consumer with increasing expected utility agrees that the dominating bundle is at least as good.

Worked Example: checking first-order stochastic dominance

Let

$$\pi = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right), \quad x = (7, 5, 1), \quad x' = (1, 0, 4).$$

To test whether

$$x \succeq_{FOSD} x',$$

it is enough to check the payoff thresholds appearing in x' .

For $\alpha = 1$:

$$\pi(x, 1) = 1, \quad \pi(x', 1) = \frac{3}{4}.$$

For $\alpha = 0$:

$$\pi(x, 0) = 1, \quad \pi(x', 0) = 1.$$

For $\alpha = 4$:

$$\pi(x, 4) = \frac{1}{2}, \quad \pi(x', 4) = \frac{1}{2}.$$

Thus

$$\pi(x, \alpha) \geq \pi(x', \alpha)$$

for all relevant thresholds, so

$$x \succeq_{FOSD} x'.$$

Therefore every consumer with increasing expected utility weakly prefers x to x' .

5.7 Risk Attitudes

Concavity of Bernoulli utility captures dislike of risk.

Theorem 5.4: Jensen inequality for expected utility

If u is concave, then for every contingent bundle x ,

$$u(E[x]) \geq E[u(x)].$$

Equivalently,

$$u(E[x]) \geq U(x).$$

Proof. This is Jensen's inequality applied to the concave function u . □

The economic meaning is crucial: if utility is concave, then receiving the expected value for sure is at least as good as facing the risky prospect itself.

Definition 5.11: Risk attitudes

Let

$$U(x) = E[u(x)].$$

Then:

- the consumer is *risk averse* if

$$u(E[x]) \geq U(x) \quad \text{for all } x;$$

- the consumer is *risk neutral* if

$$u(E[x]) = U(x) \quad \text{for all } x;$$

- the consumer is *risk loving* if

$$u(E[x]) \leq U(x) \quad \text{for all } x.$$

Theorem 5.5: Linear utility implies risk neutrality

A consumer is risk neutral if the Bernoulli utility function is affine:

$$u(x) = ax + b \quad \text{with } a > 0.$$

In particular, $u(x) = x$ is risk neutral.

Proof. If

$$u(x) = ax + b,$$

then

$$U(x) = E[u(x)] = E[ax + b] = aE[x] + b = u(E[x]).$$

So the consumer is risk neutral. □

Worked Example: same expected value, different risk

Suppose

$$\pi = \left(\frac{1}{2}, \frac{1}{2}\right).$$

Compare the safe bundle

$$x = (10^7, 10^7)$$

with the risky bundle

$$x' = (4 \times 10^7, 0).$$

Their expected values are

$$E[x] = 10^7, \quad E[x'] = 2 \times 10^7.$$

So x' has the higher expected value.

However, if the consumer has increasing and concave Bernoulli utility, expected utility need not follow expected value. A sufficiently risk-averse consumer may prefer the certain bundle x to the much riskier x' despite its lower expected value.

This is exactly why expected utility rather than expected value is the central decision criterion in the lecture.

Worked Example: certainty equivalent under concavity

Suppose

$$\pi = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right), \quad x = (4, 8, 12).$$

Then the expected value is

$$E[x] = \frac{1}{2} \cdot 4 + \frac{1}{4} \cdot 8 + \frac{1}{4} \cdot 12 = 2 + 2 + 3 = 7.$$

If u is increasing and concave, Jensen's inequality gives

$$u(7) = u(E[x]) \geq U(x).$$

So the consumer weakly prefers receiving 7 for sure to receiving the contingent bundle $(4, 8, 12)$ itself.

This is the standard exam interpretation of concavity:

$\text{concavity} \iff \text{risk aversion.}$

Common Mistake

Do not conclude that a bundle with higher expected value must be preferred. That is true only for risk-neutral consumers. Under concave expected utility, a lower-mean but less risky bundle can be preferred.

5.8 Chapter Summary**Key Formula: Lecture 5 Summary**

$x = (x_1, \dots, x_S)$	contingent consumption bundle,
$p \cdot x \leq m$	state-contingent budget,
$q \cdot \theta = \sum_{k=1}^K q_k \theta_k$	portfolio cost,
$\sum_{k=1}^K \theta_k a^k$	portfolio payoff,
$q(\tilde{a}) = q \cdot \theta$	if $\tilde{a} = \sum_{k=1}^K \theta_k a^k$,
$q(a) = p \cdot a$	state-price representation,
$E[x] = \sum_{s=1}^S \pi_s x_s$,	
$U(x) = E[u(x)] = \sum_{s=1}^S \pi_s u(x_s)$,	
$\frac{\pi_i u'(x_i)}{\pi_j u'(x_j)} = \frac{p_i}{p_j}$	interior optimum,
$x \succeq_{FOSD} x' \implies U(x) \geq U(x')$	if u is increasing,
$u(E[x]) \geq E[u(x)]$	if u is concave.

Exam Focus

The highest-yield skills from Lecture 5 are:

- represent uncertain outcomes as contingent consumption bundles;
- translate insurance or asset trading into state-contingent budgets;
- identify arbitrage and use no-arbitrage to price replicated assets;
- recover state prices from asset prices in simple two-state settings;
- use expected utility rather than expected value when risk attitudes matter;
- test first-order stochastic dominance carefully using payoff thresholds;
- interpret concavity of Bernoulli utility as risk aversion.

6 Part I Summary

This chapter consolidates the main ideas from Part I. The goal is not to re-teach Chapters 1–5 in full, but to compress the most examinable material into a revision-oriented framework. The unifying theme is that all five chapters study *choice under constraints*: first with ordinary consumption bundles, then with price changes and endowments, then across time, and finally across risky states of the world.

6.1 Core Definitions and Frameworks

Definition 6.1: Part I master framework

A Part I consumer problem can usually be written in the form

$$\max_{x \in X} U(x) \quad \text{subject to} \quad p \cdot x \leq m,$$

or, with an endowment ω ,

$$\max_{x \in X} U(x) \quad \text{subject to} \quad p \cdot x \leq p \cdot \omega.$$

The interpretation of x , p , and ω changes by topic:

- **ordinary consumption:** x is a bundle of goods;
- **endowments:** ω is the initial bundle owned;
- **intertemporal choice:** coordinates of x are dated consumptions;
- **contingent consumption:** coordinates of x are state-contingent payoffs.

Key Formula: Part I notation map

$$x = (x_1, \dots, x_L) \in \mathbb{R}_+^L, \quad p = (p_1, \dots, p_L) \in \mathbb{R}_{++}^L, \quad p \cdot x = \sum_{\ell=1}^L p_\ell x_\ell.$$

Core interpretations:

- x : chosen bundle / plan / payoff;
- p : price vector / discounted price vector / state price vector;
- m : expenditure or wealth;
- ω : endowment;
- $z = x - \omega$: net demand.

Exam Focus

A large fraction of Part I can be solved by first identifying exactly what the coordinates of x mean. Once that is clear, the problem is usually just a standard consumer-choice problem with a suitably interpreted price vector.

6.2 High-Yield Definitions and Results

6.2.1 Preferences and revealed preference

Key Formula: preference language

$x \succeq y$ weak preference, $x \succ y$ strict preference, $x \sim y$ indifference.

A preference relation is usually assumed to be:

- **complete**;
- **transitive**;
- **increasing**: if $x > y$, then $x \succ y$.

Definition 6.2: Revealed preference notation

If bundle x^t is chosen at prices p^t , then:

$$x^t R x \iff p^t \cdot x^t \geq p^t \cdot x$$

means x was affordable when x^t was chosen, and

$$x^t P x \iff p^t \cdot x^t > p^t \cdot x$$

means x was strictly cheaper than the chosen bundle.

Key Formula: WARP and GARP

If choices maximize an increasing utility function, then:

$$x^t R x \implies U(x^t) \geq U(x), \quad x^t P x \implies U(x^t) > U(x).$$

Hence:

WARP:

$$x^t R x^s \implies \text{not } (x^s P x^t).$$

GARP: if

$$x^t R^* x^s,$$

then not

$$x^s P x^t.$$

Equivalently, utility maximization with increasing utility forbids a weak revealed-preference chain in one direction together with a strict reversal.

Exam Focus

For revealed-preference questions:

1. compute each observed expenditure $p^t \cdot x^t$;
2. test cross-affordability $p^t \cdot x^s$;
3. translate into R or P ;
4. check for WARP or GARP violations.

Never start with utility before checking the affordability relations.

6.2.2 Demand response to price and expenditure changes

Key Formula: demand classifications

For Marshallian demand $x(p, m)$:

- **ordinary good 1:** $p_1 \uparrow \implies x_1 \downarrow$;
- **Giffen good 1:** $p_1 \uparrow \implies x_1 \uparrow$;
- **gross substitutes:** $p_1 \uparrow \implies x_2 \uparrow$;
- **gross complements:** $p_1 \uparrow \implies x_2 \downarrow$;
- **normal good 1:** $m \uparrow \implies x_1 \uparrow$;
- **inferior good 1:** $m \uparrow \implies x_1 \downarrow$.

Key Formula: Slutsky decomposition

If good 1's price rises from p_1 to $p'_1 > p_1$, with $p' = (p'_1, p_2)$, then

$$\Delta x_1 = x_1(p', m) - x_1(p, m),$$

$$m^s = p' \cdot x(p, m),$$

$$\Delta x_1^s = x_1(p', m^s) - x_1(p, m), \quad \Delta x_1^n = x_1(p', m) - x_1(p', m^s),$$

and

$$\Delta x_1 = \Delta x_1^s + \Delta x_1^n.$$

Always:

$$\Delta x_1^s \leq 0 \quad \text{when } p_1 \uparrow.$$

If the good is normal, then also

$$\Delta x_1^n \leq 0.$$

Therefore normal goods satisfy the law of demand:

$$p_1 \uparrow \implies x_1 \downarrow.$$

Key Formula: Hicksian substitution

For the same price increase, let $u_0 = U(x(p, m))$ and let m^h be the minimum expenditure at prices p' needed to attain utility u_0 . Then

$$\Delta x_1 = \Delta x_1^h + \Delta x_1^q,$$

where

$$\Delta x_1^h = h_1(p', u_0) - x_1(p, m) \leq 0.$$

Also

$$m^h \leq m^s.$$

6.2.3 Marginal utility, MRS, and interior optimality

Key Formula: MRS toolkit

For differentiable utility $U(x_1, x_2)$:

$$MU_1 = \frac{\partial U}{\partial x_1}, \quad MU_2 = \frac{\partial U}{\partial x_2}.$$

At an interior optimum with budget line

$$p_1 x_1 + p_2 x_2 = m,$$

the tangency condition is

$$MRS_{12} = \frac{MU_1}{MU_2} = \frac{p_1}{p_2}.$$

Exam Focus

The tangency condition is only valid for an *interior* optimum. If the tangency candidate is infeasible or gives negative demand, switch immediately to checking corners.

6.2.4 Endowments and net demand

Definition 6.3: Net demand

Given endowment ω and chosen bundle x ,

$$z = x - \omega.$$

Equivalently, if $x(p, m)$ is Marshallian demand and wealth comes from selling the endowment,

$$z(p) = x(p, p \cdot \omega) - \omega.$$

Interpretation:

$$z_\ell > 0 \Rightarrow \text{net buyer of good } \ell, \quad z_\ell < 0 \Rightarrow \text{net seller of good } \ell.$$

Key Formula: endowment budget and Walras' law

Affordability with endowments:

$$p \cdot x \leq p \cdot \omega \quad \Longleftrightarrow \quad p \cdot z \leq 0.$$

At an optimum with increasing utility,

$$p \cdot x = p \cdot \omega \quad \Longrightarrow \quad p \cdot z = 0.$$

Key Formula: endowment price-change logic

When the price of good 1 rises:

- substitution effect still satisfies

$$\Delta x_1^s \leq 0;$$

- the sign of the income effect depends on whether the consumer is a buyer or seller of good 1.

For normal demand:

Change	Status in good 1	Income effect	Definite total sign?
$p_1 \uparrow$	buyer	≤ 0	yes: $\Delta x_1 \leq 0$
$p_1 \uparrow$	seller	≥ 0	ambiguous
$p_1 \downarrow$	buyer	≥ 0	yes: $\Delta x_1 \geq 0$
$p_1 \downarrow$	seller	≤ 0	ambiguous

Exam Focus

Under endowments, the most important first step is to classify the consumer as a *net buyer* or *net seller* of the good whose price changes. Without that step, the sign of the income effect is usually wrong.

6.2.5 Intertemporal consumption**Key Formula: intertemporal budget**

For periods $t = 0, \dots, T$ and interest rate $r \geq 0$,

$$\sum_{t=0}^T \frac{x_t}{(1+r)^t} = \sum_{t=0}^T \frac{y_t}{(1+r)^t}.$$

With

$$p_t = \left(\frac{1}{1+r} \right)^t,$$

this is simply

$$p \cdot x = p \cdot y.$$

Definition 6.4: Intertemporal net demand

For income stream $y = (y_0, \dots, y_T)$ and chosen stream $x = (x_0, \dots, x_T)$,

$$z = x - y.$$

If $T = 1$, then:

- $z_0 < 0$ means saving in period 0;
- $z_0 > 0$ means borrowing in period 0;
- z_1 records the opposite side of that saving/borrowing decision in period 1.

Key Formula: interest-rate comparative statics

In the two-period case,

$$p_1 = \frac{1}{1+r}.$$

So:

- $r \uparrow$ means future consumption becomes cheaper;
- a saver (net buyer of future consumption) stays a saver;
- if $r \downarrow$, a borrower (net seller of future consumption) stays a borrower.

Key Formula: discounted utility

For

$$U(x_0, x_1) = u(x_0) + \delta u(x_1),$$

the interior first-order condition is

$$u'(x_0) = (1+r)\delta u'(x_1).$$

Hence:

- if $(1+r)\delta = 1$, then $x_0 = x_1$;
- if $(1+r)\delta > 1$, then $x_1 > x_0$;
- if $(1+r)\delta < 1$, then $x_0 > x_1$.

6.2.6 Contingent consumption, state prices, and risk**Definition 6.5: Contingent consumption and expected utility**

With S states, probabilities $\pi = (\pi_1, \dots, \pi_S)$, and contingent consumption

$$x = (x_1, \dots, x_S),$$

expected utility is

$$U(x) = \sum_{s=1}^S \pi_s u(x_s).$$

Key Formula: state prices and asset pricing

If $p = (p_1, \dots, p_S)$ is the state-price vector, then a contingent bundle x costs

$$p \cdot x.$$

If an asset has payoff vector

$$a = (a_1, \dots, a_S),$$

then its no-arbitrage price is

$$q(a) = p \cdot a.$$

More generally, if a new asset can be replicated by portfolio θ , its price must equal the cost of the replicating portfolio.

Key Formula: expected value, FOSD, and risk attitudes

Expected value:

$$E[x] = \sum_{s=1}^S \pi_s x_s.$$

Interior FOC under expected utility:

$$\frac{\pi_i u'(x_i)}{\pi_j u'(x_j)} = \frac{p_i}{p_j}.$$

First-order stochastic dominance:

$$x \succeq_{FOSD} x' \implies U(x) \geq U(x')$$

for all increasing u .

If u is concave, then

$$u(E[x]) \geq E[u(x)] = U(x).$$

Interpretation: the consumer is risk averse.

Risk classifications:

$$u(E[x]) \geq U(x) \Rightarrow \text{risk averse,}$$

$$u(E[x]) = U(x) \Rightarrow \text{risk neutral,}$$

$$u(E[x]) \leq U(x) \Rightarrow \text{risk loving.}$$

6.3 Comparison of Central Models

Topic	Choice vector	Price interpretation	Core technique
Revealed preference	ordinary bundle x	market prices p	affordability tests, R/P , WARP, GARP
Price changes	Marshallian demand $x(p, m)$	relative goods prices	Slutsky decomposition
Endowments	consumption x with endowment ω	goods prices applied to both x and ω	net demand $z = x - \omega$, buyer/seller classification
Intertemporal choice	dated consumption $(x_0, \dots, x_T)$	discounted prices $p_t = (1+r)^{-t}$	present value, saving/borrowing, Euler condition
Risk / contingent claims	state payoffs $(x_1, \dots, x_S)$	state prices p_s	expected utility, no-arbitrage pricing, FOSD

Exam Focus

A strong way to remember Part I is:

ordinary goods $\rightarrow$ endowments $\rightarrow$ time $\rightarrow$ states.

The mathematics stays almost the same. Only the economic interpretation of each coordinate changes.

6.4 Standard Problem Types and Solution Patterns

6.4.1 Type A: revealed-preference consistency

Method Pattern: WARP / GARP problems

Use the following template:

1. list each observation (p^t, x^t) ;

2. compute

$$p^t \cdot x^t \quad \text{and} \quad p^t \cdot x^s$$

for other bundles x^s ;

3. infer R or P relations;

4. look for:

$$x^t R x^s \quad \text{and} \quad x^s P x^t$$

for WARP, or

$$x^t R^* x^s \quad \text{and} \quad x^s P x^t$$

for GARP.

6.4.2 Type B: price change and demand decomposition

Method Pattern: Slutsky questions

When a price changes:

1. compute the original demand $x(p, m)$;

2. compute the compensated expenditure

$$m^s = p' \cdot x(p, m);$$

3. compute the compensated demand $x(p', m^s)$;

4. split the total effect into substitution and income effects;

5. use normal/inferior status to sign the income effect.

6.4.3 Type C: endowment comparative statics

Method Pattern: endowment questions

Always do this first:

$$z = x - \omega.$$

Then identify whether the consumer is a buyer or seller of the good whose price changes. Only after that should you analyze the substitution and income effects.

6.4.4 Type D: intertemporal optimization

Method Pattern: intertemporal questions

Standard order:

1. convert the stream budget into present-value form;
2. rewrite as a standard consumer problem with prices

$$p_t = (1 + r)^{-t};$$

3. if discounted utility is given, apply

$$u'(x_0) = (1 + r)\delta u'(x_1);$$

4. use the budget equation to solve.

6.4.5 Type E: risk and asset pricing

Method Pattern: contingent consumption / assets

When a risk problem involves insurance or asset trade:

1. write the final contingent consumption bundle;
2. identify the linear tradeoff across states;
3. infer the implied state prices or use no-arbitrage replication;
4. if expected utility is given, solve

$$\frac{\pi_i u'(x_i)}{\pi_j u'(x_j)} = \frac{p_i}{p_j}.$$

6.5 Common Mistakes and Exam Traps

Common Mistake

In revealed-preference questions, do not mix up the meanings of R and P . The lecture convention is:

$$\begin{aligned} x^t R x &\iff x \text{ was affordable when } x^t \text{ was chosen,} \\ x^t P x &\iff x \text{ was strictly cheaper than the chosen bundle.} \end{aligned}$$

Common Mistake

Do not conclude that a good is normal from a negative own-price effect. Normality is about the sign of the *expenditure* effect, not the total price effect.

Common Mistake

Do not apply the tangency condition automatically. It requires an interior optimum. If the candidate is infeasible or violates nonnegativity, check corners.

Common Mistake

In endowment problems, forgetting to classify buyer versus seller is usually fatal. The substitution effect has its usual sign, but the income effect reverses between buyers and sellers.

Common Mistake

In intertemporal questions, do not keep the raw interest-rate algebra longer than necessary. Rewrite immediately as

$$p_t = (1 + r)^{-t}$$

and reduce to a standard budget problem.

Common Mistake

In risky-choice problems, do not use expected value when the question specifies expected utility. Higher expected value does *not* imply higher welfare for a risk-averse consumer.

6.6 Integrated Worked Review Examples

Worked Review Example 1: GARP logic from a chain of choices

Suppose three bundles are observed:

$$x, y, z.$$

You find that

$$xRy, \quad yRz, \quad zPx.$$

Then

$$xR^*z$$

because there is a revealed-preference chain from x to z . Since also

$$zPx,$$

the data violate GARP.

Why? Under increasing utility maximization,

$$xR^*z \implies U(x) \geq U(z),$$

but

$$zPx \implies U(z) > U(x),$$

which is impossible.

So the correct conclusion is:

The dataset cannot be rationalized by an increasing utility function.

Worked Review Example 2: Slutsky decomposition and the law of demand

Suppose a two-good consumer has demand

$$x(p_1, p_2, m) = \left(\frac{m}{2p_1}, \frac{m}{2p_2} \right).$$

This is the Cobb–Douglas demand from $U(x_1, x_2) = x_1x_2$.

Let

$$p = (1, 1), \quad m = 8, \quad p' = (2, 1).$$

Original demand:

$$x(p, m) = \left(\frac{8}{2}, \frac{8}{2}\right) = (4, 4).$$

New uncompensated demand:

$$x(p', m) = \left(\frac{8}{4}, \frac{8}{2}\right) = (2, 4).$$

So the total change in demand for good 1 is

$$\Delta x_1 = 2 - 4 = -2.$$

Now compute Slutsky compensation:

$$m^s = p' \cdot x(p, m) = 2 \cdot 4 + 1 \cdot 4 = 12.$$

Compensated demand:

$$x(p', m^s) = \left(\frac{12}{4}, \frac{12}{2}\right) = (3, 6).$$

Hence

$$\Delta x_1^s = 3 - 4 = -1, \quad \Delta x_1^n = 2 - 3 = -1.$$

Therefore

$$\Delta x_1 = \Delta x_1^s + \Delta x_1^n = -1 + (-1) = -2.$$

Interpretation:

- substitution effect is negative;
- income effect is negative because the good is normal;
- law of demand holds.

Worked Review Example 3: endowment buyer versus seller

Suppose the same demand

$$x(p_1, p_2, m) = \left(\frac{m}{2p_1}, \frac{m}{2p_2}\right)$$

applies, and the endowment is

$$\omega = (1, 6).$$

At prices

$$p = (1, 1),$$

wealth is

$$m = p \cdot \omega = 7.$$

So demand is

$$x(1, 1, 7) = \left(\frac{7}{2}, \frac{7}{2}\right).$$

Hence net demand is

$$z = x - \omega = \left(\frac{7}{2} - 1, \frac{7}{2} - 6\right) = \left(\frac{5}{2}, -\frac{5}{2}\right).$$

So the consumer is a *buyer of good 1*.

If p_1 rises, then:

- substitution effect on good 1 is negative;

- because the consumer is a buyer of good 1, the income effect is also negative if demand is normal.

Therefore

$$\Delta x_1 \leq 0.$$

The exam lesson is that the buyer/seller classification determines the sign of the income effect in endowment problems.

Worked Review Example 4: intertemporal demand and discounting

Suppose the income stream is

$$y = (190, 11), \quad r = 0.1,$$

and utility is

$$U(x_0, x_1) = x_0 x_1.$$

Present-value wealth is

$$m = 190 + \frac{11}{1.1} = 200.$$

For this utility, intertemporal demand is

$$x(r, m) = \left(\frac{m}{2}, \frac{(1+r)m}{2} \right).$$

So

$$x(0.1, 200) = \left(100, \frac{1.1 \cdot 200}{2} \right) = (100, 110).$$

Hence the consumer saves

$$190 - 100 = 90$$

in period 0.

The revision point is that once the present value is computed, the problem is just a standard two-good demand problem with prices

$$\left(1, \frac{1}{1+r} \right).$$

Worked Review Example 5: state prices and expected-utility demand

Suppose there are two states with probabilities

$$\pi = \left(\frac{1}{3}, \frac{2}{3} \right),$$

and Bernoulli utility

$$u(x) = \ln x.$$

For log utility under expected utility,

$$x(p_1, p_2, m) = \left(\frac{\pi_1 m}{p_1}, \frac{\pi_2 m}{p_2} \right).$$

First suppose the implied state prices are

$$p = (3, 1)$$

and wealth is

$$m = 30.$$

Then

$$x(3, 1, 30) = \left(\frac{(1/3) \cdot 30}{3}, \frac{(2/3) \cdot 30}{1} \right) = \left(\frac{10}{3}, 20 \right).$$

Now suppose instead two assets generate state prices

$$p = (6, 1)$$

and the consumer has wealth

$$m = 18.$$

Then demand is

$$x(6, 1, 18) = \left(\frac{(1/3) \cdot 18}{6}, \frac{(2/3) \cdot 18}{1} \right) = (1, 12).$$

These examples show the standard review workflow:

1. recover the state prices from insurance or assets;
2. plug them into the expected-utility demand formula.

6.7 Part I Revision Checklist

Part I final checklist

Before the test, make sure you can do all of the following quickly:

1. define R , P , R^* , WARP, and GARP precisely;
2. classify goods as ordinary/Giffen, normal/inferior, substitutes/complements;
3. derive and use the Slutsky identity;
4. explain the difference between Slutsky and Hicksian compensation;
5. compute MU_1 , MU_2 , and apply

$$\frac{MU_1}{MU_2} = \frac{p_1}{p_2}$$

when appropriate;

6. form net demand

$$z = x - \omega$$

and classify buyer versus seller;

7. write any intertemporal budget in present-value form;

8. apply

$$u'(x_0) = (1 + r)\delta u'(x_1)$$

in discounted-utility problems;

9. price assets by replication or state prices;
10. compute expected value, expected utility, and identify risk aversion from concavity;
11. test first-order stochastic dominance.

Key Formula: Part I memory sheet

$$x^t R x \iff p^t \cdot x^t \geq p^t \cdot x,$$

$$x^t P x \iff p^t \cdot x^t > p^t \cdot x,$$

$$\Delta x_1 = \Delta x_1^s + \Delta x_1^n,$$

$$z = x - \omega,$$

$$p \cdot z = 0 \quad \text{at an optimum with increasing utility,}$$

$$\sum_{t=0}^T \frac{x_t}{(1+r)^t} = \sum_{t=0}^T \frac{y_t}{(1+r)^t},$$

$$u'(x_0) = (1+r)\delta u'(x_1),$$

$$U(x) = \sum_{s=1}^S \pi_s u(x_s),$$

$$\frac{\pi_i u'(x_i)}{\pi_j u'(x_j)} = \frac{p_i}{p_j},$$

$$u(E[x]) \geq E[u(x)] \quad \text{if } u \text{ is concave.}$$

7 Measuring Welfare

This chapter begins Part II by introducing monetary measures of welfare. Utility is the conceptual object that captures welfare, but utility itself is not directly observable. For policy analysis, market intervention, and social-welfare comparisons, it is therefore useful to translate changes in utility into dollar terms. The main measures in this chapter are reservation prices, gains to trade, consumer surplus, compensating variation, equivalent variation, and producer surplus.

7.1 Why Measure Welfare in Monetary Terms?

A central goal of welfare economics is to compare alternative allocations, prices, and policies. Since utility is not directly observed, economists often use monetary measures that track how much income a consumer would be willing to give up or require in compensation.

Definition 7.1: Monetary welfare measure

A *monetary welfare measure* is a dollar amount that is equivalent to a change in utility. It answers questions such as:

- How much is a consumer willing to pay for a good or opportunity?
- How much compensation is needed after a harmful price change?
- How much welfare is gained from access to a market?

Exam Focus

The point of monetary welfare measures is not to replace utility, but to express changes in utility in a common unit that can be compared with costs, benefits, taxes, subsidies, or producer gains.

7.2 Reservation Prices and Gains to Trade

The most direct monetary measure of value is a reservation price.

Definition 7.2: Reservation price

A *reservation price* is the maximum amount of money a consumer is willing to pay for a good, action, or marginal increase in consumption.

For an indivisible object, such as buying one iPhone, the reservation price is defined by indifference between buying and not buying.

Key Formula: Reservation Price for a Discrete Object

Suppose utility depends on cash holdings and ownership of an object:

$$U(\text{cash}, \text{ownership}).$$

If income is I , then the reservation price r satisfies

$$U(I - r, \text{own object}) = U(I, \text{do not own object}).$$

Hence:

- if price $p < r$, buy;
- if price $p > r$, do not buy.

For repeated units of a good, reservation prices can be defined unit by unit.

Definition 7.3: Reservation price sequence and gains to trade

Let r_n denote the reservation price for the n th unit of a good, given that the consumer already has $n - 1$ units. If the market price is p , then the consumer buys the n th unit only if

$$r_n - p > 0.$$

If the consumer buys n units, the total dollar value of net utility gains from trade is

$$\sum_{i=1}^n (r_i - p) = \left(\sum_{i=1}^n r_i \right) - np.$$

When the good is infinitely divisible, this becomes the area between the reservation price curve and the price line.

Worked Example: reservation price for nuggets

Suppose Tom has utility

$$U(\text{cash held}, \# \text{nuggets}) = \text{cash held} + 10 \times \# \text{nuggets}.$$

Then the gain from one extra nugget is always 10 units of utility, and one dollar of cash is worth 1 unit of utility. Therefore the reservation price for the first nugget is

$$\boxed{10}.$$

Because utility is linear in the number of nuggets, the marginal utility of nuggets does not change with the number already consumed. Hence the reservation price does *not* change across units:

$$r_1 = r_2 = r_3 = \dots = 10.$$

7.3 Consumer Surplus and Its Interpretation

Consumer surplus is often used to approximate the gains to trade from market participation.

Definition 7.4: Consumer surplus

Let $x(p)$ denote Marshallian demand for a good and let $P(x)$ denote inverse demand. Consumer surplus at price p is

$$CS = \int_0^{x(p)} P(q) dq - p x(p).$$

Graphically, it is the area under the inverse demand curve and above the price line, up to the chosen quantity.

For a price increase from p' to $p'' > p'$, the loss in consumer surplus is the area under Marshallian demand between the two prices:

$$\Delta CS_{\text{loss}} = \int_{p'}^{p''} x(p) dp.$$

Common Mistake

Consumer surplus is *not* generally the same as the exact utility gain from trade. It is computed from the inverse demand curve, whereas exact gains to trade are based on the reservation price curve.

7.4 Reservation Price Curve versus Inverse Demand Curve

These two curves look similar, but they are conceptually different.

Definition 7.5: Reservation price curve

A *reservation price curve* records the willingness to pay for successive marginal units of a good, holding the reference level of money available for other goods fixed.

Definition 7.6: Inverse demand curve

An *inverse demand curve* gives the price at which a total quantity x is the optimal quantity purchased when all x units are bought at that price.

The difference comes from income effects:

- along the reservation price curve, the reference cash level is held fixed when valuing marginal units;
- along inverse demand, the amount of money left over changes with the quantity chosen.

Therefore the two curves need not coincide.

Exam Focus

If the question asks whether consumer surplus is an *exact* welfare measure, the correct answer is usually “not in general.” It becomes exact only under special assumptions, especially quasi-linear utility.

7.5 Quasi-Linear Utility and Exact Consumer Surplus

The main special case where consumer surplus becomes exact is quasi-linearity.

Definition 7.7: Quasi-linear utility

A two-good utility function is *quasi-linear in x_2* if it takes the form

$$U(x_1, x_2) = v(x_1) + \alpha x_2,$$

where v is increasing and typically concave, and $\alpha > 0$ is constant.

If utility is quasi-linear in x_2 , then the marginal utility of x_2 is constant. This removes the income effect on x_1 .

Theorem 7.1: No income effect under quasi-linear utility

If

$$U(x, m) = v(x) + m,$$

where m is money left over for other goods, then demand for x is independent of income. In particular, the inverse demand curve and reservation price curve coincide.

Proof. Consider

$$\max_{x,m} v(x) + m \quad \text{subject to} \quad px + m = I.$$

Substituting $m = I - px$ gives

$$\max_x v(x) + I - px.$$

The first-order condition is

$$v'(x) = p.$$

This condition depends on x and p , but not on income I . Hence there is no income effect on x . Since willingness to pay for the marginal unit is also given by

$$v'(x),$$

the inverse demand and reservation price curves coincide. □

Theorem 7.2: Consumer surplus equals gains to trade under quasi-linear utility

If

$$U(x, m) = v(x) + m,$$

then consumer surplus is an exact measure of gains to trade:

$$CS = v(x) - v(0) - px.$$

Proof. Under quasi-linear utility, inverse demand is

$$p = v'(x).$$

Hence

$$CS = \int_0^x v'(q) dq - px = v(x) - v(0) - px.$$

This is exactly the net utility gain from obtaining x units rather than zero units. □

Worked Example: identifying quasi-linear utility

Consider the following utility functions:

$$U(x, y) = x + y, \quad U(x, y) = x + 4y, \quad U(x, y) = x + 3\sqrt{y}.$$

- $U(x, y) = x + y$ is quasi-linear in both goods, since it is linear in both.
- $U(x, y) = x + 4y$ is also quasi-linear in both goods.
- $U(x, y) = x + 3\sqrt{y}$ is quasi-linear in x , since it can be written as

$$U(x, y) = x + v(y) \quad \text{with} \quad v(y) = 3\sqrt{y}.$$

However, the zero-income-effect conclusion must be attached to the good that enters through the *nonlinear* part in the standard quasi-linear setup. For example, if

$$U(x, y) = y + 3\sqrt{x},$$

then utility is quasi-linear in y , and demand for x has zero income effect.

Common Mistake

Do not say “quasi-linear means no income effects for both goods.” The standard result is that the good appearing inside the nonlinear part has zero income effect when utility is quasi-linear in the other good.

7.6 Price Changes and Welfare: Indirect Utility, CV, and EV

Consumer surplus is convenient, but it is not always exact. Two exact monetary measures of welfare change are compensating variation and equivalent variation.

Definition 7.8: Indirect utility

The *indirect utility function* is

$$V(p, I) = \max_x U(x) \quad \text{subject to} \quad p \cdot x \leq I.$$

It gives the maximum utility attainable at prices p and income I .

Suppose the price of good 1 rises from p'_1 to p''_1 , with p_2 fixed and income I fixed.

Definition 7.9: Compensating variation and equivalent variation

For a price increase:

- *Compensating variation (CV)* is the least extra income needed at the *new* prices to restore the consumer to the *original* utility level:

$$V(p''_1, p_2, I + CV) = V(p'_1, p_2, I).$$

- *Equivalent variation (EV)* is the greatest amount of income that can be taken away at the *old* prices while leaving the consumer at the *final* utility level:

$$V(p'_1, p_2, I - EV) = V(p''_1, p_2, I).$$

For a price increase, both CV and EV are positive if they are interpreted as magnitudes of welfare loss.

Exam Focus

The difference between CV and EV is the reference point:

- **CV:** new prices, old utility;
- **EV:** old prices, new utility.

A good mnemonic is:

- *compensating* = money added after the bad change;
- *equivalent* = money taken away before the bad change.

Worked Example: indirect utility, EV, and CV for perfect complements

Suppose

$$U(x_1, x_2) = \min\{x_1, x_2\}, \quad I = 100, \quad p_2 = 10,$$

and the price of good 1 rises from

$$p'_1 = 5 \quad \text{to} \quad p''_1 = 10.$$

Since utility is Leontief, the optimal bundle satisfies

$$x_1 = x_2.$$

Using the budget constraint

$$p_1 x_1 + p_2 x_2 = I,$$

we get

$$x_1 = x_2 = \frac{I}{p_1 + p_2}.$$

Therefore the indirect utility is

$$V(p_1, p_2, I) = \frac{I}{p_1 + p_2}.$$

Initial utility:

$$V(5, 10, 100) = \frac{100}{15} = \frac{20}{3}.$$

Final utility:

$$V(10, 10, 100) = \frac{100}{20} = 5.$$

Equivalent variation: solve

$$V(5, 10, 100 - EV) = 5.$$

So

$$\frac{100 - EV}{15} = 5 \implies 100 - EV = 75 \implies EV = 25.$$

Compensating variation: solve

$$V(10, 10, 100 + CV) = \frac{20}{3}.$$

So

$$\frac{100 + CV}{20} = \frac{20}{3} \implies 100 + CV = \frac{400}{3} \implies CV = \frac{100}{3}.$$

Hence

$$\boxed{EV = 25, \quad CV = \frac{100}{3}.$$

Both are positive, as expected for a welfare-reducing price increase.

7.7 Comparing ΔCS , CV, and EV

These three measures are closely related.

Theorem 7.3: Relationship between consumer surplus, CV, and EV

For a price change in the two-good case, the consumer-surplus measure lies between compensating variation and equivalent variation:

$$\min\{|CV|, |EV|\} \leq |\Delta CS| \leq \max\{|CV|, |EV|\}.$$

When utility is quasi-linear in the other good, all three measures coincide.

Worked Example: comparing the three welfare measures

Continue the previous example with

$$U(x_1, x_2) = \min\{x_1, x_2\}, \quad I = 100, \quad p_2 = 10, \quad p_1 : 5 \rightarrow 10.$$

We already found

$$EV = 25, \quad CV = \frac{100}{3} \approx 33.33.$$

Marshallian demand for good 1 is

$$x_1(p_1) = \frac{100}{p_1 + 10}.$$

So the loss in consumer surplus from the price increase is

$$\Delta CS_{\text{loss}} = \int_5^{10} \frac{100}{p_1 + 10} dp_1 = 100 \ln \left(\frac{20}{15} \right) = 100 \ln \left(\frac{4}{3} \right) \approx 28.77.$$

Hence

$$25 = EV < \Delta CS_{\text{loss}} \approx 28.77 < CV \approx 33.33.$$

So ΔCS lies between EV and CV, exactly as theory predicts.

Common Mistake

Do not treat

$$x_1 \cdot (p_1'' - p_1')$$

as the welfare loss from a price increase. That ignores substitution effects and therefore generally overstates the true utility loss.

7.8 Producer Surplus

Welfare analysis also applies to producers. The lecture treats producer surplus as the producer-side analogue of consumer surplus.

Definition 7.10: Producer surplus

If a firm supplies quantity q at price p and has marginal cost curve $MC(q)$, then producer surplus is

$$PS = pq - VC(q) = \int_0^q (p - MC(\tilde{q})) d\tilde{q},$$

where $VC(q)$ is variable cost.

Under standard profit maximization, the competitive supply curve is the marginal-cost curve above shutdown. Hence producer surplus is the area above supply and below price, up to the quantity sold.

Worked Example: producer surplus from marginal cost

Suppose a competitive firm's marginal cost is

$$MC(q) = 2q,$$

and market price is

$$p = 10.$$

The profit-maximizing quantity satisfies

$$p = MC(q) \implies 10 = 2q \implies q = 5.$$

Producer surplus is

$$PS = \int_0^5 (10 - 2q) dq = [10q - q^2]_0^5 = 50 - 25 = 25.$$

Therefore

$$\boxed{PS = 25.}$$

Policy Application

Consumer surplus and producer surplus are central because later welfare analysis often aggregates them to study taxes, subsidies, quotas, and other market interventions. This chapter provides the monetary foundations for that aggregation.

7.9 Standard Problem Types and Solution Patterns**Method Pattern: reservation price questions**

For discrete or indivisible objects:

1. write the consumer's utility before and after purchase;
2. set them equal to identify the reservation price;
3. compare the actual price with the reservation price.

Method Pattern: quasi-linear utility

If utility is

$$U(x, m) = v(x) + m,$$

then:

1. use

$$v'(x) = p$$
 to find demand;
2. conclude there is no income effect on x ;
3. use consumer surplus as an exact welfare measure.

Method Pattern: EV/CV questions

For price-change questions:

1. derive or identify the indirect utility function $V(p, I)$;
2. compute initial and final utility levels;
3. solve

$$V(\text{new prices}, I + CV) = V(\text{old prices}, I)$$
 for CV ;
4. solve

$$V(\text{old prices}, I - EV) = V(\text{new prices}, I)$$
 for EV .

7.10 Chapter Summary**Key Formula: Lecture 7 Summary**

$$\begin{aligned}
 U(I - r, \text{own}) &= U(I, \text{not own}) && \text{reservation price,} \\
 \text{Gains to trade} &= \sum_{i=1}^n (r_i - p), \\
 CS &= \int_0^{x(p)} P(q) dq - p x(p), \\
 V(p, I) &= \max_x U(x) \text{ s.t. } p \cdot x \leq I, \\
 V(p_1'', p_2, I + CV) &= V(p_1', p_2, I), \\
 V(p_1', p_2, I - EV) &= V(p_1'', p_2, I), \\
 \min\{|CV|, |EV|\} &\leq |\Delta CS| \leq \max\{|CV|, |EV|\}, \\
 U(x, m) = v(x) + m &\Rightarrow v'(x) = p \text{ and } CS \text{ is exact,} \\
 PS &= \int_0^q (p - MC(\tilde{q})) d\tilde{q}.
 \end{aligned}$$

Exam Focus

The highest-yield takeaways from this chapter are:

- utility changes are often measured in money because utility is not directly observable;
- reservation prices measure willingness to pay for marginal utility gains;
- consumer surplus usually approximates gains to trade, but becomes exact under quasi-linear utility;
- EV and CV are exact welfare measures for price changes and account for substitution effects;
- for a price increase, EV and CV are positive magnitudes of welfare loss;
- producer surplus measures firm welfare when firms maximize profit.

8 Simple Welfare Analysis

This chapter moves from individual welfare to aggregate welfare. Chapter 7 introduced monetary measures of consumer and producer welfare. Here, those measures are summed across agents to study welfare at the market level. The main applications are welfare analysis in demand–supply diagrams, tax regimes, deadweight loss, tax incidence, and comparative statics.

8.1 Aggregating Welfare Across Consumers and Producers

Consider a single-good market. Let individual Marshallian demands be

$$q_i^D(p),$$

and individual supplies be

$$q_j^S(p).$$

The market curves are obtained by horizontal summation.

Definition 8.1: Aggregate demand and aggregate supply

The *aggregate demand* and *aggregate supply* functions are

$$Q^D(p) = \sum_i q_i^D(p), \quad Q^S(p) = \sum_j q_j^S(p).$$

Their inverse forms are denoted by

$$P_D(Q), \quad P_S(Q),$$

whenever these are well-defined.

Aggregate welfare is measured by adding up individual surpluses.

Definition 8.2: Aggregate consumer surplus, producer surplus, and social welfare

If market quantity is Q and consumers pay price p , aggregate consumer surplus is

$$CS(Q) = \int_0^Q P_D(q) dq - pQ.$$

If producers receive price p , aggregate producer surplus is

$$PS(Q) = pQ - \int_0^Q P_S(q) dq.$$

Hence *social welfare* in the single-good market is

$$SW(Q) = CS(Q) + PS(Q) = \int_0^Q (P_D(q) - P_S(q)) dq.$$

The last expression is the key one: social welfare is the area between inverse demand and inverse supply up to the traded quantity.

Theorem 8.1: Competitive equilibrium maximizes $CS + PS$

Suppose inverse demand is downward sloping and inverse supply is upward sloping. Then social

welfare

$$SW(Q) = \int_0^Q (P_D(q) - P_S(q)) dq$$

is maximized at the competitive quantity Q^* satisfying

$$P_D(Q^*) = P_S(Q^*).$$

Proof. Differentiate:

$$\frac{dSW(Q)}{dQ} = P_D(Q) - P_S(Q).$$

Thus welfare rises whenever

$$P_D(Q) > P_S(Q),$$

falls whenever

$$P_D(Q) < P_S(Q),$$

and is maximized when

$$P_D(Q) = P_S(Q).$$

Under the usual slopes, this point is unique and coincides with the competitive equilibrium quantity. $\square$

Exam Focus

The fastest welfare argument in a competitive single-good market is:

$$SW(Q) = \int_0^Q (P_D(q) - P_S(q)) dq.$$

Hence any deviation from the competitive quantity loses the positive gains from trade on units for which demand exceeds supply, or forces inefficient units for which supply exceeds demand.

Worked Example: aggregate demand and aggregate consumer surplus

Suppose there are 30 identical buyers, each with inverse demand

$$p = 100 - 2q.$$

Equivalently, each individual's demand is

$$q_i(p) = \frac{100 - p}{2} \quad \text{for } p \leq 100,$$

and $q_i(p) = 0$ for $p > 100$.

Therefore aggregate demand is

$$Q^D(p) = 30 \cdot \frac{100 - p}{2} = 1500 - 15p \quad \text{for } p \leq 100.$$

In inverse form,

$$p = 100 - \frac{Q}{15}.$$

Now suppose the market price is

$$p = 10.$$

Then aggregate quantity demanded is

$$Q = 1500 - 15(10) = 1350.$$

Aggregate consumer surplus is

$$CS = \frac{1}{2} \times (100 - 10) \times 1350 = \frac{1}{2} \times 90 \times 1350 = 60750.$$

So the correct answers are

$$Q^D(p) = 1500 - 15p \quad \text{and} \quad CS = 60750.$$

8.2 Deadweight Welfare Loss

Because the competitive equilibrium maximizes $CS + PS$, any policy that changes the traded quantity away from Q^* creates a welfare loss.

Definition 8.3: Deadweight welfare loss

If a policy reduces market quantity from the efficient level Q^* to some lower quantity Q' , the *deadweight welfare loss* is

$$DWL = \int_{Q'}^{Q^*} (P_D(q) - P_S(q)) dq.$$

This is the value of mutually beneficial trades that no longer occur.

Key Formula: Deadweight Loss under a Per-Unit Tax

If a per-unit tax t creates a constant wedge between buyer and seller prices and the market quantity falls from Q^* to Q' , then graphically

$$DWL = \frac{1}{2} t (Q^* - Q')$$

whenever the relevant region is triangular, in particular under linear demand and supply.

Theorem 8.2: No quantity distortion implies no deadweight loss

If a tax does not change the equilibrium quantity, then it does not create deadweight loss.

Proof. By definition,

$$DWL = \int_{Q'}^{Q^*} (P_D(q) - P_S(q)) dq.$$

If $Q' = Q^*$, the interval of integration is empty, so

$$DWL = 0.$$

□

Exam Focus

A standard implication is: if either demand or supply is perfectly inelastic, then a tax can be pure transfer with no deadweight loss because the traded quantity does not change.

8.3 Quantity Taxes

A quantity tax, or per-unit tax, imposes a fixed dollar tax t on each unit traded.

Definition 8.4: Quantity taxes, excise taxes, and sales taxes

A *quantity tax* is a fixed amount $t > 0$ per unit traded.

- If sellers pay it, it is an *excise tax*.
- If buyers pay it, it is a *sales tax*.

Let

$$p_b = \text{price paid by buyers}, \quad p_s = \text{price received by sellers}.$$

Then in either case,

$$p_b = p_s + t.$$

The statutory side of the tax does not matter for the equilibrium outcome when the tax is a fixed amount per unit. What matters is the wedge condition

$$p_b - p_s = t.$$

Theorem 8.3: Quantity excise and quantity sales taxes are equivalent

A per-unit excise tax of size t and a per-unit sales tax of size t generate the same equilibrium allocation and the same buyer and seller prices.

Proof. In both cases equilibrium requires

$$Q^D(p_b) = Q^S(p_s)$$

together with

$$p_b = p_s + t.$$

These two equations are identical across the two statutory tax regimes, so the equilibrium values of p_b , p_s , and Q are the same. $\square$

8.3.1 Graphical effect of a quantity excise tax

Let the original supply curve be

$$Q = S(p).$$

If the tax is levied on sellers, then when buyers pay price p , sellers receive only $p - t$. Therefore supply becomes

$$Q = S(p - t).$$

In inverse form, if the original inverse supply is

$$p = P_S(Q),$$

the post-tax inverse supply is

$$p = P_S(Q) + t.$$

So a quantity excise tax shifts inverse supply vertically upward by t .

8.3.2 Graphical effect of a quantity sales tax

Let the original demand curve be

$$Q = D(p).$$

If the tax is levied on buyers, then when sellers receive price p , buyers pay $p + t$. Therefore demand becomes

$$Q = D(p + t).$$

In inverse form, if the original inverse demand is

$$p = P_D(Q),$$

the post-tax inverse demand is

$$p = P_D(Q) - t.$$

So a quantity sales tax shifts inverse demand vertically downward by t .

Key Formula: Welfare Accounting under a Quantity Tax

With equilibrium quantity Q' , buyer price p_b , seller price p_s , and original demand and supply curves:

$$CS^{tax} = \int_0^{Q'} (P_D(q) - p_b) dq,$$

$$PS^{tax} = \int_0^{Q'} (p_s - P_S(q)) dq,$$

$$TR = tQ',$$

$$DWL = \int_{Q'}^{Q^*} (P_D(q) - P_S(q)) dq.$$

8.4 Ad-Valorem Taxes

An ad-valorem tax is proportional to value rather than fixed per unit.

Definition 8.5: Ad-valorem tax

An *ad-valorem tax* of rate τ satisfies $0 < \tau < 1$ and is proportional to price.

- Under an ad-valorem excise tax on sellers,

$$p_s = (1 - \tau)p_b.$$

- Under an ad-valorem sales tax on buyers,

$$p_b = (1 + \tau)p_s.$$

Unlike fixed quantity taxes, equal percentage excise and sales taxes are not exactly the same tax wedge, though they are approximately similar for small τ .

8.4.1 Ad-valorem excise tax

If original supply is

$$Q = S(p_s),$$

and buyers pay price p_b , sellers receive

$$p_s = (1 - \tau)p_b.$$

Hence the taxed supply curve is

$$Q = S((1 - \tau)p_b).$$

In inverse form,

$$p_b = \frac{P_S(Q)}{1 - \tau}.$$

So the inverse supply curve pivots upward.

8.4.2 Ad-valorem sales tax

If original demand is

$$Q = D(p_b),$$

and sellers receive price p_s , buyers pay

$$p_b = (1 + \tau)p_s.$$

Thus demand becomes

$$Q = D((1 + \tau)p_s),$$

or in inverse form,

$$p_s = \frac{P_D(Q)}{1 + \tau}.$$

So the inverse demand curve pivots downward.

Common Mistake

For ad-valorem taxes, the pivoted curve is useful for locating the new equilibrium, but it is *not* the correct object for measuring producer surplus or deadweight loss. Once equilibrium is found, welfare must be measured using the original supply and demand curves together with the actual buyer and seller prices.

Key Formula: Welfare Accounting under an Ad-Valorem Tax

If the post-tax equilibrium is (Q', p_b, p_s) , then:

$$CS^{tax} = \int_0^{Q'} (P_D(q) - p_b) dq,$$

$$PS^{tax} = \int_0^{Q'} (p_s - P_S(q)) dq,$$

$$TR = (p_b - p_s)Q',$$

$$DWL = \int_{Q'}^{Q^*} (P_D(q) - P_S(q)) dq.$$

8.5 Tax Incidence

A tax lowers welfare for both consumers and producers. The division of the burden is called tax incidence.

Definition 8.6: Tax incidence

For a unit tax with pre-tax equilibrium price p^* and post-tax prices (p_b, p_s) at traded quantity Q' , the tax burden borne by:

- consumers is measured by

$$(p_b - p^*)Q';$$

- producers is measured by

$$(p^* - p_s)Q'.$$

Their sum is the total tax transfer on traded units:

$$((p_b - p^*) + (p^* - p_s))Q' = (p_b - p_s)Q' = tQ'$$

under a quantity tax.

Theorem 8.4: More elastic side bears less tax incidence

All else equal:

- more elastic demand implies lower consumer tax incidence;
- more elastic supply implies lower producer tax incidence.

In the limiting cases:

- perfectly inelastic demand means consumers bear all of the tax;
- perfectly inelastic supply means producers bear all of the tax.

Proof. The less elastic side has fewer quantity adjustments available, so its price changes more under a tax wedge. Geometrically, that side experiences the larger movement between pre-tax and post-tax price. The formal linear-demand proof appears in the next subsection. $\square$

8.6 Comparative Statics with Linear Demand and Supply

The lecture gives a simple algebraic comparative-statics framework.

Definition 8.7: Linear market environment

Suppose market demand and supply are

$$Q^D(p_b) = a - bp_b, \quad Q^S(p_s) = c + dp_s,$$

with

$$b > 0, \quad d > 0.$$

Here b and d capture the price responsiveness of demand and supply in this linear specification.

8.6.1 No tax

Equilibrium satisfies

$$a - bp^* = c + dp^*.$$

Hence

$$p^* = \frac{a - c}{b + d}, \quad Q^* = \frac{ad + bc}{b + d}.$$

8.6.2 Quantity tax

Let a per-unit tax t create the wedge

$$p_b = p_s + t.$$

Then equilibrium satisfies

$$a - bp_b = c + dp_s, \quad p_b = p_s + t.$$

Theorem 8.5: Linear equilibrium under a quantity tax

Under

$$Q^D(p_b) = a - bp_b, \quad Q^S(p_s) = c + dp_s, \quad p_b = p_s + t,$$

the post-tax equilibrium is

$$p_b = \frac{a - c + dt}{b + d}, \quad p_s = \frac{a - c - bt}{b + d}, \quad Q' = \frac{ad + bc - bdt}{b + d}.$$

Proof. Substitute

$$p_b = p_s + t$$

into demand:

$$Q^D = a - b(p_s + t) = a - bp_s - bt.$$

Equilibrium requires

$$a - bp_s - bt = c + dp_s.$$

So

$$a - c - bt = (b + d)p_s,$$

hence

$$p_s = \frac{a - c - bt}{b + d}.$$

Then

$$p_b = p_s + t = \frac{a - c - bt}{b + d} + t = \frac{a - c + dt}{b + d}.$$

Finally,

$$Q' = c + dp_s = c + d \left(\frac{a - c - bt}{b + d} \right) = \frac{ad + bc - bdt}{b + d}.$$

□

Theorem 8.6: Tax incidence shares in the linear model

Under the same linear quantity-tax model,

$$\frac{p_b - p^*}{t} = \frac{d}{b + d}, \quad \frac{p^* - p_s}{t} = \frac{b}{b + d}.$$

So:

- consumers bear share

$$\frac{d}{b + d};$$

- producers bear share

$$\frac{b}{b + d}.$$

Proof. From the formulas above,

$$p_b - p^* = \frac{a - c + dt}{b + d} - \frac{a - c}{b + d} = \frac{dt}{b + d},$$

so

$$\frac{p_b - p^*}{t} = \frac{d}{b + d}.$$

Similarly,

$$p^* - p_s = \frac{a - c}{b + d} - \frac{a - c - bt}{b + d} = \frac{bt}{b + d},$$

so

$$\frac{p^* - p_s}{t} = \frac{b}{b + d}.$$

□

Theorem 8.7: Deadweight loss in the linear quantity-tax model

In the same linear model,

$$Q^* - Q' = \frac{bdt}{b + d}.$$

Hence

$$DWL = \frac{1}{2}t(Q^* - Q') = \frac{bdt^2}{2(b + d)}.$$

Proof. Subtract the quantity formulas:

$$Q^* - Q' = \frac{ad + bc}{b + d} - \frac{ad + bc - bdt}{b + d} = \frac{bdt}{b + d}.$$

Under linear demand and supply, the deadweight-loss region is a triangle with base

$$Q^* - Q'$$

and height

$$t.$$

Therefore

$$DWL = \frac{1}{2}t(Q^* - Q') = \frac{bdt^2}{2(b + d)}.$$

□

Worked Example: comparative statics of tax incidence

Suppose

$$Q^D(p_b) = a - bp_b, \quad Q^S(p_s) = c + dp_s,$$

and there is a unit tax t .

The consumer share of incidence is

$$\frac{d}{b + d}.$$

Compare the benchmark case

$$(b, d) = (5, 5),$$

for which the consumer share is

$$\frac{5}{10} = \frac{1}{2}.$$

Now consider the following cases:

$$(b, d) = (3, 3), \quad (3, 2), \quad (7, 3), \quad (3, 7).$$

Compute the consumer share in each case:

$$(3, 3) : \frac{3}{6} = \frac{1}{2},$$

$$(3, 2) : \frac{2}{5} = 0.4,$$

$$(7, 3) : \frac{3}{10} = 0.3,$$

$$(3, 7) : \frac{7}{10} = 0.7.$$

Therefore the only case in which consumers bear a strictly higher tax incidence than in the benchmark is

$$(b, d) = (3, 7).$$

8.6.3 Behavioral bias and tax collected

The lecture also studies a simple behavioral-taxation example.

Definition 8.8: Perceived tax parameter

Suppose consumers behave as if the buyer price were

$$p_b = p_s + kt,$$

where $k > 0$.

- If $0 < k < 1$, consumers underweight the tax.
- If $k > 1$, consumers overweight the tax.

With linear demand and supply,

$$Q^D = a - bp_b, \quad Q^S = c + dp_s,$$

equilibrium becomes

$$a - b(p_s + kt) = c + dp_s.$$

Theorem 8.8: Equilibrium and tax revenue with perceived-tax bias

Under the perceived-price rule

$$p_b = p_s + kt,$$

the equilibrium is

$$p_s = \frac{a - c - bkt}{b + d}, \quad p_b = \frac{a - c + dkt}{b + d}, \quad Q = \frac{bc + ad - bdk t}{b + d}.$$

Tax revenue is

$$T = tQ = \frac{bc + ad - bdk t}{b + d} t.$$

Therefore

$$\frac{\partial T}{\partial k} < 0.$$

Proof. The equilibrium formulas are obtained exactly as in the ordinary quantity-tax case, with t replaced by kt in the demand wedge.

Since

$$T(k) = \frac{bc + ad - bdk t}{b + d} t,$$

differentiate with respect to k :

$$\frac{\partial T}{\partial k} = -\frac{bdt^2}{b+d} < 0.$$

Thus greater perceived tax salience lowers quantity traded and therefore reduces tax revenue. $\square$

Worked Example: effect of underweighting and overweighting taxes

Using the previous formula,

$$T(k) = \frac{bc + ad - bdk t}{b + d} t.$$

Compare tax revenue under perceived-bias parameter k with the benchmark $k = 1$:

$$\Delta T = T(k) - T(1) = \frac{bc + ad - bdk t}{b + d} t - \frac{bc + ad - bdt}{b + d} t.$$

So

$$\Delta T = \frac{bd(1 - k)t^2}{b + d}.$$

Hence:

- if $k < 1$ (underweighting), then $\Delta T > 0$,
so more tax is collected than in the unbiased case;
- if $k > 1$ (overweighting), then $\Delta T < 0$,
so less tax is collected.

Therefore the lecture's conclusion is

Underweighting taxes raises tax revenue; overweighting taxes lowers it.

8.7 Common Mistakes and Exam Traps

Common Mistake

Do not confuse *statutory incidence* with *economic incidence*. The side legally paying the tax need not be the side that bears most of the burden.

Common Mistake

For quantity taxes, excise and sales taxes of the same dollar amount are equivalent. For ad-valorem taxes, equal percentage excise and sales taxes are not exactly identical.

Common Mistake

Do not use the pivoted supply curve under an ad-valorem tax to calculate producer surplus. It is a device for locating the new equilibrium, not the true marginal cost curve.

Common Mistake

When computing deadweight loss, do not forget tax revenue. The welfare decomposition is:

$$\text{loss in CS} + \text{loss in PS} = TR + DWL.$$

Only the part not transferred as revenue is deadweight loss.

8.8 Chapter Summary

Key Formula: Lecture 8 Summary

$$Q^D(p) = \sum_i q_i^D(p),$$

$$Q^S(p) = \sum_j q_j^S(p),$$

$$SW(Q) = \int_0^Q (P_D(q) - P_S(q)) dq,$$

$$p_b = p_s + t \quad (\text{quantity tax}),$$

$$p_s = (1 - \tau)p_b \quad (\text{ad-valorem excise tax}),$$

$$p_b = (1 + \tau)p_s \quad (\text{ad-valorem sales tax}),$$

$$CS^{tax} = \int_0^{Q'} (P_D(q) - p_b) dq,$$

$$PS^{tax} = \int_0^{Q'} (p_s - P_S(q)) dq,$$

$$TR = (p_b - p_s)Q',$$

$$DWL = \int_{Q'}^{Q^*} (P_D(q) - P_S(q)) dq.$$

Under

$$Q^D = a - bp_b, \quad Q^S = c + dp_s,$$

a quantity tax gives

$$p_b = \frac{a - c + dt}{b + d}, \quad p_s = \frac{a - c - bt}{b + d}, \quad Q' = \frac{ad + bc - bdt}{b + d},$$

consumer incidence share

$$\frac{d}{b + d},$$

producer incidence share

$$\frac{b}{b + d},$$

and

$$DWL = \frac{bdt^2}{2(b + d)}.$$

Exam Focus

The highest-yield takeaways from this chapter are:

- aggregate welfare in a single-good market is $CS + PS$;
- competitive equilibrium maximizes this welfare under the standard assumptions;
- taxes create a wedge, reduce traded quantity, and generate deadweight loss;

- the more inelastic side of the market bears more of the tax burden;
- quantity excise and sales taxes are equivalent, but ad-valorem taxes require more care;
- in tax questions, keep buyer price, seller price, tax revenue, and welfare triangles clearly separated.

9 Perfect Competition and Welfare

This chapter studies welfare in a more general market environment than the single-market partial-equilibrium framework of Chapter 8. Instead of measuring welfare by summing consumer and producer surplus in one market, we consider an exchange economy with multiple goods and multiple consumers, and use Pareto efficiency as the central efficiency concept. The chapter develops the simple exchange economy, the Edgeworth box, competitive equilibrium, the contract curve, envy-freeness, and the two fundamental welfare theorems.

9.1 The Simple Exchange Economy

We begin with a pure exchange economy: goods are redistributed across consumers, but there is no production.

Definition 9.1: Simple exchange economy

A simple exchange economy consists of:

- N consumers, indexed by $i = 1, \dots, N$;
- M goods, indexed by $j = 1, \dots, M$;
- for each consumer i , an endowment

$$\omega^i = (\omega_1^i, \dots, \omega_M^i) \in \mathbb{R}_+^M;$$

- for each consumer i , preferences represented by a utility function

$$U_i(x^i), \quad x^i = (x_1^i, \dots, x_M^i) \in \mathbb{R}_+^M.$$

There is no production, so goods cannot be transformed into one another.

The total resource of each good is the sum of individual endowments.

Definition 9.2: Aggregate resources and feasible allocations

For each good j , total resources are

$$e_j = \sum_{i=1}^N \omega_j^i.$$

An allocation

$$X = (x^1, \dots, x^N)$$

is *feasible* if for every good j ,

$$\sum_{i=1}^N x_j^i = e_j.$$

That is, the allocation must exactly distribute the economy's total endowment.

A feasible allocation need not respect individual endowments. A consumer may consume more than her own initial endowment of some good, provided other consumers consume less and total resources are preserved.

Exam Focus

In exchange-economy questions, always distinguish:

- *individual endowment* ω^i ,
- *individual allocation* x^i ,
- *aggregate resources* e .

Feasibility is about aggregate resources, not about each person staying within their original bundle.

9.2 Pareto Improvements and Pareto Efficiency

The chapter replaces the more subjective “maximize $CS + PS$ ” criterion with a more general efficiency concept.

Definition 9.3: Pareto improvement

Let

$$X = (x^1, \dots, x^N) \quad \text{and} \quad Y = (y^1, \dots, y^N)$$

be feasible allocations. We say that Y *Pareto improves on* X if

$$y^i \succeq_i x^i \quad \text{for all } i,$$

and

$$y^j \succ_j x^j \quad \text{for some } j.$$

So Y makes everyone weakly better off and at least one person strictly better off.

Definition 9.4: Pareto efficient allocation

A feasible allocation X is *Pareto efficient* if there is no other feasible allocation that Pareto improves on it.

The logic is simple: if a feasible reallocation can make someone better off without hurting anyone else, then the original allocation wastes gains from trade.

Common Mistake

If an allocation Y Pareto improves on X , it does *not* follow that Y is Pareto efficient. Another feasible allocation Z may still Pareto improve on Y .

Worked Example: checking Pareto efficiency

Suppose 10 apples and 10 oranges are split between consumers A and B. Consumer A has utility

$$U_A = \# \text{apples},$$

while consumer B has utility

$$U_B = \# \text{apples} + \# \text{oranges}.$$

Check the following allocations.

1. A gets 10 oranges, B gets 10 apples.

A's utility is 0 and B's utility is 10. This is not Pareto efficient because one can transfer some oranges from A to B:

- A does not care about oranges;
- B values oranges positively.

So B can be made strictly better off without hurting A.

2. A gets 10 apples, B gets 10 oranges.

A's utility is 10 and B's utility is 10. Any apple transferred from A to B hurts A, and A has no oranges to give B. So no Pareto improvement exists. This allocation is Pareto efficient.

3. A gets 5 apples, B gets 5 apples and 10 oranges.

A's utility is 5 and B's utility is 15. Since B already has all oranges, any move that gives A more apples must take apples away from B. That hurts B unless compensated, but A has no desirable oranges to offer. So this allocation is Pareto efficient.

4. A gets 0 apples, B gets 10 apples and 10 oranges.

A's utility is 0 and B's utility is 20. Although A is badly treated, any apple transferred from B to A lowers B's utility, and A has nothing valuable to compensate B with. Hence there is no Pareto improvement. This allocation is also Pareto efficient.

Therefore:

Allocation 1 is not Pareto efficient; allocations 2, 3, and 4 are Pareto efficient.

This example shows that Pareto efficiency and fairness are different concepts.

9.3 The Edgeworth Box

For the two-good, two-consumer case, the key graphical tool is the Edgeworth box.

Definition 9.5: Two-good, two-consumer exchange economy

Suppose there are two goods and two consumers, A and B. Let

$$\omega^A = (\omega_1^A, \omega_2^A), \quad \omega^B = (\omega_1^B, \omega_2^B).$$

A feasible allocation consists of bundles

$$x^A = (x_1^A, x_2^A), \quad x^B = (x_1^B, x_2^B),$$

satisfying

$$x_1^A + x_1^B = \omega_1^A + \omega_1^B, \quad x_2^A + x_2^B = \omega_2^A + \omega_2^B.$$

Definition 9.6: Edgeworth box

The *Edgeworth box* is the rectangle with:

$$\text{length} = \omega_1^A + \omega_1^B, \quad \text{height} = \omega_2^A + \omega_2^B.$$

Each point in the box represents a feasible allocation. The lower-left origin is consumer A's origin, while the upper-right origin is consumer B's origin.

Thus every point in the box automatically satisfies feasibility. The endowment point

$$(\omega_1^A, \omega_2^A)$$

from A's perspective simultaneously determines B's endowment from the opposite origin.

Definition 9.7: Contract curve

The *contract curve*, or *Pareto set*, is the set of feasible allocations in the Edgeworth box that are Pareto efficient.

At a smooth interior Pareto-efficient allocation, the consumers' indifference curves are tangent. Hence, for differentiable utilities,

$$MRS_A = MRS_B$$

at interior points of the contract curve.

Theorem 9.1: Tangency characterization of interior Pareto efficiency

Suppose preferences are locally nonsatiated and differentiable. If a feasible interior allocation in the Edgeworth box is Pareto efficient, then the indifference curves of A and B are tangent at that point. Equivalently,

$$MRS_A = MRS_B.$$

Proof. If the indifference curves cross rather than being tangent, then there is a local feasible direction that moves both consumers to higher indifference curves. This creates a Pareto improvement, contradicting Pareto efficiency. Therefore, at an interior Pareto-efficient point, the indifference curves must be tangent, so their slopes and hence their MRS values are equal. $\square$

Exam Focus

To identify Pareto improvements in an Edgeworth box:

1. draw both consumers' indifference curves through the allocation;
2. find the region that each consumer weakly prefers;
3. take the overlap.

If the overlap contains other feasible allocations, the original point is not Pareto efficient.

9.4 Fairness and Envy-Freeness

Efficiency alone does not address distributional concerns.

Definition 9.8: Envy and envy-freeness

In a two-consumer allocation (x^A, x^B) :

- A envies B if $x^B \succ_A x^A$ or equivalently $U_A(x^B) > U_A(x^A)$;
- B envies A if $x^A \succ_B x^B$.

The allocation is *envy-free* if neither consumer envies the other:

$$U_A(x^A) \geq U_A(x^B), \quad U_B(x^B) \geq U_B(x^A).$$

Definition 9.9: Fair allocation

A feasible allocation is called *fair* if it is both:

1. Pareto efficient, and
2. envy-free.

So fairness is stricter than efficiency. A Pareto-efficient allocation can still be highly unequal or envied.

Common Mistake

Do not equate Pareto efficiency with fairness. An allocation can be Pareto efficient simply because no Pareto improvement is possible, even if one consumer receives almost everything.

9.5 Competitive Equilibrium in the Exchange Economy

Markets now determine allocations through decentralized utility maximization.

Definition 9.10: Competitive equilibrium in a pure exchange economy

A *competitive equilibrium* consists of a price vector

$$p = (p_1, \dots, p_M)$$

and a feasible allocation

$$(x^1, \dots, x^N)$$

such that:

1. for each consumer i , x^i maximizes utility subject to the budget constraint

$$p \cdot x^i = p \cdot \omega^i;$$

2. the allocation is feasible, i.e. markets clear:

$$\sum_{i=1}^N x_j^i = \sum_{i=1}^N \omega_j^i \quad \text{for each good } j.$$

In the two-good, two-consumer Edgeworth box, the budget line passes through the endowment point and has slope

$$-\frac{p_1}{p_2}$$

from A's perspective. The same physical line is also B's budget line when drawn from B's origin.

Key Formula: Budget Line in the Edgeworth Box

Consumer A's budget line is

$$p_1 x_1^A + p_2 x_2^A = p_1 \omega_1^A + p_2 \omega_2^A.$$

Hence

$$x_2^A = \frac{p_1}{p_2} \omega_1^A + \omega_2^A - \frac{p_1}{p_2} x_1^A,$$

so the slope is

$$-\frac{p_1}{p_2}.$$

This same line also represents B's budget constraint after converting coordinates using feasibility.

Exam Focus

In exchange-equilibrium problems, only the *price ratio* matters. If

$$\frac{p_1^*}{p_2^*} = m,$$

then any positive scalar multiple of (p_1^*, p_2^*) gives the same budget-line slope and the same demand choices. One price is usually normalized to 1 as the numeraire.

9.6 How to Solve for Competitive Equilibrium

The lecture's solution method is entirely standard.

Method Pattern: Solving a 2-good, 2-consumer equilibrium

To solve a competitive equilibrium algebraically:

1. Derive each consumer's demand as a function of prices and wealth:

$$x^A(p, p \cdot \omega^A), \quad x^B(p, p \cdot \omega^B).$$

2. Write the market-clearing conditions:

$$x_1^A + x_1^B = \omega_1^A + \omega_1^B, \quad x_2^A + x_2^B = \omega_2^A + \omega_2^B.$$

3. Substitute the demand functions and solve for the equilibrium price ratio.
4. Plug the equilibrium prices back into the demand functions to get the equilibrium allocation.

Worked Example: Solving for competitive equilibrium

Consider a two-good, two-consumer exchange economy with endowments

$$\omega^A = (4, 8), \quad \omega^B = (7, 3).$$

Consumer A has utility

$$U_A(x_1, x_2) = \min\{x_1, x_2\},$$

and consumer B has utility

$$U_B(x_1, x_2) = x_1^{0.5} x_2^{0.5}.$$

Step 1: derive demands.

Consumer A has perfect-complements preferences, so she chooses

$$x_1^A = x_2^A.$$

Her wealth is

$$m_A = 4p_1 + 8p_2.$$

Hence

$$x_1^A = x_2^A = \frac{4p_1 + 8p_2}{p_1 + p_2}.$$

Consumer B has Cobb–Douglas utility with equal exponents, so his demand is

$$x_1^B = \frac{m_B}{2p_1}, \quad x_2^B = \frac{m_B}{2p_2},$$

where

$$m_B = 7p_1 + 3p_2.$$

Therefore

$$x_1^B = \frac{7p_1 + 3p_2}{2p_1}, \quad x_2^B = \frac{7p_1 + 3p_2}{2p_2}.$$

Step 2: impose market clearing.

Total endowment is

$$(11, 11).$$

Market 1 clearing requires

$$\frac{4p_1 + 8p_2}{p_1 + p_2} + \frac{7p_1 + 3p_2}{2p_1} = 11.$$

Solving gives

$$p_1 = p_2.$$

By symmetry of total resources, this also clears market 2.

Step 3: find the equilibrium allocation.

Normalize prices so that

$$p_1 = p_2 = 1.$$

Then

$$x_1^A = x_2^A = \frac{4 + 8}{2} = 6,$$

and

$$x_1^B = x_2^B = \frac{7 + 3}{2} = 5.$$

Therefore a competitive equilibrium is

$$p_1^* = p_2^*, \quad x^{A*} = (6, 6), \quad x^{B*} = (5, 5).$$

9.7 Competitive Equilibrium in the Edgeworth Box

The Edgeworth-box geometry gives a graphical proof of the welfare property of competitive equilibrium.

At a competitive equilibrium:

- each consumer maximizes utility subject to the common budget line through the endowment point, so each chosen bundle lies at a tangency between an indifference curve and that budget line;
- market clearing means the two consumers choose the same feasible point in the box;
- therefore both consumers' indifference curves are tangent to the same line at the same point.

Hence the equilibrium allocation lies on the contract curve.

Theorem 9.2: First Fundamental Welfare Theorem in the exchange economy

In the simple competitive exchange economy, a competitive equilibrium allocation is Pareto efficient.

Proof. At a competitive equilibrium in the two-good, two-consumer Edgeworth box, both consumers maximize utility on the same budget line. Therefore each consumer's indifference curve is tangent to the budget line at the equilibrium allocation. Since the allocation is feasible and both tangencies occur at the same point, the two indifference curves are tangent to each other there. A feasible interior allocation where the indifference curves are tangent admits no local Pareto improvement. Hence the equilibrium allocation is Pareto efficient. $\square$

The lecture emphasizes that this theorem is much less subjective than maximizing $CS + PS$: it does not require attaching interpersonal weights to utility.

9.8 Existence, Uniqueness, and the Role of Prices

The lecture notes that under standard convex preferences, the exchange economy has well-behaved equilibrium properties.

Theorem 9.3: Standard existence and uniqueness statement used in this course

Under standard convex preferences in the exchange economy considered in the lecture, a competitive equilibrium exists, and the equilibrium price ratio is unique.

Exam Focus

Be careful with the word *unique*. In these exchange problems, the equilibrium *price ratio* is unique, not the absolute price vector. If (p_1^*, p_2^*) is an equilibrium price vector, then so is

$$(\lambda p_1^*, \lambda p_2^*)$$

for any $\lambda > 0$.

9.9 The Second Fundamental Welfare Theorem

The contract curve usually contains many Pareto-efficient allocations. Competitive equilibrium selects one of them depending on the initial endowment point. The second welfare theorem explains how any desired Pareto-efficient allocation can be supported as a market equilibrium after suitable redistribution.

Theorem 9.4: Second Fundamental Welfare Theorem

Under the standard convexity assumptions, for any Pareto-efficient allocation, there exists a redistribution of endowments such that the resulting competitive equilibrium yields that allocation.

Proof. Take a Pareto-efficient allocation in the Edgeworth box. At such a point, the consumers' indifference curves admit a common supporting tangent line. Choose a redistributed endowment point lying on this tangent budget line. At the prices corresponding to the slope of that line, each consumer can afford the Pareto-efficient allocation and optimally chooses it, since the line is tangent to the relevant indifference curve. Because the allocation is feasible, markets clear. Hence the Pareto-efficient allocation is decentralized as a competitive equilibrium from the redistributed endowment. $\square$

So the theorem separates:

- **efficiency:** achieved by competitive trade;
- **distribution:** achieved by the initial redistribution of endowments.

This is one of the lecture's main messages: if society has a preferred Pareto-efficient allocation, it may be possible to reach it not by distorting trade itself, but by redistributing endowments first and then letting markets operate.

Exam Focus

A standard Edgeworth-box interpretation of the SWT is:

- pick a Pareto-efficient point on the contract curve;
- draw the common tangent budget line there;
- any redistributed endowment lying on that line can support that Pareto-efficient point as competitive equilibrium.

9.10 Implicit Assumptions Behind the Welfare Theorems

The welfare theorems are powerful, but they do not hold without structure.

Definition 9.11: Implicit assumptions emphasized in the lecture

The exchange model in this chapter relies on the following assumptions:

- **Complete markets:** all relevant goods can be traded.
- **No externalities:** each consumer's utility depends only on own consumption.
- **Standard convex preferences:** needed for the clean welfare-theorem results.

If external effects exist but cannot be traded, or if some markets are missing, then the competitive equilibrium need not be Pareto efficient.

Common Mistake

Do not quote the welfare theorems without mentioning their assumptions. In particular, missing markets and externalities are standard reasons why competitive equilibrium can fail to be Pareto efficient.

9.11 Standard Problem Types and Solution Patterns

Method Pattern: Checking Pareto efficiency

In allocation questions:

1. identify total resources and feasibility;
2. ask whether one consumer can be made strictly better off without making the other worse off;
3. in Edgeworth-box settings, look for overlap of upper contour sets;
4. at smooth interior points, check tangency:

$$MRS_A = MRS_B.$$

Method Pattern: Exchange equilibrium algebra

For two-good, two-consumer economies:

1. compute each consumer's wealth from endowment:

$$m_i = p_1\omega_1^i + p_2\omega_2^i;$$

2. derive Marshallian demand from each utility function;
3. impose market clearing on one or both goods;
4. solve for the price ratio and allocations.

Method Pattern: Fairness questions

If asked whether an allocation is fair:

1. test Pareto efficiency;
2. test envy-freeness:

$$U_A(x^A) \geq U_A(x^B), \quad U_B(x^B) \geq U_B(x^A).$$

Only if both hold is the allocation fair.

9.12 Chapter Summary

Key Formula: Lecture 9 Summary

$$e_j = \sum_{i=1}^N \omega_j^i,$$

$$\sum_{i=1}^N x_j^i = e_j \quad \text{for each good } j,$$

Y Pareto improves on $X \iff y^i \succeq_i x^i \forall i$ and $y^j \succ_j x^j$ for some j ,

X Pareto efficient $\iff$ no feasible allocation Pareto improves on X ,

$$p \cdot x^i = p \cdot \omega^i \quad \text{for each consumer in equilibrium,}$$

$$x_1^A + x_1^B = \omega_1^A + \omega_1^B,$$

$$x_2^A + x_2^B = \omega_2^A + \omega_2^B,$$

interior Pareto efficiency $\Rightarrow MRS_A = MRS_B$,

competitive equilibrium $\Rightarrow$ Pareto efficiency (FWT),

any Pareto-efficient allocation $\Rightarrow$ can be decentralized after redistribution (SWT).

Exam Focus

The highest-yield takeaways from this chapter are:

- feasibility in an exchange economy means total consumption equals total endowment good by good;
- Pareto efficiency is about the impossibility of Pareto improvements, not about fairness;
- the Edgeworth box turns a two-person, two-good exchange economy into a geometric object;
- the contract curve is the set of Pareto-efficient allocations;

- a competitive equilibrium is determined by utility maximization plus market clearing;
- in equilibrium, only the price ratio matters;
- the First Welfare Theorem says competitive equilibrium is Pareto efficient;
- the Second Welfare Theorem says any Pareto-efficient allocation can be supported after suitable redistribution, under the standard assumptions.

10 Social Choice

This chapter studies how a society should choose among alternatives when different individuals have different preferences. In earlier chapters, we developed efficiency concepts such as Pareto efficiency and showed that competitive equilibrium can support efficient allocations under standard assumptions. However, efficiency alone often does not determine a unique social choice. This chapter introduces social decision mechanisms, their desirable properties, Arrow's impossibility theorem, and social welfare functions. The main conceptual distinction is between:

- **ordinal aggregation:** using only rankings of alternatives;
- **cardinal aggregation:** using utility levels or utility intensities.

10.1 Alternatives and Individual Preferences

A social choice problem begins with a set of alternatives and a collection of individual preferences over those alternatives.

Definition 10.1: Alternatives

An *alternative* is one feasible option from which society may choose. Examples include:

- possible lunch venues;
- political candidates or parties;
- feasible allocations in an Edgeworth box;
- policy regimes such as different tax schedules.

Definition 10.2: Individual preferences

Let the set of alternatives be A . For individual i :

- $x \succeq_i y$ means i weakly prefers x to y ;
- $x \succ_i y$ means i strictly prefers x to y ;
- $x \sim_i y$ means i is indifferent between x and y .

When the lecture restricts attention to strict preferences for simplicity, each individual can rank all alternatives from most preferred to least preferred.

Definition 10.3: Rational preferences

Preferences are *well-defined* or *rational* if they satisfy:

- **Completeness:** for any two alternatives x, y , either $x \succeq_i y$ or $y \succeq_i x$ (or both);
- **Transitivity:** if $x \succeq_i y$ and $y \succeq_i z$, then $x \succeq_i z$.

Exam Focus

In this chapter, many examples are written using *strict* rankings only. When that happens, completeness means that the individual can rank every pair of alternatives, and transitivity means the ranking contains no cycles.

Worked Example: Checking completeness and transitivity

Suppose individual j has preferences over $\{a, b, c, d\}$:

$$a \succ_j b, \quad a \succ_j c, \quad a \succ_j d, \quad b \succ_j c, \quad c \succ_j d.$$

These comparisons imply:

$$a \succ_j b \succ_j c \succ_j d,$$

and therefore also

$$b \succ_j d, \quad a \succ_j d, \quad a \succ_j c.$$

So the ranking is complete and transitive.

Now suppose individual k has:

$$a \succ_k b, \quad a \succ_k c, \quad a \succ_k d, \quad b \succ_k c, \quad d \succ_k b, \quad c \succ_k d.$$

Then we have

$$b \succ_k c, \quad c \succ_k d, \quad d \succ_k b,$$

which forms a cycle. Hence transitivity fails.

So:

j 's preferences are complete and transitive, while k 's are not transitive.

10.2 The Social Choice Problem

Once individual preferences are specified, the next problem is to aggregate them into a social ranking.

Definition 10.4: Social choice problem

A social choice problem consists of:

- a set of individuals $\{1, 2, \dots, n\}$;
- a set of alternatives A ;
- for each individual i , a preference ordering over A .

The objective is to use these individual preferences to generate a social preference or social choice.

Definition 10.5: Social decision mechanism

A *social decision mechanism* (SDM) is a rule that takes individual preference orderings

$$\succ_1, \dots, \succ_n$$

as inputs and outputs a social preference ordering

$$\succ^*.$$

The key point is that an SDM uses only individuals' rankings of alternatives. It does *not* use utility intensities.

Exam Focus

A social decision mechanism is an *ordinal* aggregator. It uses rankings such as

$$x \succ_i y \succ_i z,$$

not numerical statements such as “*i* cares twice as much about *x* as about *y*.” This distinction becomes central when the chapter moves to social welfare functions.

10.3 Examples of Social Decision Mechanisms

10.3.1 Pairwise majority voting

Definition 10.6: Pairwise majority voting

For each pair of alternatives (x, y) , compare how many individuals prefer x to y and how many prefer y to x . The socially preferred alternative is the one that wins the majority vote in that pairwise comparison.

Worked Example: Pairwise majority voting

Suppose three individuals have rankings:

$$\text{Harry: } X \succ Y \succ Z, \quad \text{Ron: } Z \succ Y \succ X, \quad \text{Hermione: } Z \succ X \succ Y.$$

Compare pairwise:

- X vs Y : Harry and Hermione prefer X , so

$$X \succ^* Y.$$

- X vs Z : Ron and Hermione prefer Z , so

$$Z \succ^* X.$$

- Y vs Z : Ron and Hermione prefer Z , so

$$Z \succ^* Y.$$

Thus the social ordering is

$$Z \succ^* X \succ^* Y.$$

10.3.2 Borda count

Definition 10.7: Borda count

Under the Borda count, each individual assigns rank-points to alternatives. With m alternatives, the top-ranked alternative gets 1 point, the next gets 2 points, and so on until the last gets m points. The social ranking orders alternatives from lowest total point score to highest.

Worked Example: Borda count

Suppose 100 individuals fall into three preference types:

$$40\% : X \succ Y \succ Z, \quad 25\% : Z \succ Y \succ X, \quad 35\% : Z \succ X \succ Y.$$

With three alternatives, ranks 1, 2, and 3 correspond to 1, 2, and 3 points.

Compute total points:

$$X : 40(1) + 25(3) + 35(2) = 40 + 75 + 70 = 185,$$

$$Y : 40(2) + 25(2) + 35(3) = 80 + 50 + 105 = 235,$$

$$Z : 40(3) + 25(1) + 35(1) = 120 + 25 + 35 = 180.$$

Lower points are better, so the social ranking is

$$Z \succ^* X \succ^* Y.$$

10.3.3 Plurality voting**Definition 10.8: Plurality voting**

Plurality voting ranks first the alternative that is placed top by the largest number of individuals. After assigning that rank, remove the chosen alternative and repeat the procedure among the remaining alternatives.

Worked Example: Plurality voting

Using the same preference profile:

$$40\% : X \succ Y \succ Z, \quad 25\% : Z \succ Y \succ X, \quad 35\% : Z \succ X \succ Y,$$

the top-choice shares are:

$$X : 40\%, \quad Y : 0\%, \quad Z : 60\%.$$

So Z is ranked first. Remove Z and compare X and Y :

- the first group prefers X to Y ;
- the second group prefers Y to X ;
- the third group prefers X to Y .

Thus X beats Y by 75% to 25%.

So the social ranking is

$$Z \succ^* X \succ^* Y.$$

10.4 Why Social Decision Mechanisms Can Fail

An arbitrary rule that outputs a social ranking need not respect individual preferences or even produce a well-defined ordering.

Common Mistake

Do not assume that every rule taking rankings as inputs is a sensible social decision mechanism. The chapter's central message is precisely that many apparently reasonable rules fail some basic desiderata.

10.4.1 Condorcet cycles**Definition 10.9: Condorcet cycle**

A *Condorcet cycle* occurs when pairwise majority comparisons produce a cycle such as

$$X \succ^* Y, \quad Y \succ^* Z, \quad Z \succ^* X.$$

This means the social ranking is not transitive.

Worked Example: Condorcet cycle under pairwise majority voting

Suppose three individuals have preferences:

$$\text{Type 1: } X \succ Z \succ Y, \quad \text{Type 2: } Z \succ Y \succ X, \quad \text{Type 3: } Y \succ X \succ Z.$$

Then:

- Y beats X because types 2 and 3 prefer Y to X :

$$Y \succ^* X.$$

- X beats Z because types 1 and 3 prefer X to Z :

$$X \succ^* Z.$$

- Z beats Y because types 1 and 2 prefer Z to Y :

$$Z \succ^* Y.$$

Hence

$$Y \succ^* X, \quad X \succ^* Z, \quad Z \succ^* Y,$$

which is a cycle. Therefore pairwise majority voting does not produce a well-defined transitive social ordering in this case.

10.4.2 Vote splitting and spoiling

Some mechanisms are sensitive to the presence of other alternatives.

Worked Example: Plurality voting and vote splitting

Suppose preference types are:

$$40\% : X \succ Y \succ Z, \quad 25\% : Z \succ Y \succ X, \quad 35\% : Y \succ X \succ Z.$$

Under plurality voting, the top-choice shares are:

$$X : 40\%, \quad Y : 35\%, \quad Z : 25\%.$$

So plurality ranks

$$X \succ^* Y \succ^* Z.$$

However, a direct comparison between X and Y gives:

- the first type prefers X to Y ;
- the second and third types prefer Y to X .

So 60% prefer Y to X .

Thus plurality makes X the winner even though a majority would prefer Y to X in a head-to-head contest. The reason is that Z splits support away from Y .

10.5 Desirable Properties of Social Decision Mechanisms

The lecture focuses on three benchmark properties.

Definition 10.10: Pareto property for an SDM

An SDM satisfies *Pareto* if whenever every individual prefers x to y , the social ranking also prefers x to y :

$$x \succ_i y \text{ for all } i \implies x \succ^* y.$$

Definition 10.11: Universal Domain

An SDM satisfies *Universal Domain* if for every possible profile of well-defined individual preferences, the resulting social preference ordering is itself well-defined.

Definition 10.12: Independence of Irrelevant Alternatives

An SDM satisfies *Independence of Irrelevant Alternatives* (IIA) if for any two alternatives A and B , the social ranking of A versus B depends only on individuals' rankings of A versus B , and not on how they rank other alternatives.

Exam Focus

To check IIA:

- keep every individual's ranking of A versus B fixed;
- change rankings involving other alternatives;
- see whether the social ranking of A versus B changes.

If it changes, IIA fails.

10.6 How to Check Whether a Property Holds

A property must hold for the *entire multiverse* of possible preference profiles.

Method Pattern: verifying SDM properties

To show a property *does not hold*, it is enough to give a single counterexample.

To show a property *does hold*, one must give a general logical argument explaining why no counterexample can exist.

Worked Example: Pairwise majority voting and the three properties

Consider pairwise majority voting.

Pareto: If every individual prefers x to y , then in the pairwise vote between x and y , every individual votes for x . Hence

$$x \succ^* y.$$

So pairwise majority voting satisfies Pareto.

Universal Domain: The Condorcet-cycle example gives well-defined individual preferences but produces a non-transitive social ordering. Hence pairwise majority voting does *not* satisfy Universal Domain.

IIA: In pairwise majority voting, the social ranking of x versus y depends only on how individuals rank x against y in the pairwise contest. Changing rankings involving other alternatives does not affect that contest. Hence pairwise majority voting satisfies IIA.

Therefore:

Pairwise majority voting satisfies Pareto and IIA, but not Universal Domain.

Worked Example: Pairwise minority voting

Define the pairwise minority voting rule as follows: for each pair of alternatives, the socially preferred alternative is the one that *loses* the majority vote.

Check the three properties.

Pareto: If everyone prefers x to y , then majority voting selects x , so minority voting selects y . Hence the social preference contradicts unanimous preference. So Pareto fails.

Universal Domain: Since pairwise minority voting simply reverses pairwise-majority outcomes, a Condorcet cycle under majority voting becomes another cycle under minority voting. Hence Universal Domain fails.

IIA: For any pair (x, y) , the minority-voting outcome depends only on individuals' rankings of x versus y . So IIA holds.

Therefore:

Pairwise minority voting satisfies IIA only; Pareto and Universal Domain both fail.

10.7 Arrow's Impossibility Theorem

The three conditions above are all plausible. Arrow's theorem shows that they are jointly incompatible with non-dictatorial social choice once there are at least three alternatives.

Definition 10.13: Dictatorship

An SDM is a *dictatorship* if there exists some individual d such that for every pair of alternatives x, y ,

$$x \succ_d y \implies x \succ^* y.$$

That is, the social ordering always follows the dictator's preferences.

Theorem 10.1: Arrow's Impossibility Theorem

Suppose there are at least three alternatives. If a social decision mechanism satisfies:

1. Pareto,
2. Universal Domain,
3. Independence of Irrelevant Alternatives,

then it must be a dictatorship.

The lecture uses this theorem as a negative result: there is no perfect democratic ordinal rule satisfying all three desiderata for all preference profiles.

Common Mistake

Arrow's theorem does *not* say that every voting rule is a dictatorship. It says that any SDM satisfying all three listed properties on the full domain of preferences must be dictatorial.

Exam Focus

A common interpretation is:

With at least three alternatives, one cannot have a social decision mechanism that is simultaneously fully general, respects unanimity, ignores irrelevant alternatives, and is non-dictatorial.

So at least one condition must be weakened if one wants a non-dictatorial rule.

10.8 Social Welfare Functions

One way to escape Arrow's impossibility is to use more information than ordinal rankings alone.

Definition 10.14: Social welfare function

A *social welfare function* (SWF) is a function

$$W(u_1, u_2, \dots, u_n)$$

that takes individuals' utility levels as inputs and outputs a real number used to rank social outcomes. The function is increasing in each argument.

This differs from an SDM:

- an SDM aggregates *ordinal* preference rankings;
- an SWF aggregates *cardinal* utilities, and so can reflect preference intensities.

Definition 10.15: Common examples of SWFs

Common examples include:

- **Utilitarian:**

$$W(u_1, \dots, u_n) = \sum_{i=1}^n u_i;$$

- **Weighted utilitarian:**

$$W(u_1, \dots, u_n) = \sum_{i=1}^n \alpha_i u_i, \quad \alpha_i > 0;$$

- **Rawlsian:**

$$W(u_1, \dots, u_n) = \min\{u_1, \dots, u_n\}.$$

Exam Focus

An SWF can be sensitive to *preference intensities*. This is exactly what an IIA-style ordinal SDM rules out. That is one reason the SWF approach can bypass Arrow's impossibility.

10.9 Utility Possibility Set and Social Optima

To connect social welfare to Pareto efficiency, the lecture introduces the utility possibility set.

Definition 10.16: Utility possibility set

The *utility possibility set* is the set of utility vectors

$$(u_A, u_B)$$

that can be achieved by feasible allocations in the economy.

Definition 10.17: Utility possibility frontier

The *utility possibility frontier* (UPF) is the boundary of the utility possibility set consisting of efficient utility pairs.

A social optimum is obtained by maximizing a social welfare function over the utility possibility set.

Definition 10.18: Social optimum

A feasible allocation is *socially optimal* for a given SWF if it maximizes

$$W(u_1, \dots, u_n)$$

over the feasible set of utility outcomes.

Graphically, a social optimum occurs where the highest attainable iso-welfare curve touches the utility possibility frontier.

Theorem 10.2: Social optimum implies Pareto efficiency

If an allocation is socially optimal for some increasing social welfare function, then it is Pareto efficient.

Proof. Suppose a socially optimal allocation were not Pareto efficient. Then there would exist another feasible allocation that makes everyone weakly better off and at least one person strictly better off. Since

the social welfare function is increasing in each individual's utility, the second allocation would yield strictly higher social welfare. This contradicts social optimality. Hence every social optimum must be Pareto efficient. $\square$

Theorem 10.3: Pareto efficiency implies social optimality for some SWF

Under the regularity conditions assumed in the lecture, every Pareto-efficient allocation is socially optimal for some social welfare function.

The lecture emphasizes this as a positive converse: one need not fear that a Pareto-efficient allocation is unsupported by all welfare criteria. Some SWF can always rationalize it.

10.10 Weighted Utilitarianism and Supporting Pareto-Efficient Points

For two consumers, weighted utilitarian welfare takes the form

$$W(u_A, u_B) = \gamma_A u_A + \gamma_B u_B, \quad \gamma_A, \gamma_B > 0.$$

Its iso-welfare curves are straight lines.

Key Formula: Iso-Welfare Slope under Weighted Utilitarianism

For

$$W(u_A, u_B) = \gamma_A u_A + \gamma_B u_B,$$

an iso-welfare curve satisfies

$$\gamma_A u_A + \gamma_B u_B = \bar{W}.$$

Hence

$$u_B = \frac{\bar{W}}{\gamma_B} - \frac{\gamma_A}{\gamma_B} u_A,$$

so the slope is

$$-\frac{\gamma_A}{\gamma_B}.$$

At a smooth social optimum, this slope equals the slope of the utility possibility frontier.

Worked Example: Which weights support a given social optimum?

Suppose point X on the utility possibility frontier has tangent slope

$$-\frac{1}{2}.$$

A weighted utilitarian social welfare function supports X if

$$-\frac{\gamma_A}{\gamma_B} = -\frac{1}{2} \implies \frac{\gamma_A}{\gamma_B} = \frac{1}{2}.$$

So any positive weights in the ratio

$$\gamma_A : \gamma_B = 1 : 2$$

work. Hence both

$$(\gamma_A, \gamma_B) = (1, 2) \quad \text{and} \quad (\gamma_A, \gamma_B) = (2, 4)$$

support X .

Now suppose point Y has tangent slope

$$-2.$$

Then

$$-\frac{\gamma_A}{\gamma_B} = -2 \implies \frac{\gamma_A}{\gamma_B} = 2.$$

So any positive weights in the ratio

$$\gamma_A : \gamma_B = 2 : 1$$

work. Hence both

$$(\gamma_A, \gamma_B) = (2, 1) \quad \text{and} \quad (\gamma_A, \gamma_B) = (4, 2)$$

support Y .

Therefore:

X is supported by weights proportional to $(1, 2)$,

Y is supported by weights proportional to $(2, 1)$.

10.11 Relation to the Welfare Theorems

Chapter 9 established that, under the standard assumptions:

- competitive equilibrium is Pareto efficient;
- every Pareto-efficient allocation can be supported by a suitable redistribution of endowments followed by competitive trade.

This chapter adds:

- every social optimum is Pareto efficient;
- every Pareto-efficient allocation is socially optimal for some SWF.

Combining the two chapters gives a bridge:

Under the standard assumptions and in the absence of externalities, one can connect competitive equilibria, Pareto-efficient allocations, and social optima.

Policy Application

The important limitation is that society must still choose *which* social welfare function to use. The mathematical existence of an SWF supporting a Pareto-efficient point does not by itself solve the normative problem of deciding which efficient point is morally or politically preferable.

10.12 Common Mistakes and Exam Traps

Common Mistake

Do not confuse a social decision mechanism with a social welfare function. An SDM takes rankings as input; an SWF takes utility levels as input.

Common Mistake

Arrow's theorem is about SDMs defined on ordinal preference profiles. It does not directly rule out social welfare functions based on utility levels.

Common Mistake

Do not say that Pareto-efficient allocations are automatically fair or socially optimal for every SWF. The theorem says they are socially optimal for *some* SWF.

Common Mistake

When checking IIA, do not change individuals' rankings of the pair being tested. The point is to keep their ranking of A versus B fixed and change only preferences involving other alternatives.

10.13 Chapter Summary**Key Formula: Lecture 10 Summary**

$x \succeq_i y$ weak preference,

$x \succ_i y$ strict preference,

$x \sim_i y$ indifference,

Pareto for SDM: $x \succ_i y \forall i \implies x \succ^* y$,

Universal Domain : all well-defined individual preferences yield a well-defined social ranking,

IIA : social ranking of A vs B depends only on rankings of A vs B ,

$W(u_1, \dots, u_n)$ social welfare function,

$W(u_1, \dots, u_n) = \sum_i \alpha_i u_i$ weighted utilitarian SWF,

iso-welfare slope = $-\frac{\gamma_A}{\gamma_B}$,

social optimum $\implies$ Pareto efficiency,

Pareto efficiency $\implies$ social optimum for some SWF.

Exam Focus

The highest-yield takeaways from this chapter are:

- know the definitions of weak preference, strict preference, indifference, completeness, and transitivity;
- know the difference between SDMs and SWFs;
- be able to test Pareto, Universal Domain, and IIA;
- know why pairwise majority voting fails Universal Domain;
- understand Arrow's Impossibility Theorem and what it does *not* say;
- understand why social optima lie on the Pareto frontier;
- be able to connect weighted utilitarian slopes

$$-\frac{\gamma_A}{\gamma_B}$$

to tangencies on the utility possibility frontier.

11 Utilitarianism

This chapter gives a mathematical argument for why utilitarianism can be a reasonable social welfare criterion. The key idea is that if utility is quasi-linear in money and society can implement balanced money transfers, then the policy problem separates into:

1. **efficiency:** choose the allocation that maximizes the total utility generated by goods;
2. **distribution:** use money transfers to redistribute that utility pie.

Under these conditions, Pareto-efficient policies are exactly those built from utilitarian allocations.

11.1 Mathematical Logic for Economic Statements

Lecture 11 begins with a short review of mathematical logic because many economic results in this course take the form of conditional statements.

Definition 11.1: Conditional statement, contrapositive, and converse

A *conditional statement* is a proposition of the form

$$A \rightarrow B,$$

read as “If A , then B .”

Its *contrapositive* is

$$\neg B \rightarrow \neg A,$$

and its *converse* is

$$B \rightarrow A.$$

Theorem 11.1: Conditional statement and contrapositive are equivalent

The statements

$$A \rightarrow B \quad \text{and} \quad \neg B \rightarrow \neg A$$

are logically equivalent.

Proof. A conditional statement fails only in the case where A is true and B is false. But this is exactly the case where $\neg B$ is true and $\neg A$ is false, so the contrapositive also fails in precisely the same states of the world. Therefore the two statements always have the same truth value. $\square$

Common Mistake

Do not confuse the *contrapositive* with the *converse*. In general,

$$A \rightarrow B$$

does *not* imply

$$B \rightarrow A.$$

Worked Example: Using a contrapositive

Consider the statement:

$$\text{If } x + y = 1000, \text{ then either } x \geq 500 \text{ or } y \geq 500.$$

A direct proof must consider all cases with $x + y = 1000$. It is easier to prove the contrapositive:

$$\text{If } x < 500 \text{ and } y < 500, \text{ then } x + y \neq 1000.$$

But if both x and y are less than 500, then

$$x + y < 1000.$$

So the contrapositive is true, and hence the original statement is also true.

Exam Focus

In this lecture, several results are proved most cleanly by contraposition:

$$A \rightarrow B \iff \neg B \rightarrow \neg A.$$

This is often easier than proving the original implication directly.

11.2 Rawlsian and Utilitarian Social Welfare

Lecture 10 introduced social welfare functions. Lecture 11 contrasts two central examples.

Definition 11.2: Rawlsian and utilitarian social welfare functions

For utility vector

$$u = (u_1, \dots, u_n),$$

two common social welfare functions are:

- **Rawlsian:**

$$W_R(u) = \min\{u_1, \dots, u_n\};$$

- **Utilitarian:**

$$W_U(u) = \sum_{i=1}^n u_i.$$

The Rawlsian criterion puts maximal emphasis on the worst-off individual, while the utilitarian criterion puts equal weight on increasing the sum of utilities.

Policy Application

The lecture's goal is not to prove that utilitarianism is always morally correct. Rather, it gives a precise economic argument showing that utilitarian allocations have a special efficiency property once money transfers and quasi-linear utility are introduced.

11.3 Social Choice with Money Transfers

The lecture now enriches the social choice problem by allowing monetary transfers alongside allocations of goods.

Definition 11.3: Allocation, transfer scheme, and policy

A *transfer scheme* is a vector

$$t = (t_1, \dots, t_n),$$

where:

- $t_i > 0$ means individual i receives money;

- $t_i < 0$ means individual i pays money.

A *policy* is a pair

$$(A, t),$$

consisting of:

- an allocation A of goods;
- a monetary transfer scheme t .

Definition 11.4: Balanced transfers

A transfer scheme is *budget balanced* if

$$\sum_{i=1}^n t_i = 0.$$

Thus money is only redistributed across individuals; no outside surplus or deficit is created.

Theorem 11.2: Why balanced transfers are needed

If unrestricted transfer schemes are allowed, Pareto efficiency of policies becomes trivial: any policy can be Pareto improved upon by giving every individual additional money.

Proof. Suppose (A, t) is any policy. If transfers need not balance, choose $\varepsilon > 0$ and consider

$$t'_i = t_i + \varepsilon \quad \text{for all } i.$$

If utility is increasing in money, then every individual is strictly better off under (A, t') than under (A, t) . Hence (A, t) cannot be Pareto efficient. Therefore meaningful analysis requires balanced transfers. $\square$

Common Mistake

Do not confuse an *allocation* with a *policy*. In this chapter:

- an allocation concerns only goods;
- a policy concerns goods *and* transfers.

Pareto efficiency is applied to policies, not just to allocations.

11.4 Quasi-Linear Utility and Willingness to Pay

The central assumption is that individuals value money directly and linearly.

Definition 11.5: Quasi-linear utility with transfers

Suppose that if allocation A is chosen and individual i receives transfer t_i , utility is

$$U_i(A, t_i) = v_i(A) + t_i,$$

where:

- $v_i(A)$ is the utility derived from the allocation itself;
- t_i is money, valued one-for-one in utility.

This means money can be used to transfer utility across individuals in a direct way.

Theorem 11.3: Willingness to pay under quasi-linear utility

Suppose the status quo is allocation A , and the new allocation is B . If moving from A to B requires individual i to pay c_i , then i prefers the change if and only if

$$c_i \leq v_i(B) - v_i(A).$$

Hence

$$v_i(B) - v_i(A)$$

is individual i 's maximum willingness to pay to move from A to B .

Proof. Individual i weakly prefers the new policy if

$$U_i(B, -c_i) \geq U_i(A, 0).$$

Using quasi-linearity,

$$v_i(B) - c_i \geq v_i(A),$$

which is equivalent to

$$c_i \leq v_i(B) - v_i(A).$$

□

Worked Example: Movie ticket allocation with transfers

Suppose Joseph and Ivy both value a movie ticket, with willingness to pay

$$WTP_J < WTP_I.$$

If Joseph initially gets the ticket and no money changes hands, that allocation is Pareto efficient *without* transfers: giving the ticket to Ivy would hurt Joseph.

Now allow money transfers. Give the ticket to Ivy and let Ivy pay Joseph an amount x such that

$$WTP_J \leq x \leq WTP_I.$$

Then:

- Joseph weakly prefers losing the ticket and receiving x ;
- Ivy weakly prefers getting the ticket and paying x ;
- at least one of them is strictly better off unless the inequalities are equalities at both ends.

So the initial allocation is Pareto improved upon.

Hence, once transfers are allowed, Pareto efficiency requires the ticket to go to the person with the highest willingness to pay.

11.5 Utilitarian Allocations

Since transfers only change the distribution of utility, the relevant measure of the size of the “utility pie” generated by an allocation is the sum of $v_i(A)$ across individuals.

Definition 11.6: Utilitarian allocation

An allocation A is *utilitarian* if it maximizes

$$\sum_{i=1}^n v_i(A)$$

over all feasible allocations.

Equivalently, society weakly prefers allocation A to allocation B under utilitarianism if and only if

$$\sum_{i=1}^n v_i(A) \geq \sum_{i=1}^n v_i(B).$$

Exam Focus

In this chapter, utilitarianism is applied to *allocations*, not policies. Money transfers are dealt with separately through the vector t .

11.6 Pareto Efficiency of Policies with Money Transfers

We now define Pareto efficiency for policies, subject to balanced transfers.

Definition 11.7: Pareto improvement and Pareto efficiency for policies

A policy (B, t') Pareto improves on policy (A, t) if

$$U_i(B, t'_i) \geq U_i(A, t_i) \quad \text{for all } i,$$

and

$$U_j(B, t'_j) > U_j(A, t_j) \quad \text{for some } j.$$

A policy is *Pareto efficient* if there is no other feasible policy with balanced transfers that Pareto improves on it.

The key result of the lecture is that under quasi-linear utility and balanced transfers, Pareto-efficient policies and utilitarian allocations coincide.

Theorem 11.4: Utilitarianism and Pareto efficiency with money transfers

Suppose:

- each individual has quasi-linear utility

$$U_i(A, t_i) = v_i(A) + t_i;$$

- only balanced transfer schemes are allowed:

$$\sum_{i=1}^n t_i = 0.$$

Then a policy (A, t) is Pareto efficient for some balanced transfer scheme t if and only if A is a utilitarian allocation.

Proof. We prove both directions by contraposition.

Direction 1. If A is utilitarian, then (A, t) is Pareto efficient for some t .

It is enough to prove the contrapositive:

If (A, t) is not Pareto efficient for any balanced t , then A is not utilitarian.

Take any balanced t . If (A, t) is not Pareto efficient, then there exists another balanced policy (B, t') such that

$$v_i(B) + t'_i \geq v_i(A) + t_i \quad \text{for all } i,$$

with strict inequality for at least one individual. Summing across all individuals gives

$$\sum_{i=1}^n (v_i(B) + t'_i) > \sum_{i=1}^n (v_i(A) + t_i).$$

Because both transfer schemes are balanced,

$$\sum_{i=1}^n t'_i = \sum_{i=1}^n t_i = 0,$$

so

$$\sum_{i=1}^n v_i(B) > \sum_{i=1}^n v_i(A).$$

Therefore A does not maximize total utility and is not utilitarian.

Direction 2. If (A, t) is Pareto efficient for some balanced t , then A is utilitarian.

Again prove the contrapositive:

If A is not utilitarian, then for any balanced t , (A, t) is not Pareto efficient.

If A is not utilitarian, then there exists some feasible allocation B such that

$$\sum_{i=1}^n v_i(B) > \sum_{i=1}^n v_i(A).$$

Take any balanced transfer scheme t . Define a transfer vector

$$\hat{t}_i = v_i(A) + t_i - v_i(B).$$

Then for each individual i ,

$$U_i(B, \hat{t}_i) = v_i(B) + \hat{t}_i = v_i(A) + t_i = U_i(A, t_i).$$

So each individual is indifferent between (A, t) and $(B, \hat{t})$.

Now sum these transfers:

$$\sum_{i=1}^n \hat{t}_i = \sum_{i=1}^n v_i(A) + \sum_{i=1}^n t_i - \sum_{i=1}^n v_i(B) = \sum_{i=1}^n v_i(A) - \sum_{i=1}^n v_i(B) < 0,$$

since t is balanced and B yields strictly higher total utility. Thus $\hat{t}$ runs a budget surplus. Let

$$S = - \sum_{i=1}^n \hat{t}_i > 0.$$

Redistribute this surplus equally by defining

$$t'_i = \hat{t}_i + \frac{S}{n}.$$

Then

$$\sum_{i=1}^n t'_i = 0,$$

so t' is balanced. Also,

$$U_i(B, t'_i) = U_i(B, \hat{t}_i) + \frac{S}{n} = U_i(A, t_i) + \frac{S}{n} > U_i(A, t_i)$$

for every individual i . Hence (B, t') Pareto improves on (A, t) , so (A, t) is not Pareto efficient.

This proves both directions. □

Theorem 11.5: Any balanced transfer supports a utilitarian allocation as Pareto efficient

Under the assumptions of Theorem 11.4, if A is utilitarian, then

$$(A, t)$$

is Pareto efficient for *every* balanced transfer scheme t .

Proof. If some balanced policy (B, t') Pareto improved on (A, t) , then by the summation argument in the proof of Theorem 11.4,

$$\sum_{i=1}^n v_i(B) > \sum_{i=1}^n v_i(A),$$

contradicting that A is utilitarian. □

11.7 Interpretation

The theorem shows that, under the lecture's assumptions, the policy problem splits cleanly into two parts.

Policy Interpretation: Efficiency then distribution

Under quasi-linear utility and balanced transfers:

1. choose the allocation that maximizes

$$\sum_i v_i(A),$$

i.e. the largest utility pie;

2. then use money transfers to determine how that pie is shared.

So any non-utilitarian allocation is Pareto dominated by some policy built from a utilitarian allocation.

Graphically, the lecture interprets this as follows:

- without transfers, different allocations generate a utility possibility frontier;
- with balanced transfers and quasi-linear utility, the frontier expands along lines of slope -1 , because utility can be shifted across individuals through money;
- the outer frontier is attained by utilitarian allocations.

Exam Focus

The strongest intuition is:

Different allocations determine the *size* of the utility pie. Transfers determine only the *division* of that pie.

Under quasi-linearity, any division of a smaller pie is Pareto dominated by some division of a larger pie.

11.8 Worked Review Examples**Worked Example: Which utility forms satisfy the utilitarianism argument?**

Suppose policies involve allocations (x_1^i, x_2^i) and transfers t_i . In which of the following cases does the lecture's utilitarianism argument apply?

$$(a) \quad U_i = x_1^i x_2^i + t_i$$

$$(b) \quad U_i = x_1^i x_2^i$$

$$(c) \quad U_i = x_1^i x_2^i t_i$$

$$(d) \quad U_i = 0.5x_1^i + 0.5x_2^i + 0.5t_i$$

The argument requires utility to be quasi-linear in money, so that money enters additively with a constant marginal utility.

- (a) Yes. This is of the form

$$U_i = v_i(x) + t_i$$

with

$$v_i(x) = x_1^i x_2^i.$$

- (b) No. There is no monetary-transfer term.
- (c) No. Money enters multiplicatively, so utility is not quasi-linear in money.
- (d) Yes. This is still quasi-linear in money:

$$U_i = v_i(x) + \alpha t_i$$

with constant $\alpha = 0.5 > 0$. Since the coefficient on money is constant and common, the same logic applies after a positive rescaling of utility.

Therefore the correct choices are

(a) and (d).

Worked Example: Which allocations can be part of a Pareto-efficient policy?

Consider a two-person, one-good economy with balanced money transfers (t_A, t_B) and utilities

$$U_A = x_A + t_A, \quad U_B = x_B + t_B.$$

Suppose the feasible allocations are:

$$(a) \quad (x_A, x_B) = (4, 4), \quad (b) \quad (x_A, x_B) = (5, 3), \quad (c) \quad (x_A, x_B) = (3, 5).$$

Because

$$v_A(x) = x_A, \quad v_B(x) = x_B,$$

the utilitarian sum for each allocation is

$$x_A + x_B.$$

For all three allocations,

$$x_A + x_B = 8.$$

Hence all three allocations maximize the utilitarian objective over the listed feasible set.

By Theorem 11.4, each utilitarian allocation can be part of a Pareto-efficient policy with some balanced transfers. In fact, by Theorem 11.5, any balanced transfers work.

So the correct answer is

(a), (b), and (c).

11.9 Critical Assumptions and Limitations

The lecture emphasizes two critical assumptions.

Definition 11.8: Critical assumptions behind the utilitarian result

The utilitarian-efficiency argument requires:

1. **Quasi-linear preferences in money:**

$$U_i(A, t_i) = v_i(A) + t_i.$$

2. **Correct implementation of balanced transfers:** society must be able to choose and enforce the needed transfer scheme after selecting the utilitarian allocation.

If these fail, then the theorem may no longer hold.

Common Mistake

Do not say that utilitarianism is always the best policy rule. The lecture's result is explicitly conditional:

quasi-linear utility + balanced transfers + correct implementation.

Without these, the equivalence between Pareto-efficient policies and utilitarian allocations can fail.

Policy Application

Even when the utilitarian allocation is clear, society may still disagree over transfers. The theorem resolves the efficient *allocation* question, but not the distributive question of how the gains from that allocation should be shared.

11.10 Chapter Summary

Key Formula: Lecture 11 Summary

$$\begin{aligned}
A \rightarrow B &\iff \neg B \rightarrow \neg A, \\
U_i(A, t_i) &= v_i(A) + t_i, \\
\sum_{i=1}^n t_i &= 0 \quad (\text{balanced transfers}), \\
\text{WTP}_i(A \rightarrow B) &= v_i(B) - v_i(A), \\
A \text{ utilitarian} &\iff \sum_{i=1}^n v_i(A) \geq \sum_{i=1}^n v_i(B) \quad \text{for all feasible } B.
\end{aligned}$$

Under quasi-linear utility and balanced transfers:

$$(A, t) \text{ is Pareto efficient for some balanced } t \iff A \text{ is utilitarian.}$$

Indeed, if A is utilitarian, then

$$(A, t)$$

is Pareto efficient for every balanced transfer vector t .

Exam Focus

The highest-yield takeaways from this chapter are:

- know the difference between a conditional statement, its contrapositive, and its converse;
- know the difference between an allocation and a policy;
- know why balanced transfers are imposed;
- know how quasi-linear utility makes willingness to pay equal to utility differences;
- know the theorem:

$$\text{Pareto-efficient policy with transfers} \iff \text{utilitarian allocation};$$

- know that the result depends critically on quasi-linear utility and the feasibility of implementing the right transfers.

12 Part II Summary

Part II develops a sequence of welfare frameworks that become progressively broader. Lecture 7 begins at the level of the individual and asks how utility changes can be measured in money. Lecture 8 aggregates consumer and producer welfare in a single market and studies taxes, tax incidence, and deadweight loss. Lecture 9 then moves beyond partial equilibrium to the exchange economy, where Pareto efficiency and competitive equilibrium become the central concepts. Lecture 10 asks how society should rank alternatives or utility outcomes, introducing social decision mechanisms, social welfare functions, and the utility possibility frontier. Finally, Lecture 11 shows that once utility is quasi-linear in money and balanced transfers are feasible, Pareto-efficient policies correspond to utilitarian allocations.

This summary is designed for revision rather than first exposure. The emphasis is therefore on the main objects, the highest-yield results, the links across topics, and the standard patterns behind common exam questions.

12.1 Part II at a Glance

Part II in one conceptual chain

The broad logic of Part II is:

1. measure individual welfare in money;
2. aggregate welfare in a market;
3. generalize efficiency beyond a single market;
4. distinguish efficiency from social choice;
5. show when transfers make utilitarianism the efficiency criterion for policies.

In short, Part II moves from *how much one person gains or loses*, to *how a market performs*, to *which allocations are efficient*, to *which efficient allocations society may want*, and finally to *when money transfers let us separate efficiency from distribution*.

12.2 Core Frameworks and Definitions

12.2.1 Monetary welfare at the individual level

Lecture 7 asks how utility changes can be expressed in dollars.

Definition 12.1: Core monetary welfare measures

The main individual monetary welfare measures are:

- *reservation price*: the maximum amount a consumer is willing to pay for a good or marginal unit;
- *consumer surplus*:

$$CS = \int_0^{x(p)} P(q) dq - p x(p);$$

- *compensating variation (CV)*: the income needed at the *new* prices to restore the *old* utility level;
- *equivalent variation (EV)*: the income removable at the *old* prices that leaves the consumer at the *new* utility level;

- *producer surplus*:

$$PS = \int_0^q (p - MC(\tilde{q})) d\tilde{q}.$$

The key distinction is that consumer surplus is usually convenient but approximate, whereas CV and EV are exact welfare measures for price changes. Consumer surplus becomes exact under quasi-linear utility.

12.2.2 Aggregate welfare in a single market

Lecture 8 aggregates consumer and producer welfare.

Definition 12.2: Single-market social welfare

In the standard one-good market with inverse demand $P_D(Q)$ and inverse supply $P_S(Q)$, social welfare is

$$SW(Q) = CS(Q) + PS(Q) = \int_0^Q (P_D(q) - P_S(q)) dq.$$

The competitive quantity maximizes this expression under the standard downward-sloping demand and upward-sloping supply assumptions.

This framework is still partial equilibrium: it is powerful for one-market tax questions, but it is not yet the general notion of efficiency used in exchange economies.

12.2.3 Pareto efficiency in the exchange economy

Lecture 9 replaces $CS + PS$ with Pareto efficiency as the central efficiency concept.

Definition 12.3: Feasibility, Pareto improvement, and Pareto efficiency

In a pure exchange economy:

- an allocation is *feasible* if total consumption equals total endowment good by good;
- a *Pareto improvement* makes everyone weakly better off and at least one person strictly better off;
- an allocation is *Pareto efficient* if no feasible Pareto improvement exists.

In the two-good, two-consumer case, the Edgeworth box is the main graphical tool, and at a smooth interior Pareto-efficient point,

$$MRS_A = MRS_B.$$

This chapter separates efficiency from fairness: a Pareto-efficient allocation need not be envy-free or otherwise equitable.

12.2.4 Social choice and social welfare

Lecture 10 asks how society ranks alternatives or utility outcomes.

Definition 12.4: SDM, SWF, and social optimum

A *social decision mechanism* (SDM) takes individuals' rankings over alternatives as input and outputs a social ranking. A *social welfare function* (SWF) takes utility levels as input and assigns a social value

$$W(u_1, \dots, u_n).$$

A *social optimum* is a feasible allocation that maximizes the chosen SWF over the feasible utility set.

This is the chapter where the course explicitly turns from positive efficiency analysis to normative aggregation.

12.2.5 Policies with transfers

Lecture 11 adds money transfers to the allocation problem.

Definition 12.5: Allocation, policy, and balanced transfers

An *allocation* specifies who gets the goods. A *policy* specifies both an allocation A and transfers $t = (t_1, \dots, t_n)$. Transfers are *balanced* if

$$\sum_{i=1}^n t_i = 0.$$

Under quasi-linear utility,

$$U_i(A, t_i) = v_i(A) + t_i,$$

so money can redistribute utility directly across individuals.

This is the final conceptual step of Part II: once balanced transfers are feasible and utility is quasi-linear in money, one can separate the choice of the allocation from the choice of who ultimately bears the gains and losses.

12.3 Key Results from Part II

Key Formula: Part II Master Summary

$$\begin{aligned}
CS &= \int_0^{x(p)} P(q) dq - p x(p), \\
V(p, I) &= \max_x U(x) \text{ s.t. } p \cdot x \leq I, \\
V(\text{new prices}, I + CV) &= V(\text{old prices}, I), \\
V(\text{old prices}, I - EV) &= V(\text{new prices}, I), \\
SW(Q) &= \int_0^Q (P_D(q) - P_S(q)) dq, \\
DWL &= \int_{Q'}^{Q^*} (P_D(q) - P_S(q)) dq, \\
\text{interior Pareto efficiency} &\Rightarrow MRS_A = MRS_B, \\
\text{competitive equilibrium} &\Rightarrow \text{Pareto efficiency}, \\
\text{social optimum} &\Rightarrow \text{Pareto efficiency}, \\
W(u_A, u_B) = \gamma_A u_A + \gamma_B u_B &\Rightarrow \text{iso-welfare slope} = -\frac{\gamma_A}{\gamma_B}, \\
U_i(A, t_i) = v_i(A) + t_i, \sum_i t_i = 0 &\Rightarrow (A, t) \text{ is Pareto efficient} \iff A \text{ is utilitarian.}
\end{aligned}$$

The main examinable results can be organized as follows.

Theorem 12.1: High-yield results across Lectures 7–11

The central results of Part II are:

1. Under quasi-linear utility,

$$U(x, m) = v(x) + m,$$

demand for x has no income effect and consumer surplus is an exact welfare measure.

2. In the standard single-good market, the competitive quantity maximizes

$$CS + PS.$$

Taxes reduce quantity, create a wedge between buyer and seller prices, and generally generate deadweight loss.

3. In the exchange economy, a competitive equilibrium allocation is Pareto efficient, and under the standard convexity assumptions any Pareto-efficient allocation can be decentralized after suitable redistribution of endowments.
4. For any increasing SWF, a social optimum is Pareto efficient; conversely, under the regularity assumptions emphasized in the course, each Pareto-efficient point can be supported by some SWF.
5. No non-dictatorial SDM can simultaneously satisfy Universal Domain, the Pareto property, and IIA while always producing a transitive social ranking.
6. Under quasi-linear utility in money and balanced transfers, Pareto-efficient policies are exactly those built from utilitarian allocations.

Exam Focus

A useful way to memorize the architecture of Part II is to attach one question to each lecture:

- **Lecture 7:** how do I measure welfare?
- **Lecture 8:** what quantity is efficient in a market?
- **Lecture 9:** what allocation is Pareto efficient?
- **Lecture 10:** which Pareto-efficient point should society choose?
- **Lecture 11:** when do transfers make utilitarianism the relevant efficiency criterion?

12.4 Cross-Topic Comparisons

12.4.1 Consumer surplus versus EV and CV

Lecture 7 contains one of the most important distinctions in the course.

- CS is computed from Marshallian demand and is usually easy to obtain.
- EV and CV are defined through indirect utility and are exact.
- In general,

$$\min\{|CV|, |EV|\} \leq |\Delta CS| \leq \max\{|CV|, |EV|\}.$$
- Under quasi-linear utility, all three coincide.

This distinction matters again in Lecture 11, because quasi-linearity is exactly the assumption that turns willingness to pay into a clean monetary measure of utility difference.

12.4.2 Market welfare versus Pareto efficiency

Lecture 8 and Lecture 9 use different welfare languages.

- In Lecture 8, welfare is

$$CS + PS,$$
 which is especially useful in a single-market demand–supply diagram.
- In Lecture 9, welfare is not summarized by a single market-area formula. Instead, efficiency means the absence of feasible Pareto improvements.

The conceptual link is that both frameworks are trying to identify unrealized gains from trade. In Lecture 8 this appears as lost surplus triangles. In Lecture 9 it appears as allocations inside the Edgeworth-box improvement lens rather than on the contract curve.

12.4.3 Efficiency versus fairness versus social choice

Part II repeatedly warns against collapsing these concepts into one another.

- A Pareto-efficient allocation need not be fair.
- An envy-free allocation need not be the competitive-equilibrium allocation from the given endowment.
- A social optimum depends on the chosen SWF.
- An SDM need not produce a transitive or stable ranking, as Condorcet cycles show.

So the course does *not* say that efficiency settles all normative questions. Instead, it shows exactly where efficiency analysis ends and social-choice or distributive judgments begin.

12.4.4 Redistribution before trade versus transfers after allocation

Lecture 9 and Lecture 11 solve a similar problem in different ways.

- In Lecture 9, the Second Welfare Theorem says that if society wants a particular Pareto-efficient allocation, it may be supported by first redistributing endowments and then allowing competitive trade.
- In Lecture 11, if utility is quasi-linear in money, society can instead choose the utilitarian allocation and then use balanced money transfers to determine the final distribution of welfare.

The common logic is the separation of *efficiency* from *distribution*. The technical difference is that Lecture 9 works through endowment redistribution in the exchange economy, whereas Lecture 11 works through utility-transferable money.

12.4.5 SDMs versus SWFs

Lecture 10 introduces two very different aggregation objects.

- An SDM uses *ordinal rankings* over alternatives.
- An SWF uses *utility levels* and therefore supports geometric analysis using the utility possibility frontier.

Arrow's theorem is about SDMs on ordinal preference profiles. It does not directly invalidate the SWF analysis used later in Lecture 10 or in Lecture 11.

12.5 Standard Problem Types and Solution Patterns

Method Pattern: Welfare-measure questions

If the question is about reservation price, CS , EV , or CV :

1. identify whether the object is discrete, continuous, or a price change;
2. if exact monetary welfare is needed, derive indirect utility and solve for EV or CV ;
3. check whether utility is quasi-linear, because that may let you use consumer surplus exactly;
4. keep the reference point straight:

$$CV = \text{new prices, old utility}, \quad EV = \text{old prices, new utility}.$$

Method Pattern: Tax and deadweight-loss questions

If the question is about a per-unit or ad-valorem tax:

1. define buyer price p_b and seller price p_s ;
2. write the wedge condition;
3. solve demand equals supply using the correct post-tax prices;
4. compare with the no-tax benchmark;
5. compute tax revenue, incidence, and deadweight loss separately.

Method Pattern: Exchange-economy questions

If the question is about Pareto efficiency or competitive equilibrium in an exchange economy:

1. write total resources and feasibility conditions;
2. derive each consumer's demand from utility and endowment wealth;
3. solve for the equilibrium price ratio using market clearing;
4. if checking efficiency, use either the improvement-lens logic or

$$MRS_A = MRS_B$$

at smooth interior allocations;

5. if fairness is asked, test envy-freeness separately.

Method Pattern: Social-choice questions

If the question is about an SDM or SWF:

1. distinguish immediately whether the object is an SDM or an SWF;
2. for an SDM, compute the social ranking methodically and then test properties such as Pareto, Universal Domain, and IIA;
3. for an SWF over a utility possibility set, look for the highest attainable iso-welfare curve on the frontier;
4. for weighted utilitarianism in two dimensions, match the frontier slope to

$$-\frac{\gamma_A}{\gamma_B}.$$

Method Pattern: Lecture 11 utilitarianism questions

If the question involves money transfers:

1. distinguish allocation from policy;
2. check that transfers are balanced;
3. check whether utility is quasi-linear in money:

$$U_i(A, t_i) = v_i(A) + t_i;$$

4. compute willingness to pay as

$$v_i(B) - v_i(A);$$

5. identify the allocation that maximizes

$$\sum_i v_i(A).$$

12.6 Common Mistakes and Exam Traps

Common Mistake

Do not confuse the reservation price curve with the inverse demand curve. They coincide only under special assumptions, especially quasi-linear utility.

Common Mistake

Do not confuse the statutory side of a quantity tax with its economic incidence. The side that formally remits the tax need not be the side that bears more of the burden.

Common Mistake

Do not say that a Pareto-efficient allocation is automatically fair. Pareto efficiency rules out mutually beneficial reallocation; it does not say anything by itself about envy, equality, or justice.

Common Mistake

Do not confuse an SDM with an SWF. Arrow's theorem concerns SDMs defined on ordinal rankings, not every possible welfare criterion used in economics.

Common Mistake

Do not confuse the *contrapositive*

$$A \rightarrow B \iff \neg B \rightarrow \neg A$$

with the *converse*

$$B \rightarrow A.$$

Lecture 11 uses contraposition repeatedly, and losing this distinction can collapse an otherwise correct proof.

Common Mistake

Do not confuse an allocation with a policy in Lecture 11. Pareto efficiency there is applied to policies (A, t) , not merely to allocations A in isolation.

12.7 Integrated Worked Review Examples

Worked Review Example 1: When is consumer surplus exact?

Suppose a consumer has quasi-linear utility

$$U(x, m) = 2\sqrt{x} + m,$$

income is high enough that the interior optimum is affordable, and the price of x rises from

$$p' = 1 \quad \text{to} \quad p'' = 2.$$

Because utility is quasi-linear in money, the optimal quantity solves

$$\frac{d}{dx}(2\sqrt{x} - px) = 0 \implies \frac{1}{\sqrt{x}} = p.$$

Hence

$$x(p) = \frac{1}{p^2}.$$

So

$$x(1) = 1, \quad x(2) = \frac{1}{4}.$$

The loss in consumer surplus from the price increase is

$$\Delta CS_{\text{loss}} = \int_1^2 x(p) dp = \int_1^2 \frac{1}{p^2} dp = \left[-\frac{1}{p} \right]_1^2 = \frac{1}{2}.$$

Under quasi-linear utility, consumer surplus is exact, so

$$\Delta CS = EV = CV = \frac{1}{2}.$$

This is the key revisional lesson from Lecture 7: whenever the utility function is quasi-linear in money, the distinction between CS , EV , and CV disappears.

Worked Review Example 2: Linear tax incidence and deadweight loss

Suppose

$$Q^D = 20 - p_b, \quad Q^S = p_s,$$

and a per-unit tax of

$$t = 4$$

is imposed.

Without tax:

$$20 - p = p \implies p^* = 10, \quad Q^* = 10.$$

With tax:

$$p_b = p_s + 4, \quad 20 - p_b = p_s.$$

Substituting gives

$$20 - (p_s + 4) = p_s \implies 16 = 2p_s \implies p_s = 8.$$

Hence

$$p_b = 12, \quad Q' = 8.$$

So buyers' price rises by

$$12 - 10 = 2,$$

while sellers' price falls by

$$10 - 8 = 2.$$

Thus the tax burden is shared equally in this case.

Deadweight loss is

$$DWL = \frac{1}{2} \times 4 \times (10 - 8) = 4.$$

Therefore

$$p_b = 12, \quad p_s = 8, \quad Q' = 8, \quad DWL = 4.$$

This example consolidates the Lecture 8 pattern: write the wedge condition carefully, solve for the post-tax equilibrium, then keep incidence and deadweight loss conceptually separate.

Worked Review Example 3: Competitive equilibrium, efficiency, and fairness

Consider a two-good, two-consumer exchange economy with total resources

$$(12, 12),$$

endowments

$$\omega^A = (4, 8), \quad \omega^B = (8, 4),$$

and utilities

$$U_A(x_1, x_2) = x_1^{1/2} x_2^{1/2}, \quad U_B(x_1, x_2) = x_1^{1/2} x_2^{1/2}.$$

Normalize prices so that

$$p_2 = 1, \quad p_1 = p.$$

Each consumer has Cobb–Douglas demand with equal expenditure shares, so

$$x_1^i = \frac{m_i}{2p}, \quad x_2^i = \frac{m_i}{2},$$

where

$$m_A = 4p + 8, \quad m_B = 8p + 4.$$

Market clearing for good 1 requires

$$\frac{4p + 8}{2p} + \frac{8p + 4}{2p} = 12,$$

so

$$\frac{12p + 12}{2p} = 12 \implies 12p + 12 = 24p \implies p = 1.$$

Thus

$$m_A = m_B = 12,$$

and each consumer chooses

$$x^{A*} = x^{B*} = (6, 6).$$

Hence the competitive equilibrium allocation is

$$\boxed{x^{A*} = x^{B*} = (6, 6)}.$$

Because this is a competitive equilibrium, Lecture 9 implies it is Pareto efficient. It is also envy-free here because both consumers receive the same bundle and have identical preferences. The revisional point is that these are distinct properties:

competitive equilibrium $\Rightarrow$ Pareto efficiency,

but envy-freeness is an additional fairness test, not part of the First Welfare Theorem itself.

Worked Review Example 4: Condorcet cycles under pairwise majority voting

Suppose three voters have preferences

$$\text{Voter 1: } X \succ Z \succ Y, \quad \text{Voter 2: } Z \succ Y \succ X, \quad \text{Voter 3: } Y \succ X \succ Z.$$

Pairwise comparisons give:

- Y beats X , because voters 2 and 3 prefer Y to X ;
- X beats Z , because voters 1 and 3 prefer X to Z ;

- Z beats Y , because voters 1 and 2 prefer Z to Y .

Therefore

$$Y \succ^* X, \quad X \succ^* Z, \quad Z \succ^* Y.$$

So the social ranking cycles:

$$Y \succ^* X \succ^* Z \succ^* Y.$$

This example captures the core Lecture 10 warning: a seemingly natural voting rule can fail to produce a transitive social ranking on an unrestricted domain.

Worked Review Example 5: Utilitarian allocation with balanced transfers

Suppose a single indivisible ticket must be assigned to either Ann or Ben. Their willingness to pay is

$$WTP_A = 10, \quad WTP_B = 16.$$

Without transfers: if Ann initially has the ticket, giving it to Ben harms Ann, so the initial allocation may already be Pareto efficient relative to allocations alone.

With balanced transfers and quasi-linear utility: giving the ticket to Ben and making Ben pay Ann an amount x with

$$10 \leq x \leq 16$$

makes Ann weakly willing to give up the ticket and Ben weakly willing to receive it.

Hence any transfer in that interval supports a Pareto improvement, and the allocation that gives the ticket to Ben is the utilitarian one because it maximizes total willingness to pay.

So the efficient policy is built from the allocation

ticket to Ben

together with a balanced transfer x from Ben to Ann.

This is the core Lecture 11 insight: once quasi-linear money transfers are feasible, the efficient allocation sends the object to the person with the highest willingness to pay, while transfers determine distribution.

12.8 Lecture 11 Final Synthesis

Lecture 11 is the endpoint of Part II, so it is worth stating its conceptual role explicitly. Chapters 7–10 gradually build the machinery needed for the final theorem.

- Lecture 7 develops monetary welfare measures and shows why quasi-linearity matters.
- Lecture 8 shows how monetary measures can be aggregated into market welfare.
- Lecture 9 shows that efficiency can be defined without relying on a single-market surplus diagram.
- Lecture 10 distinguishes efficiency from the separate problem of social ranking.
- Lecture 11 then proves that under quasi-linear utility in money and balanced transfers, the efficient policy problem collapses to a utilitarian allocation problem.

Efficiency then distribution

The final message of Part II is:

1. choose the allocation that maximizes

$$\sum_{i=1}^n v_i(A);$$

2. then use balanced transfers to determine how the resulting utility pie is divided.

Thus, under the Lecture 11 assumptions, different allocations determine the *size* of the pie, while transfers determine only its *division*.

Exam Focus

If an exam question asks why utilitarianism appears in Lecture 11, the correct answer is not simply “because it is morally attractive.” The lecture’s actual argument is structural:

- utility is quasi-linear in money;
- transfers are balanced;
- willingness to pay equals utility difference in money terms;
- therefore Pareto-efficient policies are exactly those based on utilitarian allocations.

12.9 Part II Revision Checklist

What you should be able to do before the exam

By the end of revision for Part II, you should be able to:

1. compute reservation prices, consumer surplus, producer surplus, EV , and CV , and state when consumer surplus is exact;
2. solve linear tax problems with buyer price, seller price, tax revenue, incidence, and deadweight loss all clearly distinguished;
3. interpret an Edgeworth box, identify Pareto improvements, and solve basic two-good, two-consumer competitive-equilibrium problems;
4. explain the First and Second Welfare Theorems and state the assumptions under which they are used in the course;
5. distinguish SDMs from SWFs, identify Condorcet-type failures, and interpret tangency on the utility possibility frontier;
6. state Arrow’s theorem accurately without overclaiming what it proves;
7. distinguish allocations from policies in Lecture 11, compute willingness to pay under quasi-linear utility, and identify utilitarian allocations;
8. explain clearly how Part II separates efficiency questions from distributional or social-choice questions.

Key Formula: Part II Final Recall Sheet

$$\begin{aligned}
CS &= \int_0^{x(p)} P(q) dq - px(p), \\
PS &= \int_0^q (p - MC(\tilde{q})) d\tilde{q}, \\
V(\text{new}, I + CV) &= V(\text{old}, I), \\
V(\text{old}, I - EV) &= V(\text{new}, I), \\
SW(Q) &= \int_0^Q (P_D(q) - P_S(q)) dq, \\
DWL &= \int_{Q'}^{Q^*} (P_D(q) - P_S(q)) dq, \\
\text{interior PE} &\Rightarrow MRS_A = MRS_B, \\
\text{CE} &\Rightarrow \text{PE}, \\
\text{PE} &\Rightarrow \text{social optimum for some SWF}, \\
\text{weighted utilitarian slope} &= -\frac{\gamma_A}{\gamma_B}, \\
U_i(A, t_i) &= v_i(A) + t_i, \quad \sum_i t_i = 0, \\
(A, t) \text{ PE policy} &\iff A \text{ utilitarian}.
\end{aligned}$$

A Formula & Key Results Reference

- Chapter 1: Budget sets, direct and strict revealed preference, WARP.
- Part I: Slutsky decomposition, endowment-budget formulas, intertemporal constraints, asset-pricing identities, and uncertainty results.
- Part II: Welfare measures, surplus formulas, Pareto efficiency criteria, and social choice results.

B Notation

symbol	meaning	context
$x \in \mathbb{R}_+^L$	consumption bundle	Consumer theory chapters
$p \in \mathbb{R}_{++}^L$	price vector	Consumer theory chapters
m	expenditure / wealth level	Budget constraints
$B(p, m)$	budget set	Choice under prices and expenditure
$\succeq, \succ, \sim$	weak preference, strict preference, indifference	Preference theory
R, P	directly revealed preference, strictly directly revealed preference	Revealed preference